Device Management application

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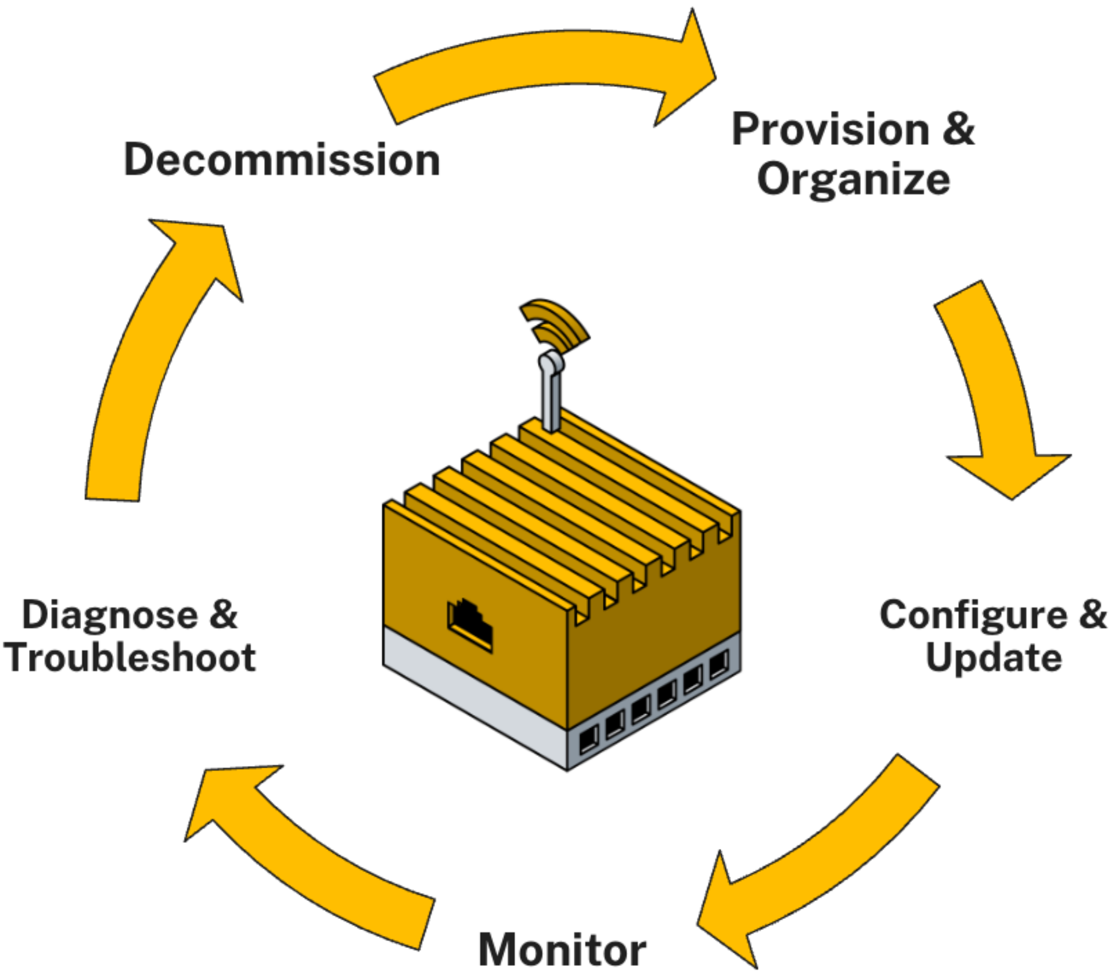
INTRODUCTION

Device management is essential for building a robust and scalable foundation for any IoT solution, bridging the gap between your IoT devices and the (business) application layer.

In order to operate IoT systems efficiently, devices need to be monitored and managed across their entire lifecycle from installation and setting up their connection up until their decommissioning and replacement. This process inherits significant complexity due to, for example, geographical distribution, the heterogeneity and constraints of used devices as well as the sheer number of devices being part of the deployment.

In addition, device management plays an important role in ensuring compliance with the highest security standards, such as managing device certificates and providing over-the-air update mechanisms to prevent devices from becoming non-compliant or even compromised.

Cumulocity's Device Management application provides a unified control pane from which all functionalities can be accessed that are required to monitor and operate devices, independent of the hardware architecture or protocol the devices are using. As this functionality is provided via an intuitive web UI, it can be used without requiring in-depth technical knowledge. By abstracting device specifics and allowing to address multiple devices at once, the effort of maintaining device fleets at scale can be significantly reduced:



Provision & organize

Onboard new devices either by registering each device individually or via bulk registration, and organize them by using groups. These can either be based on individual selection of devices (static groups) or based on any property (smart groups). Additionally, the Digital Twin Manager application can be used to model complex hierarchies and add metadata in order to create holistic representations of your devices and assets.

Configure & update

Setup devices by defining device profiles that contain everything the device needs for being fully operational including firmware, software packages as well as configuration files. Single and bulk operations allow for efficient update campaigns once a newer version of a rolled-out artifact is available to keep the fleet up to date. Device credentials and certificates can directly be managed within the UI to ensure secure operations.

Monitor

The devices list contains all devices that are connected to the tenant. The list can be customized to show all relevant properties like the connectivity status, and includes search and filter options to quickly identify the correct device(s). When selecting a device, an info page for the particular device is displayed which can be customized in order to hold all information that is relevant for the specific device. Depending on the operations supported by the device, additional tabs are available that allow you to interact with the device. To provide transparency across the entire device fleet, views are available to see all events and alarms that recently occurred, the status of operations that got triggered, and the current location of each device.

Diagnose & troubleshoot

In case a device is not behaving as expected, additional information can be retrieved in the form of log files. The relevant timeframe and log type can be further specified in the application. To resolve issues, the device can be restarted or remotely accessed to fix the problem without having to send out a service technician. Additionally, shell commands can be sent via SMS as a fallback in case the internet connection is broken. In case the issues are related to the services running on top of the devices and not to the device itself, the software management functionality can be used to monitor, update and reinstall any service.

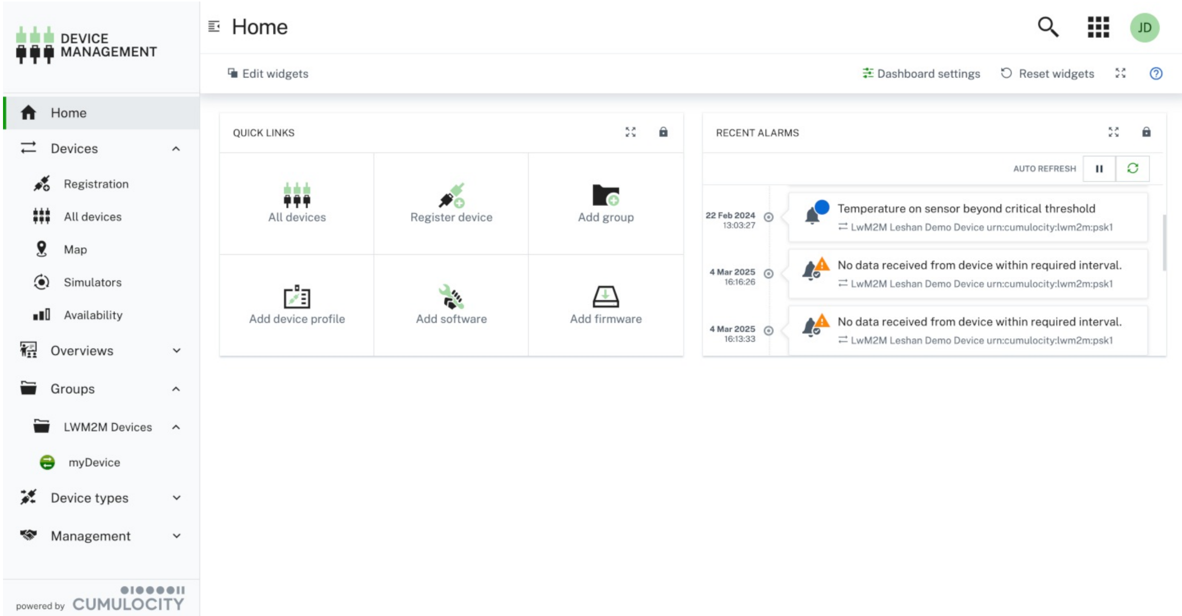
Decomission

At one point in time, the device will reach its end of life. The digital representation of the device can either be permanently deleted if not needed anymore or the device can be replaced. The replacement wizard allows you to map the digital representation to another physical device, so that physical components can be replaced without losing the historical data of the asset that might still be operational.

Visit the Device integration section to learn more about the specifics on how to connect and integrate your devices into Cumulocity's device management suite. With thin-edge.io you can directly benefit from the aforementioned functionality as it comes with full support out-of-the-box.

HOME DASHBOARD

The Home screen of the Device Management application is a dashboard which shows data for the tenant.



The data shown on the Home dashboard is shared by all users of the tenant. By default, the Home dashboard includes recent alarms and quick links. The Home dashboard can be edited and designed individually according to your needs. You can add, remove or change widgets which are displayed here.

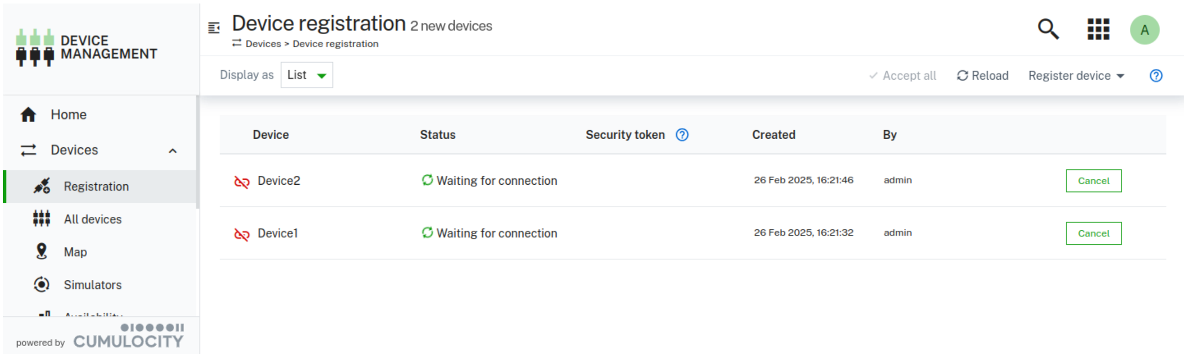
For details on editing a dashboard, refer to Working with dashboards. The Device Management application dashboard works just like the Cockpit dashboard.

To reset the Home dashboard to its default, click **Restore dashboard** at the right of the top menu bar.

REGISTERING DEVICES

OVERVIEW

In the **Device registration** page all devices are displayed which are currently in the registration process.



The following information is shown for each device:

- Device name specified in the registration process
- Status of the device (see below)
- Creation date
- Tenant from which the device was registered

The devices may have one of the following status:

- **Waiting for connection** - The device has been registered but no device with the specified ID has tried to connect.
- **Pending acceptance** - There is communication from a device with the specified ID, but the user doing the registration must still explicitly accept it so that the credentials are sent to the device.
- **Accepted** - The user has allowed the credentials to be send to the device.
- **Blocked** - The device registration has been blocked due to the exceeded limit of failed attempts.

INFO

If a device registration is **blocked**, you will need to delete it first and then create it again.

Devices can be connected to your Cumulocity account in different ways.

To register devices, you can select one of the following options:

- **Single device registration** - to manually connect one device or several devices one by one.
- **Bulk device registration** - to register larger amounts of devices in one step.

Microservice developers can also use the Extensible device registration and implement a custom registration form that blends seamlessly into the UI.

INFO

The following descriptions apply to the general device registration processes. If you subscribe to specific protocol integrations, you will see additional protocol-specific options (for example, for LWM2M or OPC UA). A full list of supported protocols can be found in Device integration. It also contains descriptions for the protocol specific registration processes.

RELATED TOPICS

- Device management & connectivity > Device integration for step-by-step instructions on registering devices via REST or MQTT, and for details on registering devices using various standard protocol types.
- The New device requests API for REST API methods concerning the creation of new devices.

SINGLE DEVICE REGISTRATION

Cumulocity offers single device registration to connect devices manually one by one.

TO CONNECT A DEVICE MANUALLY

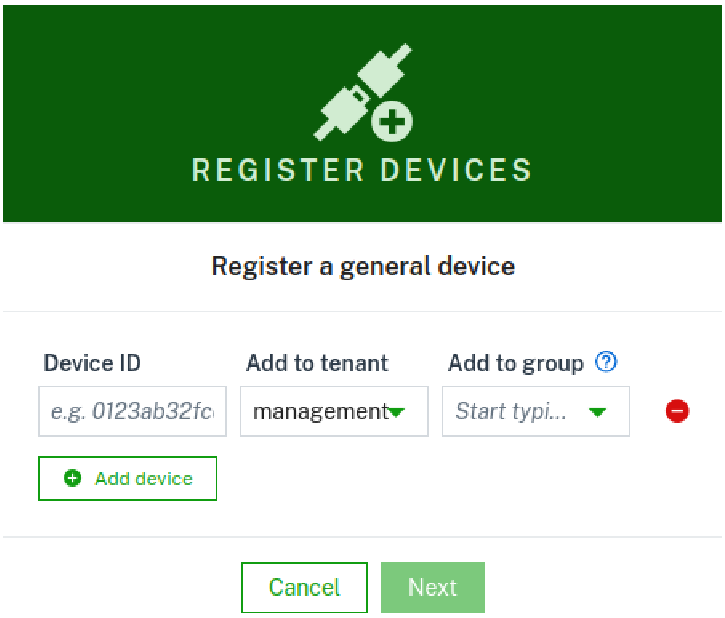
INFO

Depending on the type of device you want to connect, not all steps of the following process may be relevant.

1. Click **Registration** in the **Devices** menu of the navigator.
2. In the **Device registration** page, click **Register device** at the right of the top bar and from the dropdown menu select **Single registration** > **General**. The **Register devices** dialog box will be displayed.
3. In the **Device ID** field, enter a unique ID for the device. To determine the ID, consult the device documentation. In case of mobile devices the ID usually is the IMEI (International Mobile Equipment Identity) often found on the back of the device.
4. Optionally, select a group to assign your device to after registration, see also Grouping devices.
5. Click **Add device** to register one more device. Again, enter the device ID and optionally select a group. This way, you can add multiple devices in one step.
6. Click **Next** to register your device(s).

INFO

In an Enterprise tenant, the Management tenant may also directly select a tenant to which the device will be added from here. Note that since the Management tenant does not have access to the subtenant’s inventory you can either register devices to a tenant OR to a group, not both.



After successful registration the device(s) will be listed in the Device registration page with the status “Waiting for connection”.

Turn on the device(s) and wait for the connection to be established.

Once a device is connected, its status will change to “Pending acceptance”.

INFO

The **Pending acceptance** screen might differ depending on the security token policy.

Click **Accept** to confirm the connection. The status of the device will change to “Accepted”.

INFO

In case of any issues, consult the documentation applicable for your device type in the Cumulocity Partner Devices Ecosystem or look up the manual of your device.

SECURITY TOKEN POLICY

Configure the security token policy to reduce the risk of devices which are not yet registered being taken over by threat actors, for example, by guessing their serial numbers.

INFO

The feature requires READ permission for “Option management”. If the permission is missing, the security token policy defaults to OPTIONAL.

Cumulocity supports the following values for the security token policy:

- IGNORED - Even if a device requires secure registration, Cumulocity will ignore that requirement.
- OPTIONAL - If a device requires secure registration, Cumulocity will request an additional security token from the user.
- REQUIRED - All devices connected to Cumulocity must use a security token during registration.

The policy can be configured by setting the following tenant option with one of the values listed above, for example:

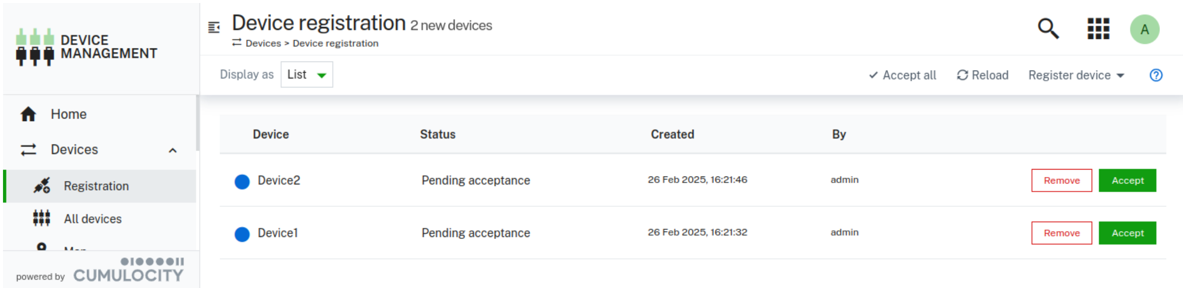
```
{
 "category": "device-registration",
 "key": "security-token.policy",
 "value": "IGNORED"
}
```

INFO

The **Pending acceptance** screen might differ depending on the security token policy.

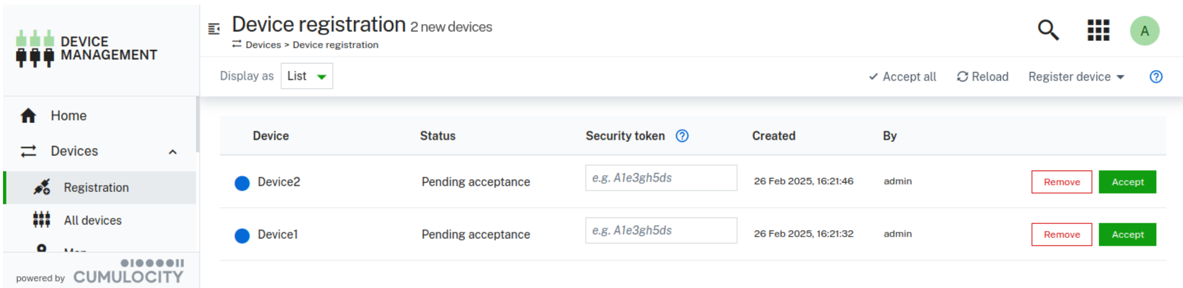
Ignored security token policy

With a value of IGNORED for the security token policy, a device connected to Cumulocity can be accepted without any token validation:



Optional security token policy

The list of device registrations is presented in the image below. Note that the input for security token is displayed for all devices.



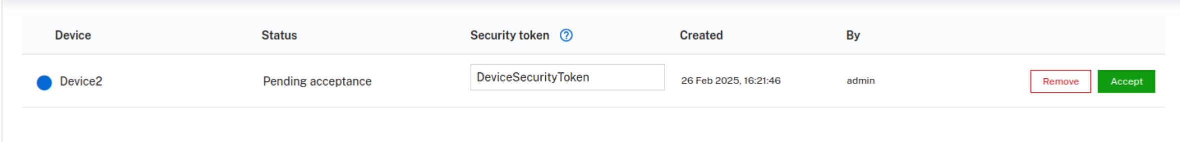
Registration without using a security token

When a device connected to Cumulocity doesn't use a security token, the registration can proceed without providing any value in the security token input.

If a security token is provided for a device which is connected insecurely, it will be accepted and the token will be ignored.

Registration using a security token

When a device connected to Cumulocity does use a security token, the registration can be completed only if the user provides a token matching the one sent by the device on establishing the connection.



In the case of providing an incorrect token, an error message will be displayed informing about a mismatch between the value used by the device and the value provided via the user interface.

After a certain amount of failed attempts, the registration will reach the blocked state, indicated by a corresponding error message. The blocked registration must be removed before the next attempt to connect the device.

Limited usage of “Accept all” feature

The **accept all** feature is supported for devices connected to Cumulocity without the usage of a security token.

For any device which uses a security token, the **accept all** feature is not available and will display a warning message. The details of the warning message provide the list of devices which could not be accepted automatically.

Such devices must be accepted manually by providing the correct **Security token** value and clicking **Accept**.

Required security token policy

In this mode any device connected to Cumulocity must use a security token on establishing the connection and the user must enter the same token when accepting the device.

The procedure of accepting devices is the same as described in Optional security token policy.

While in this mode, any devices connecting to Cumulocity without a security token will be blocked and it won’t be possible to complete their registration.

BULK DEVICE REGISTRATION

To connect larger amounts of devices, Cumulocity offers the option to bulk-register devices, that means, to register larger amounts of devices by uploading a CSV file.

INFO

There is no restriction on the number of devices that you can bulk-register but the more devices you add the slower the creation and operation gets.

TO BULK-REGISTER DEVICES

1. Click **Registration** in the **Devices** menu of the navigator.
2. In the **Device registration** page, click **Register device** at the right of the top bar and from the dropdown menu select **Bulk registration** > **General**. The **Bulk device registration** dialog box will be displayed.
3. Click the Plus button to select or drag-and-drop the CSV file you want to upload.

Depending on the format of the uploaded CSV file, one of the following registration types will be processed:

- Simple registration
- Full registration

INFO

Bulk registration creates an elementary representation of the device. Then, the device must update it to a full representation with its own status.

A separator is automatically obtained from the CSV file. Valid separator values are: `\t` (tabulation mark), `;` (semicolon) and `,` (comma).

Simple registration

The CSV file contains two columns: ID;PATH, where ID is the device identifier, for example, serial number, and PATH is a slash-separated list of group names (path to the group where the device should be assigned to after registration).

```
ID;PATH
Device1;Group A
Device2;Group A/Group B
```

After the file is uploaded, all required new groups will be created, new registrations will be created with status “Waiting for connection”, and the normal registration process must be continued (see above).

Full registration

The CSV files must contain at least the IDs as device identifier and the credentials of the devices.

In addition to these columns the file can also contain other columns like ICCID, NAME, TYPE as shown in the following example:

```
ID;CREDENTIALS;TYPE;NAME;ICCID;IDTYPE;PATH;SHELL;AUTH_TYPE
006064ce800a;5ecur3_p455w0rd;c8y_Device;Sample_Device1;+491555555;c8y_Serial;bulk group/subgroup1;1;BASIC
006064ce8077;AbcD1234!1234AbcD;c8y_Device;Sample_Device2;+491555555;c8y_Serial;bulk group/subgroup2;1;BASIC
```

To connect the devices, they are pre-registered with the relevant information. More specific, each device will be configured as follows:

- Username - the username for accessing Cumulocity must have the format <tenant>/device_<id>, where <tenant> refers to the tenant from which the CSV file is imported and <id> refers to the respective value in the CSV file.
- Password - the unique password for each device to access Cumulocity equals the value “Credentials” in the CSV file.
- Device in managed object representation - fields TYPE, NAME, ICCID, IDTYPE, PATH, SHELL in the CSV file.

After the data is imported, you will get feedback on the number of devices that were pre-registered as well as on any potential errors that may have occurred.

For your convenience, we provide CSV template files for both bulk registration types (simple/full) which you can download from the registration wizard to view or copy the structure.

INFO

If the device with the given identifier already exists, it will be updated with the data from the CSV file.

TO IMPORT CSV DATA IN MICROSOFT EXCEL

1. In Microsoft Excel, switch to the **Data** tab.
2. In the **Data** tab, select **From Text** in the top menu bar.
3. Select the CSV file you want to import by browsing for it (in this case the template file that you have downloaded from the Cumulocity platform).
4. In Step 1 of the **Text Import Wizard**, leave the default settings and click **Next**.
5. In Step 2 of the **Text Import Wizard**, select **Semicolon** as delimiter and click **Finish**.

For further information on the file format and accepted CSV variants, also refer to Create a bulk device credentials request in the Cumulocity OpenAPI Specification.

INFO

In an Enterprise tenant you may also register devices across multiple tenants by adding a **Tenant** column to the spreadsheet and importing the CSV file from the Management tenant. Contact your Operations team for further support.

EXTENSIBLE DEVICE REGISTRATION

To address the growing number of IoT protocols and certain restrictions in the general single or bulk device registration, Cumulocity provides an extensible device registration feature.

The general concept is based on extending the device registration using a metadata-based approach. Microservices and agents that implement current device registrations can add custom forms to the device registration wizard by providing simple descriptions of the required registration attributes. The metadata is then used by the UI to render a corresponding device registration wizard.

There are two possible ways to extend the UI:

- with a single device registration form
- with a bulk device registration form

REQUIREMENTS

Extensible device registration requires application extensions to be defined, and the microservice to implement the predefined endpoints used for getting device registration metadata and creating the device.

ADVANTAGES OF EXTENDED DEVICE REGISTRATION

The extended device registration provides the following advantages:

- **Extensibility of the device registration wizard**: You can easily add own forms to the device registration wizard in the Device Management application UI. The values to be entered in the user-specified forms can be freely customized by the device integration developers.
- **Support for bulk registration using custom CSV**: You can customize the bulk registration and hence implement support for CSV files of a different format.
- **No UI code changes required**: You do not need to write UI Angular code. This keeps the amount of integration work as little as possible. The device integration developer must only subscribe a microservice that provides an own wizard, and the wizard shows up automatically.

EXTENSION ENABLING

As a first step to extend the device registration flow you must define extensions in the application representation.

INFO

The `extensions` fragment can be placed either as root level or inside the `manifest` fragment of the application representation.

There are two types of extensions:

- Single device registrations type: `extensibleDeviceRegistration` , for example:

```
"extensions": [
  {
    "name": "<extension name>",
    "description": "<description>",
    "type": "extensibleDeviceRegistration"
  },
  ...
]
```

- Bulk device registration type: `extensibleBulkDeviceRegistration` , for example:

```
"extensions": [
  {
    "name": "<extension name>",
    "description": "<description>",
    "type": "extensibleBulkDeviceRegistration"
  },
  ...
]
```

SINGLE DEVICE REGISTRATION

After enabling the `extensibleDeviceRegistration` extension type, the **Devices** > **Register device** menu in the Device Management application is extended with an entry corresponding to the extension `name` property.

From now on, everything will be rendered based on data provided via the custom microservice. The added menu entry opens a window which fetches the form definition using the following endpoint:

```
GET /service/<contextPath>/deviceRegistration/metadata&lang=<user-language>
```

Make use of the `lang` query parameter in your microservice to respond with the already translated JSON Schema metadata. See also Limitations.

The UI automatically takes the contextPath for the GET request from the application definition of the microservice:

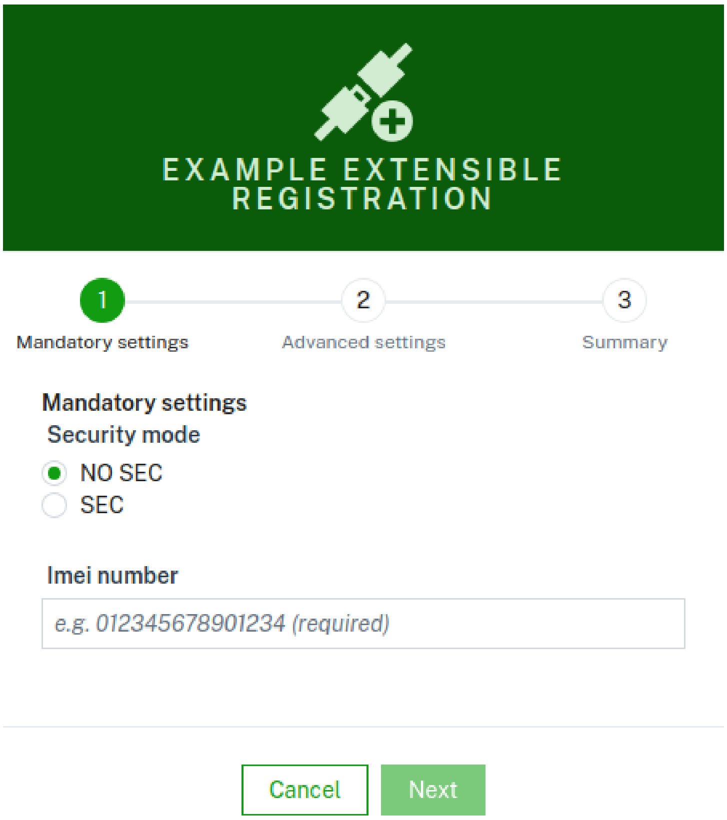
```
{
  "contextPath": "<relative path>",
  "availability": "MARKET",
  "type": "MICROSERVICE",
  "name": "<agent name>",
  ...
}
```

Example metadata definition:

```
{
  "c8y_DeviceRegistration": {
    "title": "Example extensible registration",
    "description": "The required and optional properties to register and setup devices",
      "pages": [
          {
              "$schema": "https://json-schema.org/draft/2020-12/schema",
              "type": "object",
              "title": "Mandatory settings",
              "properties": {
                  "security": {
                    "default": "NO_SEC",
                    "type": "string",
                    "title": "Security mode",
                    "enum": [
                      "NO_SEC",
                      "SEC"
                    ]
                  },
                  "imei": {
                      "examples": [
                          "012345678901234"
                      ],
                      "type": "string",
                      "title": "Imei number"
                  }
              },
              "required": [
                  "imei",
                  "security"
              ]
          },
          {...}
      ]
  }
}
```

The important part is the `pages` array which contains steps of the wizard that the modal is going to render accordingly to the JSON Schema definition: https://json-schema.org/.

As a result the following wizard will be displayed:



In the final step all data collected via the wizard will be sent back to the microservice using the following REST endpoint:

`POST /service/<contextPath>/deviceRegistration`

```
{
  "imei": "012345678901234",
  "security": "NO_SEC",
  ...
}
```

The form is able to send anything defined via JSON Schema to the microservice. The Microservice provides the form definition and is responsible for the proper handling of the submitted data.

API specification

The device integration microservices must implement the following REST endpoints:

```
GET /service/<contextPath>/deviceRegistration/metadata?lang=<user-language>
Accept: application/json
```

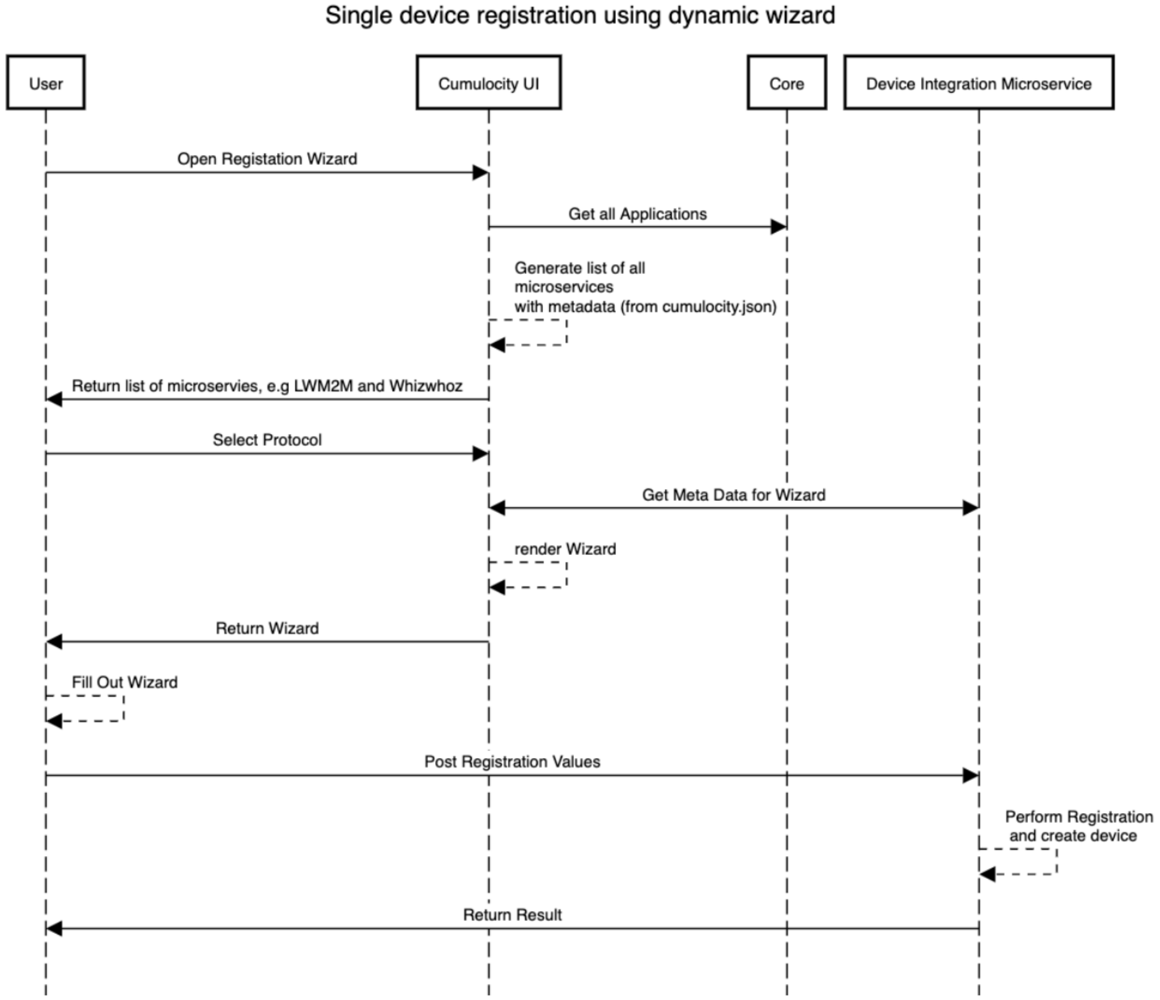
Returns the metadata in the vocabulary of the JSON Schema.

```
POST /service/<contextPath>/deviceRegistration
Content-type: application/json
```

Creates a single device based on the collected data. Sends application/json with key-value pairs.

Single device registration flow diagram

The following diagram visualizes the single device registration flow:



BULK DEVICE REGISTRATION

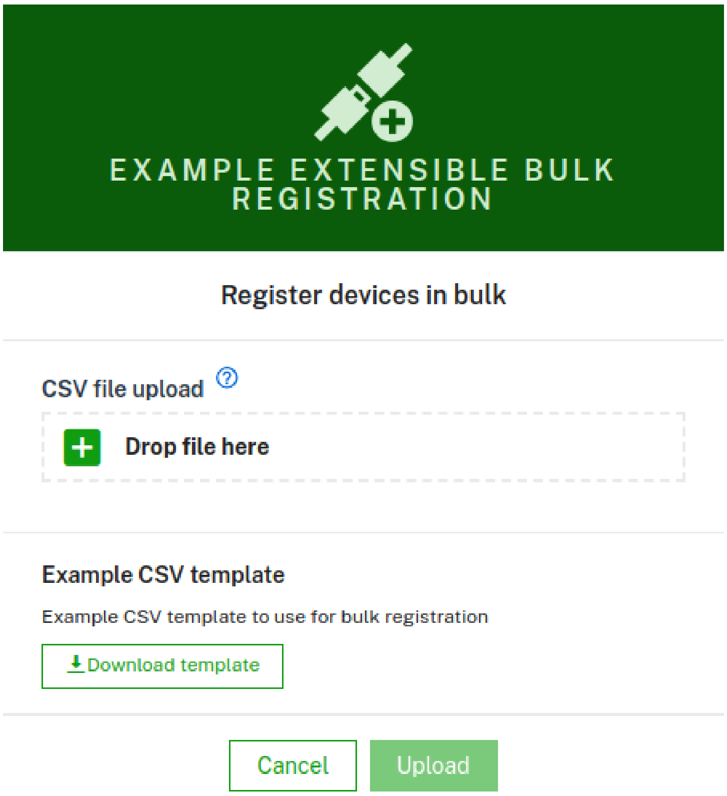
Many device integrations require the registration of many devices at the same time. Currently, all protocols must rely on the bulk registration mechanism of the platform, which often either requires too many fields or requires custom fields to be added. The latter ones can however so far not be validated, as the core directly creates devices – and microservices and agents have no control over the properties being written to the managed objects.

After enabling the `extensibleBulkDeviceRegistration` extension type, the Device management & connectivity > Devices > Register device `Bulk device registration` modal is displayed with an extended wizard entry corresponding to the extension `name` property.

Additionally, the microservice provides the title of the wizard step and example bulk file(s):

```
{
 "c8y_DeviceRegistration": {
   "title": "<title>",
   "description": "<description>",
   "bulk": {
     "title": "<bulk form title>",
     "exampleFileUrls": [{
       "title": "<example title>",
       "description": "<example description>",
       "url": "<publicly-reachable-URL>"
     }]
   }
 }
}
```

As a result the following wizard will be displayed:



API specification

The device integration microservices must implement the following REST endpoints:

```
GET /service/<contextPath>/deviceRegistration/metadata?lang=<user-language>
Accept: application/json
```

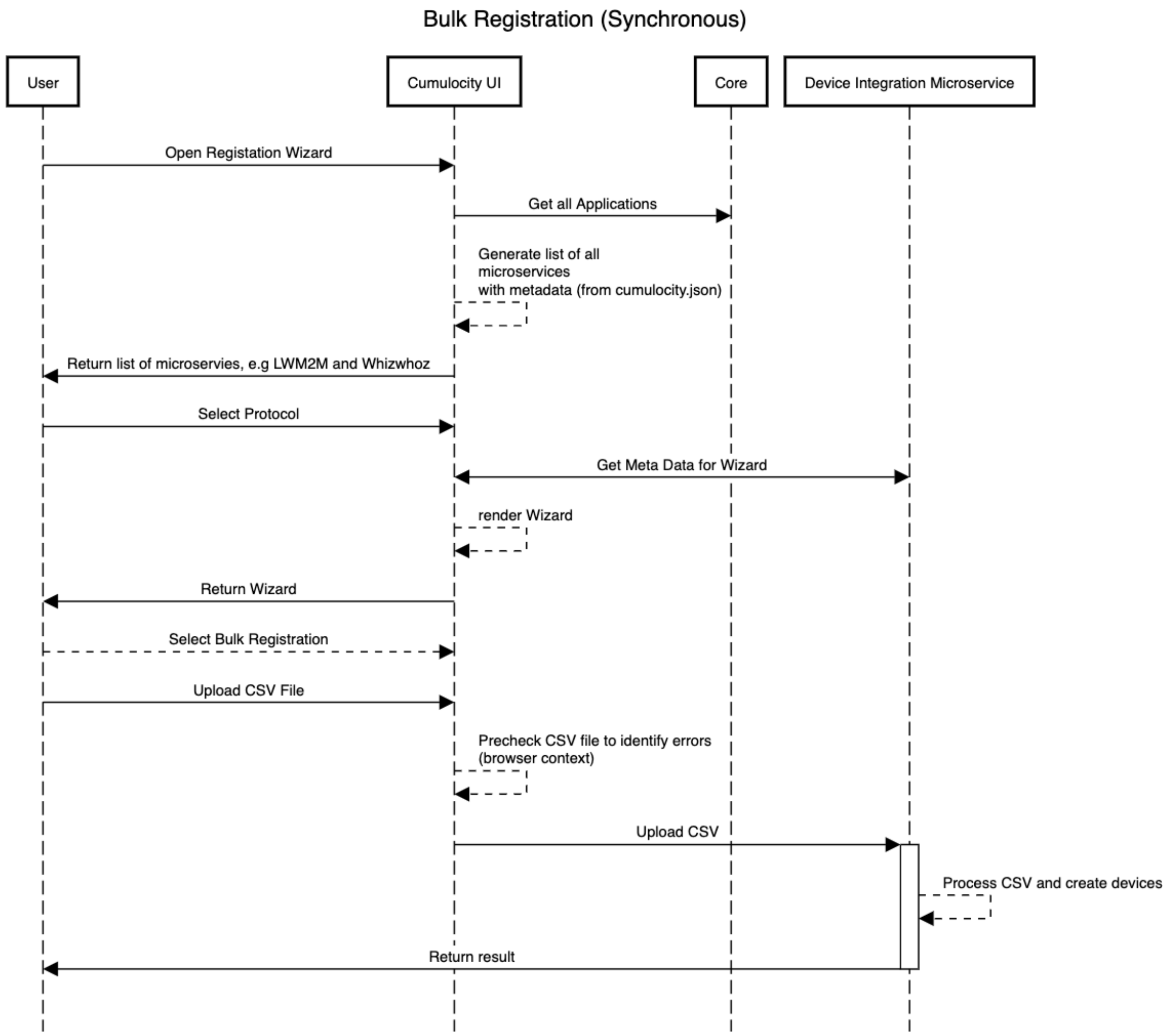
Returns the metadata in the vocabulary of the JSON Schema.

```
POST /service/<contextPath>/deviceRegistration/bulk`
Content-Type: multipart/form-data; boundary=boundary

--boundary
Content-Disposition: form-data; name="file"; filename="<input csv file>"
Content-Type: text/csv

--boundary--
```

Sends multipart form-data of the csv file type.

Bulk flow diagram

The following diagram visualizes the bulk device registration flow:



LIMITATIONS

- The concept does not allow the microservice to hook into deregistrations/decommissioning of a device:
  - Any device integration microservice must check if a device was deleted, for example to perform garbage collection.
- Interactive steps (like “Bluetooth Coupling”) can not be implemented as of now. The reason here is that the concept assumes that only static properties are required for a device registration, not some process that requires user interactions.
  - In theory, the meta-data based approach can easily be extended in a way that the JSON used to render the form is not static, but dynamically generated by the microservice. For example, you could specify a “page” that draws the next specification dynamically from an endpoint after posting all given user inputs there.
  - We step back from such a solution for now, because it complicates the data model and the requirements to be fulfilled by the microservice.
- No custom validations in the UI besides input validations by the JSON Schema vocabulary
- No custom styling possible
- Internationalization is not handled by the UI. As microservice developer it is your responsibility to provide already translated JSON Schema metadata

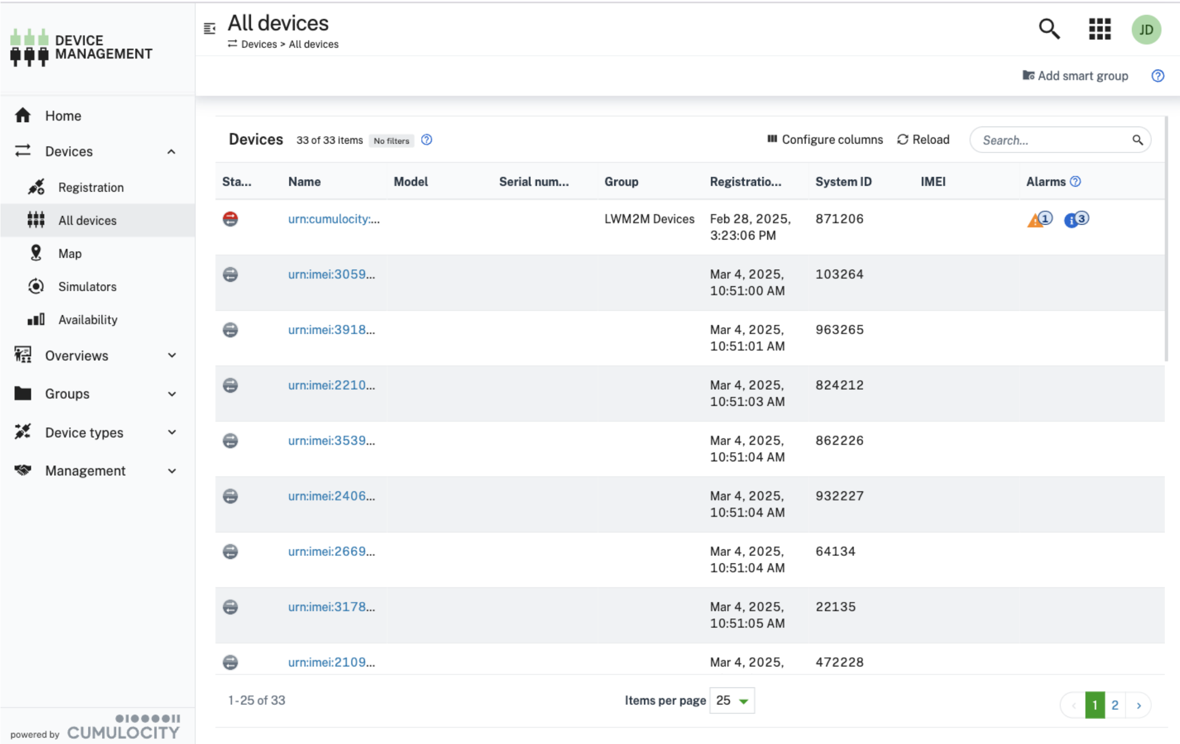
VIEWING ALL DEVICES

RELATED TOPICS

- The managed objects API for REST API methods concerning managed objects (devices or groups of devices).

To view all devices connected to your account, click **All devices** in the **Devices** menu in the navigator.

A detailed device list will be displayed.



DEFAULT COLUMNS

For each device, the device list shows the following information provided in columns:

Column	Description
Status	An icon representing the connection status. For details, see Connection monitoring.
Name	Unique name of the device.
Model	Model type of the device. Not always displayed, depends on browser width.

Serial Number	Serial number of the device. Not always displayed, depends on browser width.
Group	Group the device is assigned to, if any.
Registration Date	Date when the device was registered to your account.
System ID	System ID of the device.
IMEI	IMEI of the device.
Alarms	The alarm status of the device, showing number and type of alarms currently unresolved for the device. Only includes alarms for the parent device. See Working with alarms for further information on working with alarms.

For users with global roles the devices list displays the items in a paginated manner. You can select the number of items per page and directly jump to any page. Users with inventory roles see up to 50 items initially. If there are more than 50 devices available more devices are loaded as you scroll down the list.

CONFIGURING COLUMNS

The columns shown in the device list may be configured to your needs.

TO SHOW/HIDE STANDARD COLUMNS

1. In the table header, click **Configure columns**.
2. In the resulting dropdown, select/clear the checkboxes for all columns as required.

The device list will reflect your changes and only show the selected columns.

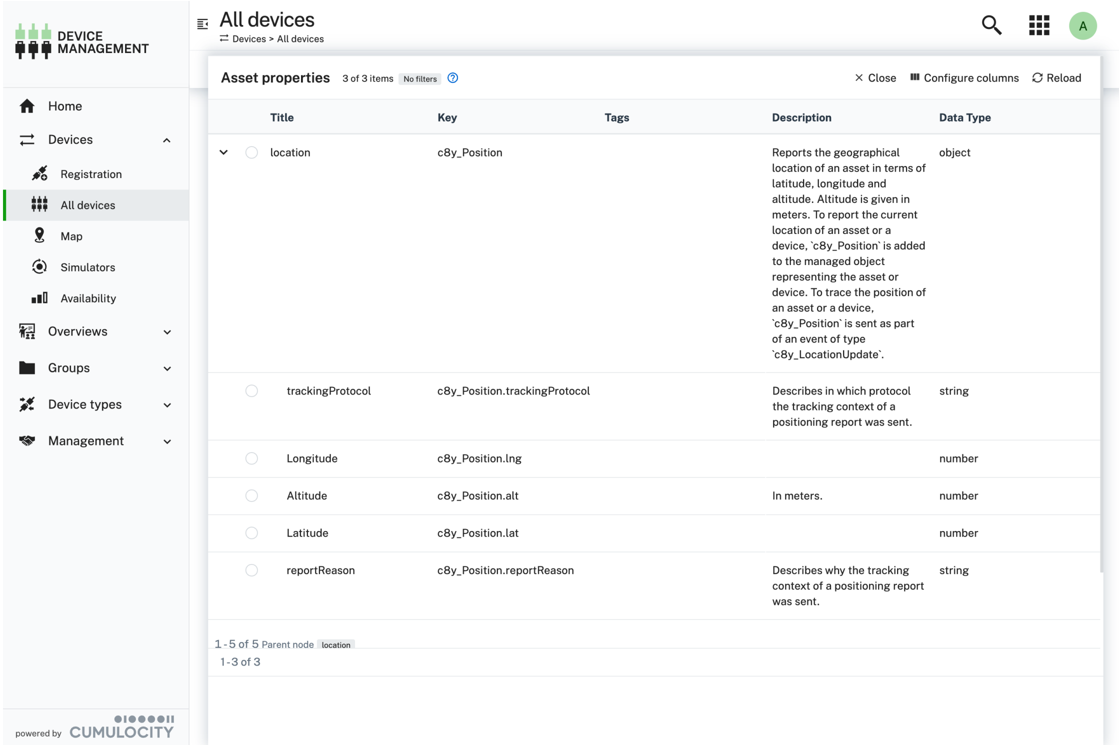
TO ADD CUSTOM COLUMNS

The device grid offers the option to add custom columns to display selected additional device properties. There are two alternative ways for adding custom columns.

With Digital Twin Manager

If the **Digital Twin Manager** application is available on your tenant, you can use pre-configured asset properties to add them as custom columns. Read how to create asset properties in the Asset models > Asset properties section.

1. In the **Configure columns** dropdown, click **Add custom column**. An **Asset properties** modal will open that lets you browse through all declared asset properties.

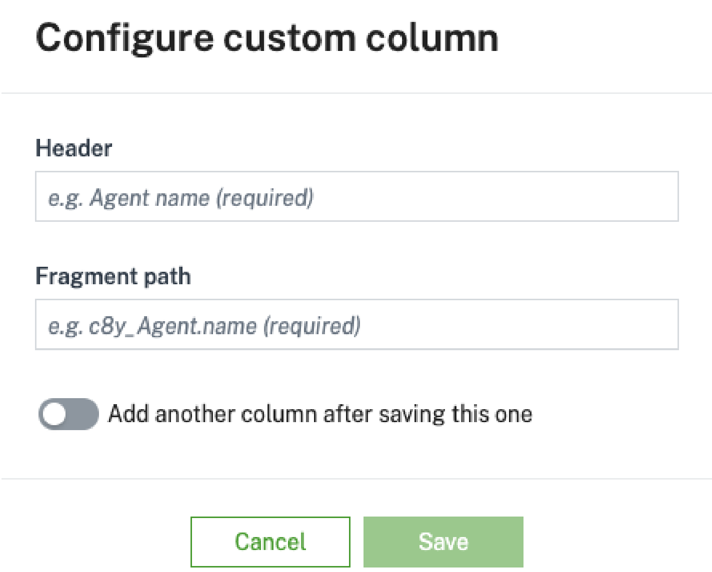


2. Browse through the list to find the property you want to display. You can filter the list by **Title**, **Key** and **Tags**. An arrow in the first column indicates that the property is a complex property. Click on the arrow to expand the list of its nested properties.
3. Check the radio button and the new column will be added and displayed in the device list.

By defining a custom property path

If the **Digital Twin Manager** application is not available on your tenant, you can still create custom columns by manually providing the path to the property you want to display.

1. In the **Configure columns** dropdown, click **Add custom column**.



2. In the **Header** field, enter a header for the new custom column.
3. In the **Fragment path** field, enter the property of the device to be shown. Nested properties will be accepted. However, for nested properties its only possible to select Cumulocity standard fragments like `c8y_Mobile.mcc` .
4. Switch the **Add another column after saving this one** toggle to active to create another custom column right after saving the current one without leaving the dialog.
5. Click **Save**.

The new column will be added and displayed in the device list.

INFO

While standard columns can only be shown/hidden as required, custom columns may be deleted permanently.

TO FILTER DEVICES

The device list offers a filtering functionality to filter devices in the list for specific criteria.

Filtering is available on every column. Click the filter icon next to the name of the column you want to set a filter for.

Specify your filter options in the dropdown list.

Most columns represent text fields. You can filter these columns by simply entering an arbitrary text into the textbox as in the search field. Click **+ Add next** to add another textbox if you want to filter for more than one term.

Apart from filtering for text there are several other options:

- In case of date fields (for example **Registration date**), you specify a date range to filter for.
- In the **Status** column you can filter for various criteria representing the send, push or maintenance status of the device.
- In the **Alarm** column the filtering options you may select correspond to the alarm types (critical, major, minor, warning, no alarms).
- For custom columns, select **Only rows where value is defined** to filter based on whether the fragment exists, or specify one or more filter terms the entry must match.

Finally, click **Apply** to carry out the filtering.

The devices list will now only display devices matching the filtering options.

Sorting is available on every column. Click the sort icon ↑↓ in the respective column header once to sort the entries in ascending order and twice to sort the entries in descending order. Click the sort icon ↑↓ once more to remove sorting for this column.

Click **Clear filters** in the table header if you want to clear all filters and view all devices.

INFO

If you select to sort a text field, for example, device name, in ascending or descending order, keep in mind that the resulting alphabetical sorting is based on ASCII/UTF: A < B < … < Z < … < a < b … < z. Names starting with lower case letters will be sorted below all names with uppercase letters or vice versa.

TO REPLACE A DEVICE

The device list supports the replacement of a physical device with another by preserving the original device representation in the platform including all its generated data. A wizard guides you through this process. Access the wizard by hovering over the device you want to replace in the device list and click the **Replace device** button.

INFO

Currently LWM2M devices are not supported and the **Replace device** button is not available for this type of devices.

In the **Replacement device** step you can see the list of devices available in the platform excluding the device which you are going to replace. Select a replacement device from the list.

INFO

In order for a particular replacement device to show up in the list, you must register it through the corresponding registration process for this device.

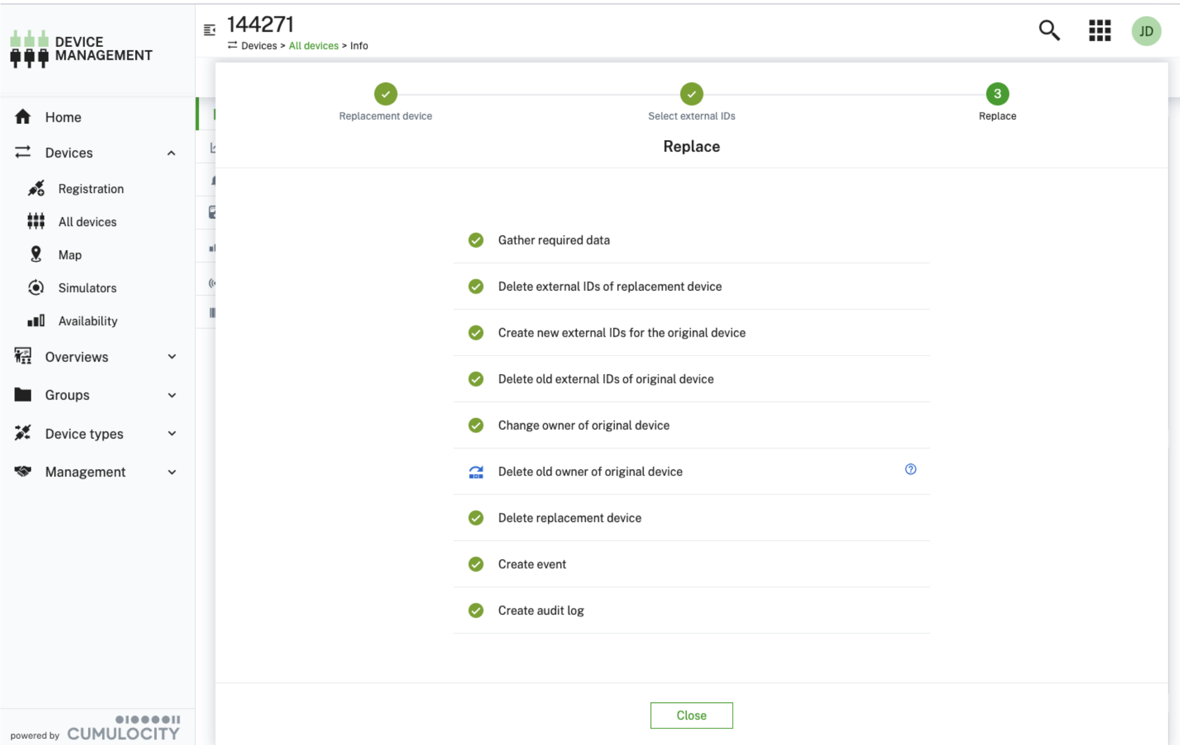
The **Select external IDs** step lists the external IDs assigned to the replacement device. Select those you want assigned to the device representation of the replaced device. You must select at least one external ID. This links the data which your new physical devices sends to the original device representation object.

Before proceeding with the actual replacement, you must confirm this action. Click **Replace** in the **Replace device** dialog.

IMPORTANT

The replacement process removes the replacement device from the Cumulocity database including all the data generated for it so far.

The **Replace** step displays the progress of the replacement process and the result of every step executed.



If a step has been skipped you will find an icon with a tooltip which contains the reason why the step has been skipped.

If the execution of a given step has failed, you can see the error details by expanding the step details. Note that an error in one of the steps will not prevent the next steps from being executed. The wizard will try to execute as many steps as possible and provide information about failed steps. If you are able to resolve the reason for the failure(s), you can retry all failed steps by clicking the **Retry** button in the bottom bar. If you wish to retry a single failed step you can only do so by clicking the **Retry this step** button form the step actions dropdown.

The **Close** button brings you back the the updated device list. The replacement device will have been removed from Cumulocity while all data sent by the new physical device will be linked to the platform representation of the original device. This ensures that the physical replacement remains transparent to users.

In order to keep track of the replacements done for a given device an event and an audit log are created for every replacement.

TO DELETE A DEVICE FROM THE LIST

1. Hover over the row of the device you want to delete.
2. Click the delete icon at the right of the row.
3. Confirm the device removal. Optionally, select whether to delete child devices of the device or to delete the associated device owner. Note that it is not possible to select both options.

The device will be permanently deleted from the platform.

IMPORTANT

Deleting a device means to remove the device from the Cumulocity database including all its generated data. Alternatively, you can arrange all retired devices in one group, see Grouping devices). This ensures that all reports remain correct. To prevent alarms from being raised for the retired devices, disable connection monitoring. Deleting a device does not delete the data of its child devices.

GROUPING DEVICES

RELATED TOPICS

- Application enablement & solutions > Cockpit > Managing assets > Assets hierarchy for information on the asset hierarchy, assets and groups.
- Platform administration > Standard tenant administration > Managing permissions > Inventory roles on how to assign inventory roles to groups of devices.
- The managed objects API for REST API methods concerning managed objects (devices or groups of devices).
- The bulk operations API for REST API methods concerning bulk operations.

Devices can be grouped according to a particular use case. A device can be located in multiple groups and groups themselves can again be part of multiple groups.

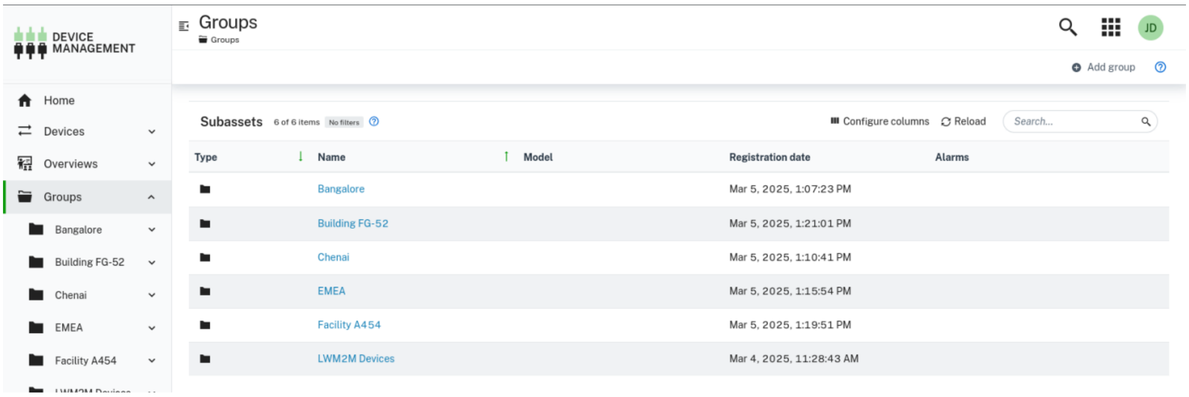
Cumulocity distinguishes between top-level groups and subgroups:

- **Top-level groups** are shown in the **Group** menu in the navigator at top-level.
- **Subgroups** can be used to further subdivide top-level groups.

MANAGING GROUPS

TO VIEW GROUPS

Click **Groups** in the navigator to see all groups in a list format.



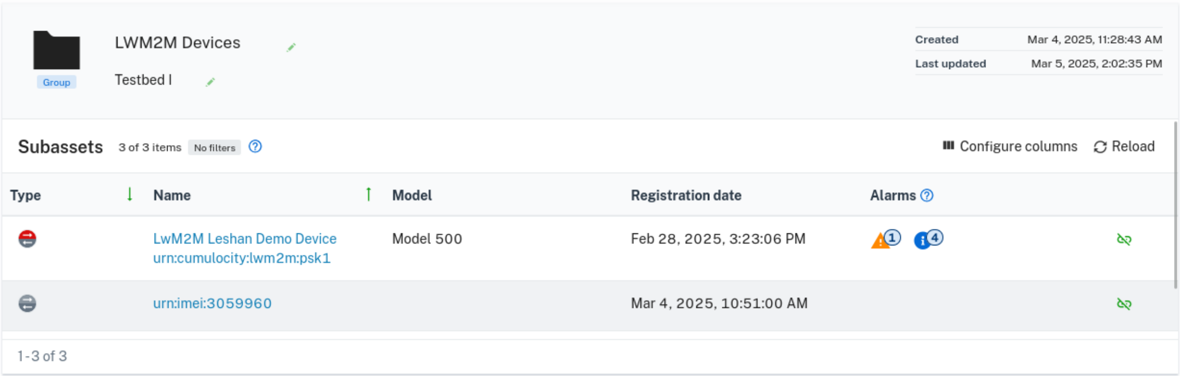
For each group, various information is provided, for example the type and name. Click **Configure columns** at the right, to add or remove columns and customize the view to your preference. See also Configuring columns.

To filter the groups for certain criteria, hover over the column headers and click the respective filter icon .

See also To filter devices.

Note that this function only creates a temporary filter. For permanent filters, you can use the smart groups function.

Click a group to view its details.



Subassets page

At the top of the **Subassets** page, the name and the description of the group is displayed (editable), followed by the information when the group was created and last updated.

Below, all assets assigned to the group are listed. For each asset, various information is displayed, for example the type and name. As with the top-level groups list, you can add or remove columns and customize the list to your preference, or you can apply filters to filter the list for certain criteria.

Morover, you can assign devices, see To assign devices to a group.

TO ADD A GROUP

1. Click **Add group** at the right of the top menu bar.
2. In the resulting dialog box, enter a unique group name and an optional description and click **Next**.
3. In the list, select the devices you want to add. You may apply filters to reduce the number of displayed devices.
4. Click **Create** to create the new group.

The new group will be added to the groups list.

INFO

A group can be created with “0” devices in it.

To add a new group as a child of an existing group, navigate to its **Subassets** page and click **Add Group** in the top menu bar.

TO EDIT A GROUP

1. In the navigator, click a group to open it.
2. In the **Subassets** page, you can edit the name and description of the group.

For further information on permissions, see Managing permissions.

TO DELETE A GROUP

Hover over the respective entry you want to delete and click the delete icon at the right.

MANAGING DEVICES IN GROUPS

TO ASSIGN DEVICES TO A GROUP

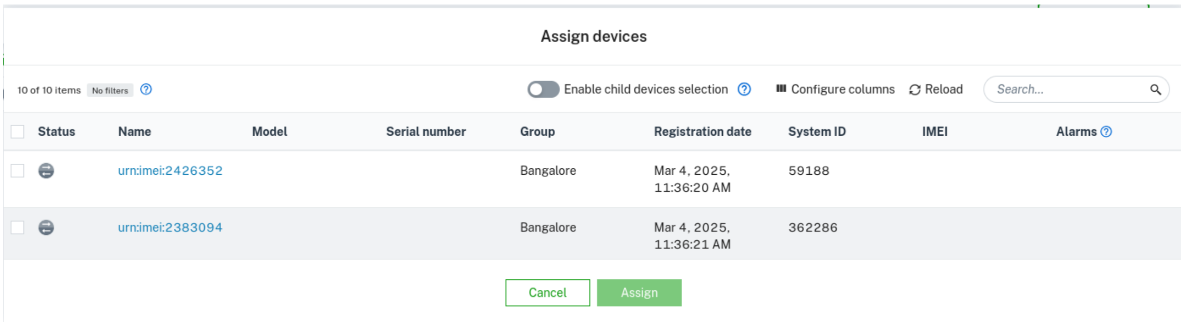
You can assign devices to groups in several ways.

From the group perspective

You can quickly assign devices to groups by using the drag and drop functionality in the navigator, see Restructuring groups and devices.

Moreover, you can assign devices performing the following steps:

1. In the navigator, select a group from the **Group** menu and then open the **Subassets** page.
2. Click **Assign devices** at the right of the top menu bar.
3. In the list, select the devices you want to add. You may apply filters to reduce the number of displayed devices.
4. Click **Assign** to assign the selected devices.



The devices will be assigned to the selected group and shown as subassets in the **Subassets** page.

From the device perspective

1. Select a device from the device list and open it.
2. In the **Info** tab, scroll down to the **Groups assignment** card. From the dropdown field, select the group you want to assign the device to. You can also directly enter a group name here or you can enter just parts of a name to filter the list for it and only show the matching group names.
3. Click **Assign**.

The device will be assigned to the selected group.

If you search for a group by its name which does not exist yet, a **New** button will appear so that you can create a new group with this name from here and assign the device to that group.

INFO

In order to create a new group, the user must have the permissions ROLE_INVENTORY_CREATE and ROLE_INVENTORY_ADMIN.

TO UNASSIGN A DEVICE

Hover over the respective device you want to unassign and click the unassign icon at the right.

Unassigning a device does not delete the device, subdevices or any associated data. The device is only removed from this group.

TO DELETE A DEVICE

Hover over the respective device you want to delete and click the delete icon at the right.

The device will be permanently deleted.

TO VIEW THE DEVICE DETAILS

To display the details of a particular device, click its name.

The device details for the device will be displayed.

RESTRUCTURING GROUPS AND DEVICES

You can easily restructure groups, subgroups and devices by a drag & drop functionality.

TO MOVE A GROUP

1. In the navigator, select a group which you want to move to another group.
2. Drag and drop it to the desired group.
3. In the resulting dialog box, confirm the operation.

TO MOVE OR ADD A DEVICE

1. In the navigator, select the group or device which you want to move or add to another group.
2. Drag and drop it to the desired group.
3. In the resulting dialog box, select if you want to move or add the device.

USING SMART GROUPS

Smart groups are groups dynamically constructed based on filtering criteria. This type of group can be used, for example, for bulk upgrades of devices of a certain type to a new software or firmware version.

> INFO
>
> Smart groups are only available in the Device Management application and not visible in the Cockpit application.

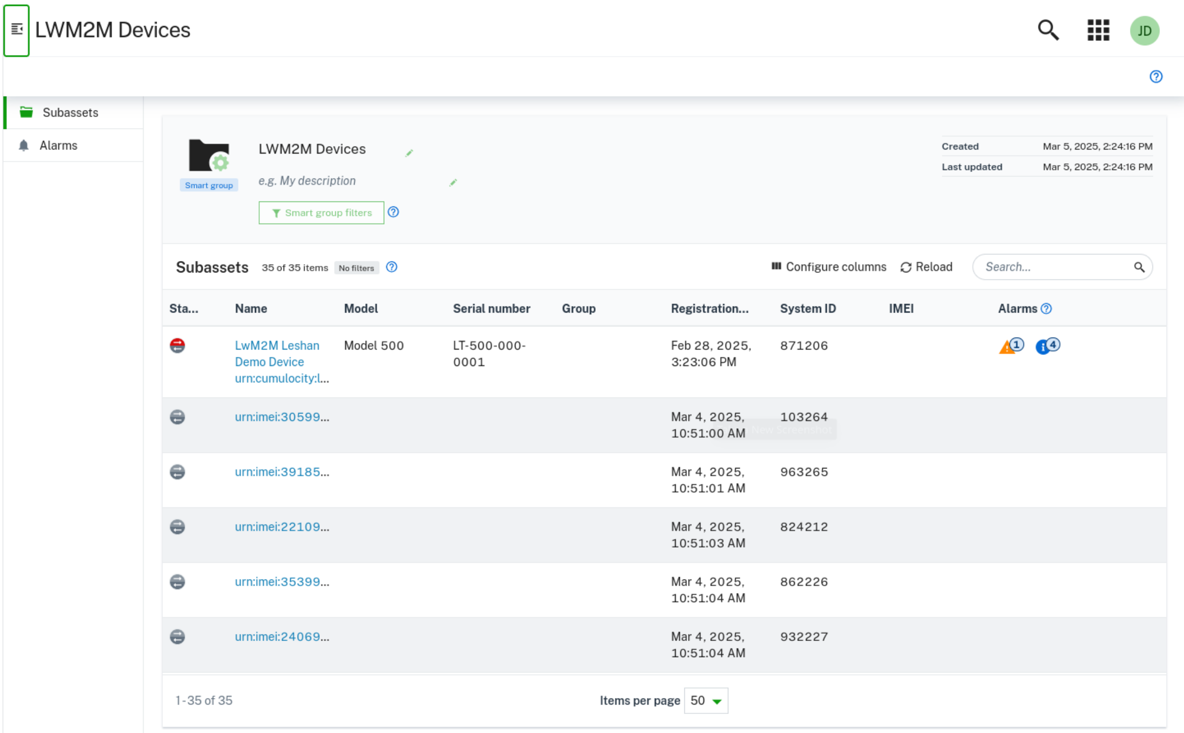
Smart groups can be created from the device list.

TO CREATE A SMART GROUP

1. To open the device list, click **All devices** in the navigator.
2. Filter the devices in the list to select the desired devices. See To filter devices for details on filtering.
3. Click **Create smart group** at the right of the top menu bar.
4. Enter a name for the group and click **Create**.

The new group will appear as a top-level group in the **Groups** menu of the navigator. Smart groups can be distinguished by a small cogwheel in the folder icon .

Below the smart group name and description you can see the filter criteria which have been applied on building the smart group. You can edit the filter settings here and adjust your selection.



TO DELETE A SMART GROUP

Hover over the respective entry you want to delete and click the delete icon at the right.

IMPORTANT

Deleting a smart group is irreversible.

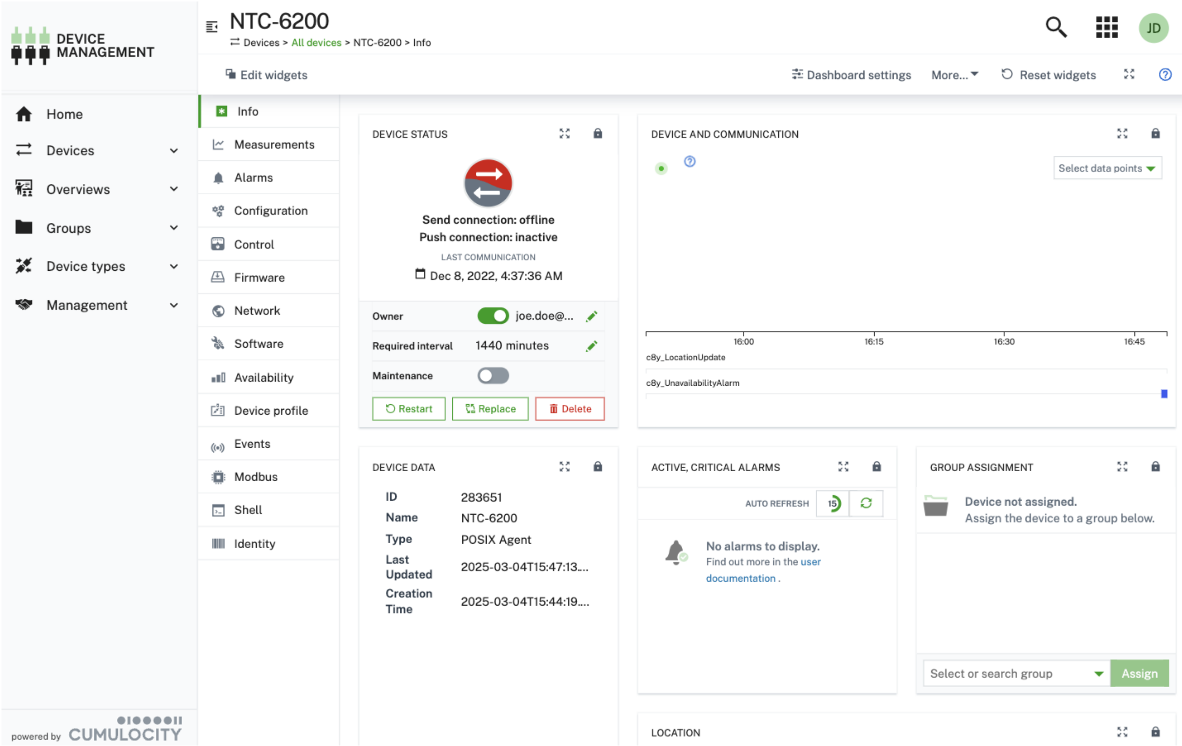
VIEWING DEVICE DETAILS

RELATED TOPICS

References to related content in the documentation are mentioned in the details descriptions below.

For each device, detailed information is available. The kind of information actually provided for a device depends on the device type, device usage and the configuration of your user interface.

To view detailed information on the device, click a device in the device list.



The device details are divided into tabs. The number of tabs is dynamic and depends on the available information, that means, tabs are only displayed if the kind of information is available for the particular device. For a detailed description of the operations and fragments related to each device detail tab see the fragment library.

Initially the **Info** tab is shown, which offers general information on a device and is available for each device.

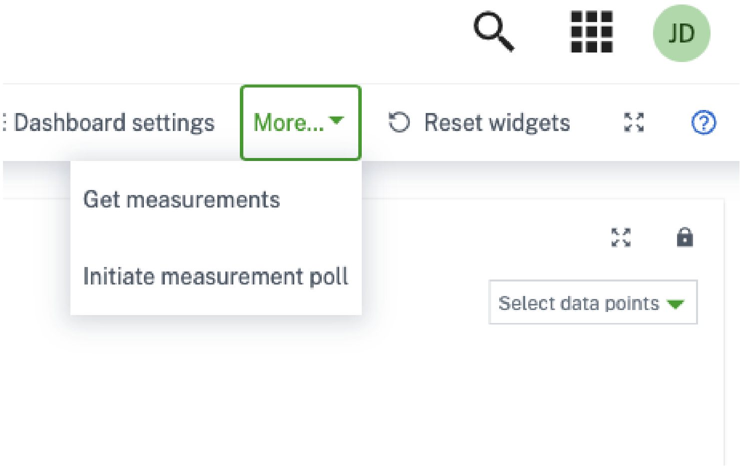
Each device at least shows the following tabs: **Info**, **Alarms**, **Control**, **Events**, **Availability**, **Identity**.

INFO

Several individual tabs, which you do not find listed here, may be described in a different context in another section of the Cumulocity documentation. Use the Search function to switch to the relevant sections. A detailed description on the **Modbus** tab, for example, can be found in Cloud fieldbus.

Below the name, a list of breadcrumbs is displayed. If the device is part of an asset hierarchy (such as a group), you can use the breadcrumbs to easily navigate up that hierarchy. Since devices can be part of multiple hierarchies, several rows of breadcrumbs may be shown.

Depending of the type and usage of a device, further actions are provided in an context menu when clicking **More…** at the right of the top menu bar.



Details on these additional menu items are provided where required.

ALARMS

The **Alarms** tab provides information on the alarms of a device. See Working with alarms for detailed information on alarms.

AVAILABILITY

The Availability tab offers availability monitoring for machines, see Availability for more information.

CHILD DEVICES

The **Child devices** tab shows a list of devices connected to the currently displayed device. For example, if you look at a gateway, the tab lists all machines connected to the gateway.

For details provided in the child device list, see Viewing all devices.

CONFIGURATION

The **Configuration** tab allows you to configure the parameters and initial settings of your device. Depending on the device, possible configurations are:

- Text-based configurations
- Binary-based configuration snapshots

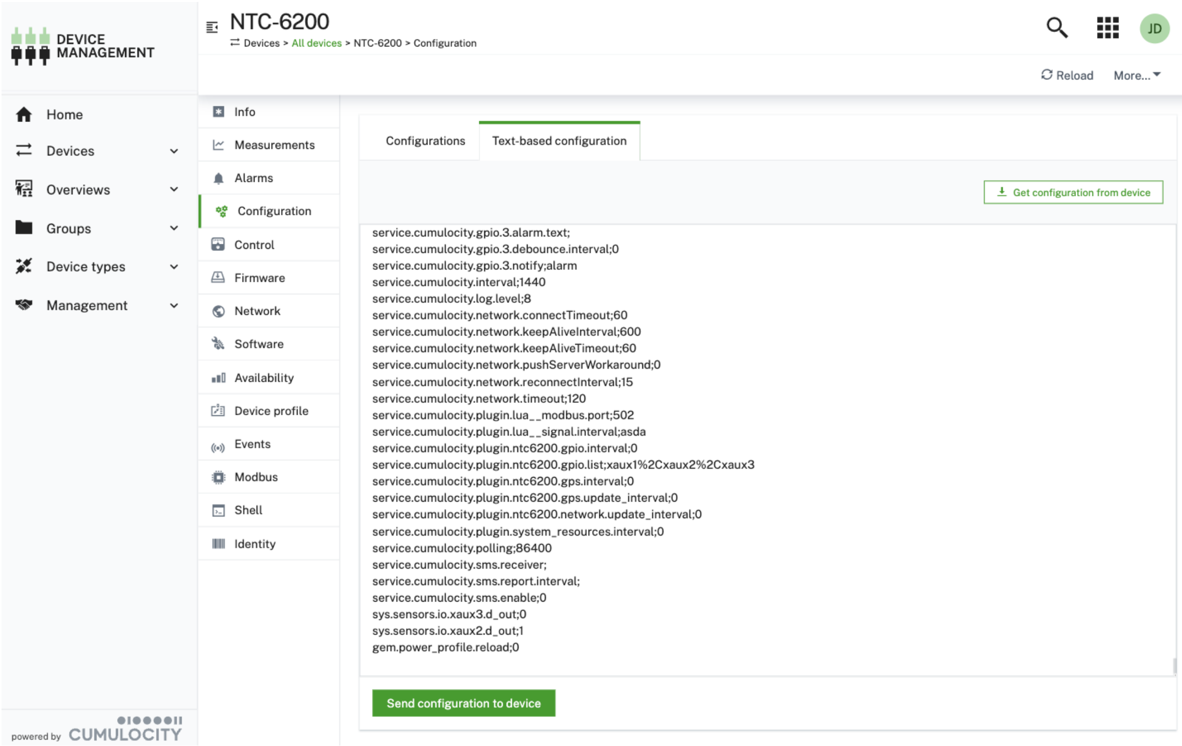
For more details on managing binary-based configuration snapshots, see Managing configurations.

To request the current text-based configuration snapshot

1. Navigate to the **Configuration** tab.
2. Select one of the device-supported configurations in the list.
3. Click **Get snapshot from device**.

To add or edit a text-based configuration snapshot

1. In the **Configuration** tab, you can manually add or edit the device configuration in the text field.
2. Click **Send configuration to device** to save your edits.

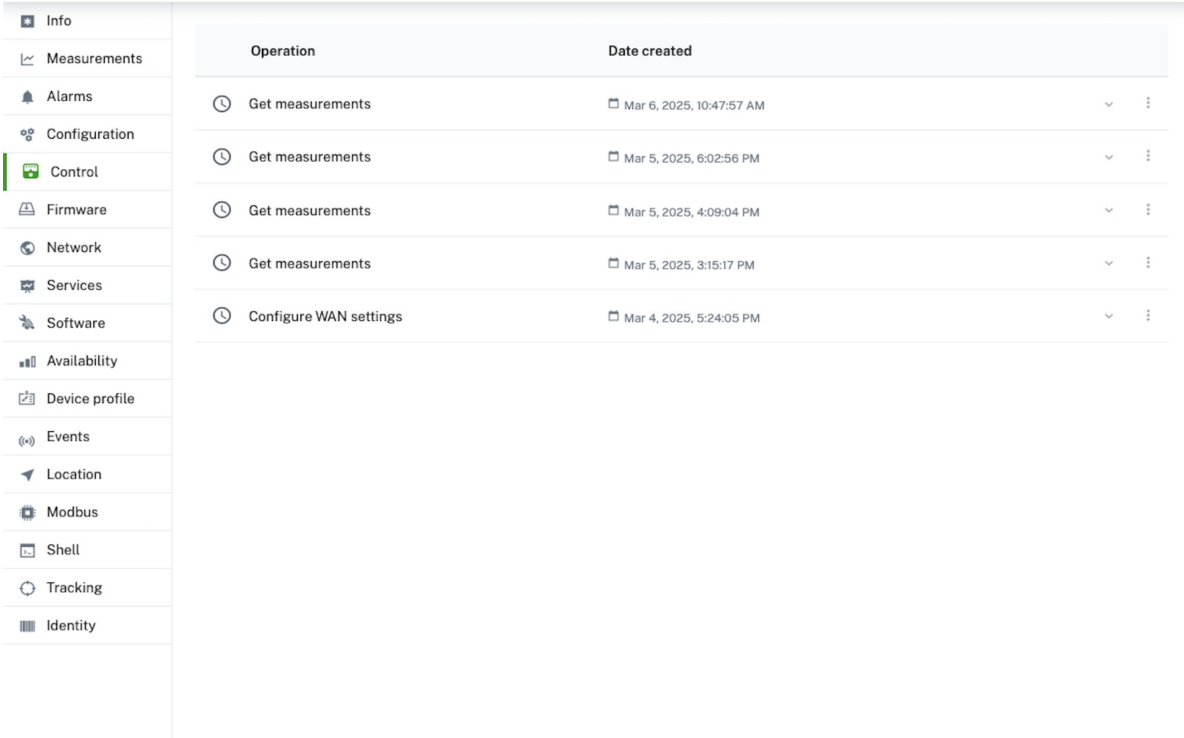


INFO

If a device supports both text-based and binary-based configuration the **Configuration** tab shows a subtab for each configuration type.

CONTROL

The **Control** tab lists the operations being sent to a device. See Working with operations for detailed information on operations.



DEVICE PROFILE

See Applying device profiles to devices for more information on how to apply device profiles to a device.

EVENTS

The **Events** tab displays events related to a device. This enables low-level troubleshooting of a device. See Troubleshooting devices for detailed information.

FIRMWARE

See Managing firmware for more information on how to manage and update the firmware on a device.

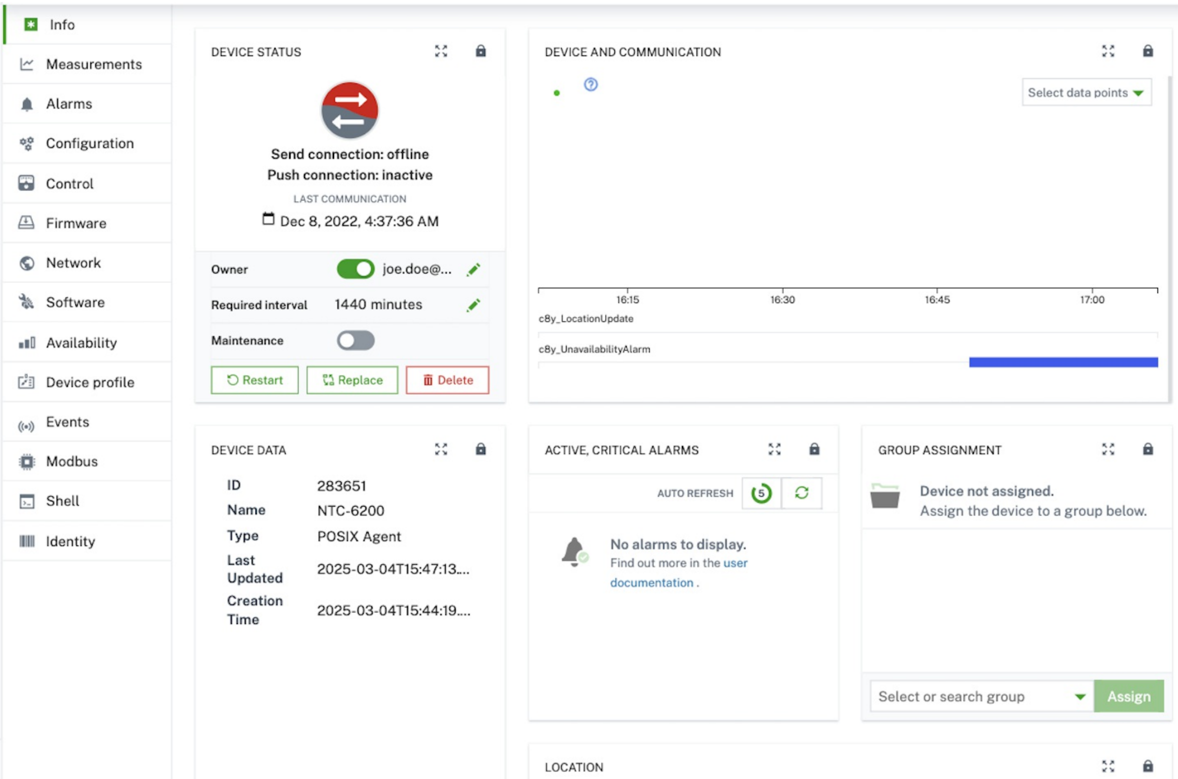
IDENTITY

Cumulocity can associate devices and assets with multiple external identities. For example, devices can often be identified by the IMEI of their modem, by a micro-controller serial number or by an asset tag. The **Identity** tab lists all the identities recorded for a particular device.

This is useful, for example, when you have non-functional hardware and must replace the hardware without losing the data that was recorded. Just connect the new hardware to your account and modify the identity entry of the old hardware, to contain the identity of the new hardware.

INFO

The **Info** tab summarizes management-relevant device information in a dashboard.

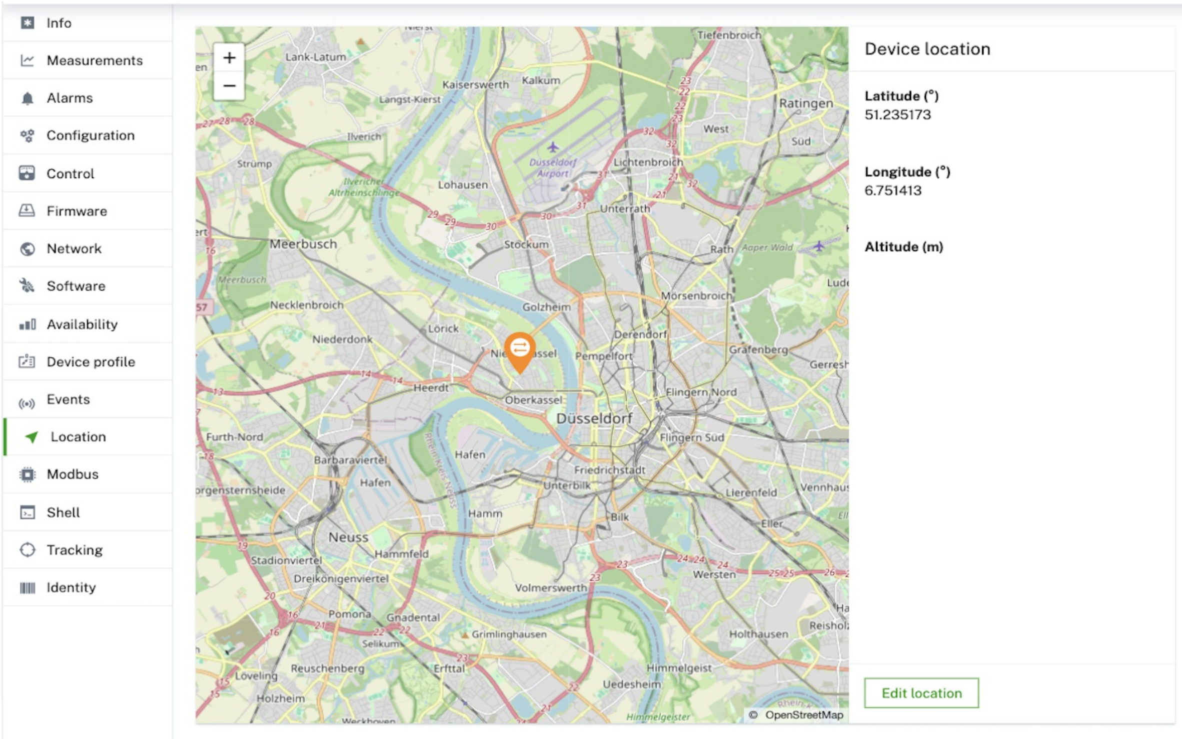


The information is provided on the following cards:

Card	Description
Notes	Provides optional notes to inform about current activities. Notes usually may only be edited by an administrator. To add or edit a note, click **Edit**, enter your note or your modifications in the text box and save your edits by clicking the green checkmark at the right of the text box.
Device status	Displays connection-relevant information, as described in detail in Connection monitoring.
Device and communication	Shows a data point graph displaying real-time data on particular measurements. Drag the x-axis to move the data point time measurement. To zoom in select a time period, double click to zoom out of the graph. For details on data point graphs, refer to Working with data explorer. The following measurements may be shown here: **Data points**: c8y_Battery.level, c8y_SignalStrength.rssi, c8y_MemoryMeasurement.Used, c8y_CPUMeasurement.Workload, c8y_NetworkStatistics.Upload, c8y_SignalStrength.RCSP, c8y_SignalStrength.ber, c8y_SignalStrength.ECN0, c8y_NetworkStatistics.Download, c8y_MemoryMeasurement.Total **Alarms**: c8y_UnavailabilityAlarm **Events**: c8y_LocationUpdate
Device data	Displays general information on the device (ID, name, type, owner, last update). The fields **Name** and **Type** can be edited. Below the general device information, the card shows status information for active alarms, availability and connection (not editable). Moreover, various device-specific information like hardware and firmware is displayed here (editable).
Active, critical alarms	Shows the active critical alarms for the device.
Groups assignment	Displays the groups the device belongs to. Moreover you can add the device to groups here or unassign it from groups, see Grouping devices.
Location	Shows the location of a device on a map as reported by the device or as manually set, see Location.

LOCATION

The **Location** tab by default shows the location of a device on a map and as coordinates, as reported by the device. For devices that do not report a location you may manually set the location. Simply place the “pin” in the correct place of the displayed map.



The **Location** tab also shows when a device contains `c8y_Position` property. When you send a new `c8y_Position` event, you can set the same `c8y_Position` fragment on the device and it will automatically mark its position on the map.

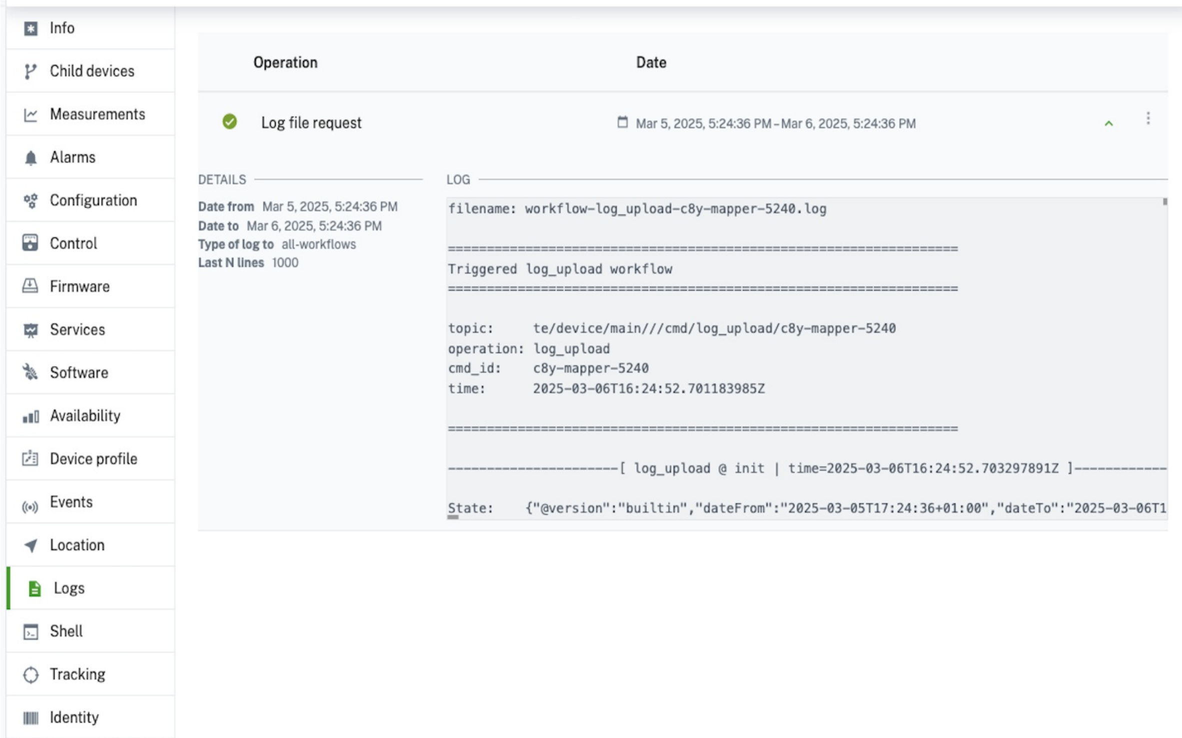
LOGS

In the **Logs** tab you can manage log information from devices.

To request log information

1. In the **Logs** tab, click **Request log file** at the right of the top menu bar.
2. In the resulting dialog box, specify a date and time range for the log information.
3. Select the type of log from the dropdown field. The supported logs listed are usually device-specific.
4. Optionally, specify a text filter. For example, if you enter “Users”, only lines including the term “Users” will appear in the returned log information.
5. Specify the maximum number of lines to be returned (counted from the end). The default value is 1000.
6. Click **Request log file**.

The log information will be requested from the device.



Requesting a log from a device may take some time.

After the log has been transferred from the device to Cumulocity, it will be listed in the **Logs** tab. The row in the list shows the requested log time range.

Click on the entry in the list to show the entire log information.

To download a log

Hover over a row and click the download icon , to download the log excerpt to your file system.

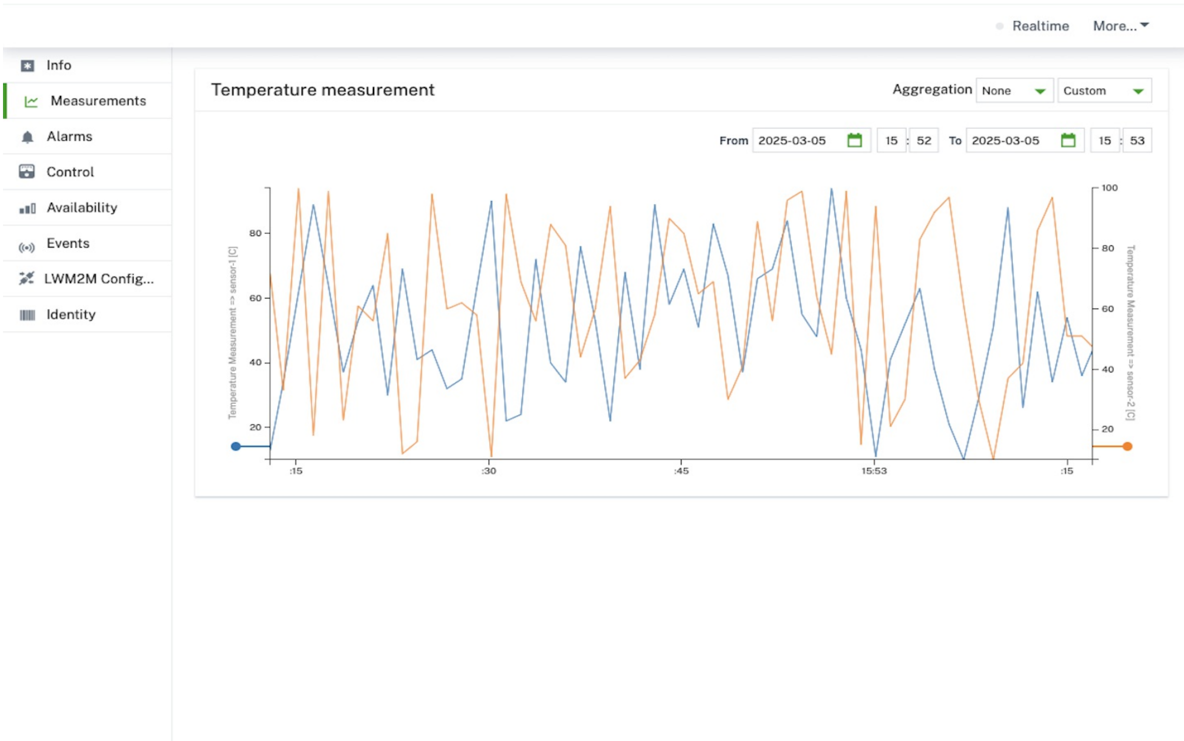
To delete a log

Hover over a row and click the delete icon , to delete the log information.

MEASUREMENTS

The **Measurements** tab provides a default visualization of numeric data provided by the device in the form of charts. Charts are grouped into types of measurements, which can contain multiple graphs or “series”.

The screenshot below, for example, shows a chart for temperature measurement with two different series.



If a chart contains measurements with different series, one Y-axis is rendered per series. In the example above, temperature data is recorded from two sensors namely “sensors-1” and “sensor-2” having the same unit as °C. Here measurements from different sensors are categorized as separate “Series” data. The measurements from the respective sensors are stored using separate series names (same as the sensor names) and hence, two axes are rendered here. Only one Y-axis is rendered if the measurements belong to the same series.

To see detailed information about the measured values, hover over the chart. A tooltip will be displayed with detailed information on the measurement next to your cursor (the tooltip will “snap” to the closest measurement).

INFO

We recommend you to have max. 20 series per measurement for optimal performance and readability of a single graph in the Device Management application (the graph displays all available series). If you need to display only a few series from the measurement, we advise you to use Data explorer in Cockpit where you can select series to be shown in the graph.

Time range and aggregation

By default, charts show the raw data of the last hour. To change the time range on the X-axis, open the “Last hour” dropdown menu at the top right and select a time range.

If you increase the time range, the value in the **Aggregation** field will automatically switch to “hourly” or “daily”. The chart now shows ranges instead of individual raw data points. For “hourly”, the chart will show a range of the minimum and maximum value measured in one hour. For “daily”, the chart will show the minimum and maximum value measured over one day. Likewise, the tooltips will now show ranges of values instead of individual values.

This enables you to get an efficient overview over larger time periods. A graph will only show 5.000 data points per graph maximum to avoid overloading your desktop browser. If you select a fine focus resulting in more than 5.000 data points, a warning message will be shown: “Truncated data. Change aggregation or select shorter date range.”

Clicking **Realtime** will enable real-time user interface updates of the graphs as new data flows into the system from the connected devices.

Measurement format

In order to see measurement graphs, the device must send measurements in a specified fragment format.

```
"fragment_name" : {
 "series_name" : {
  "value" : ...
  "unit" : ...
 }
}
```

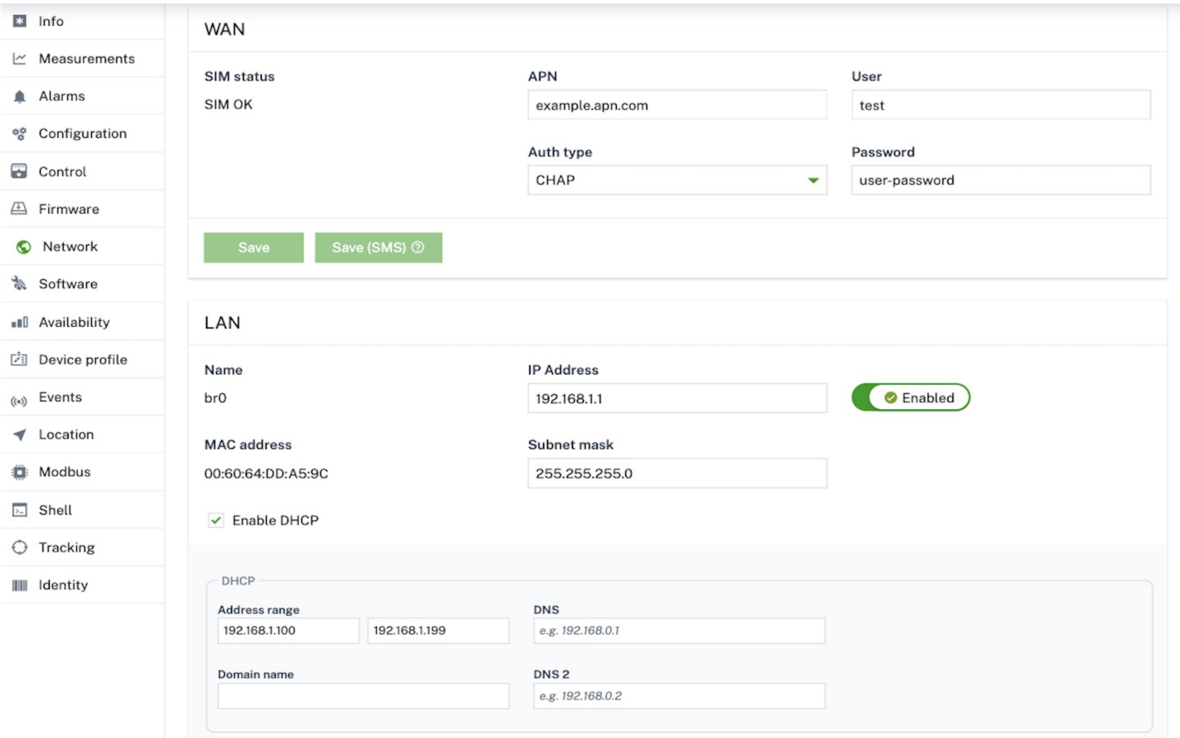
Example:

```
"c8y_SpeedMeasurement": {
      "Speed": { "value": 1234, "unit": "km/h" }
}
```

`"Fragment_name"` and `"series_name"` can be replaced by different valid JSON property names, but no whitespaces and special characters like [ ],* are allowed. The structure must be exactly as above, two-level deep JSON object.

NETWORK

In the **Network** tab, mobile network (WAN) and local area network (LAN) parameters can be viewed and configured.



The WAN parameters in the user interface correspond to the first profile stored in the router. These parameter can be configured remotely or via SMS.

> **INFO**
>
> For SMS configuration, the router must be configured to accept SMS commands.

To configure WAN parameters

1. Enter the Access Point Name (APN).

2. Enter the username and the password of your account in the platform to which you wish to establish a connection.
3. Select the authentication type.
4. Click **Save** to save your settings.

To configure LAN parameters

To configure LAN parameters, simply enter **IP address** and **Subnet mask**.

INFO

Name and **Mac address** fields are not configurable.

To configure DHCP parameters

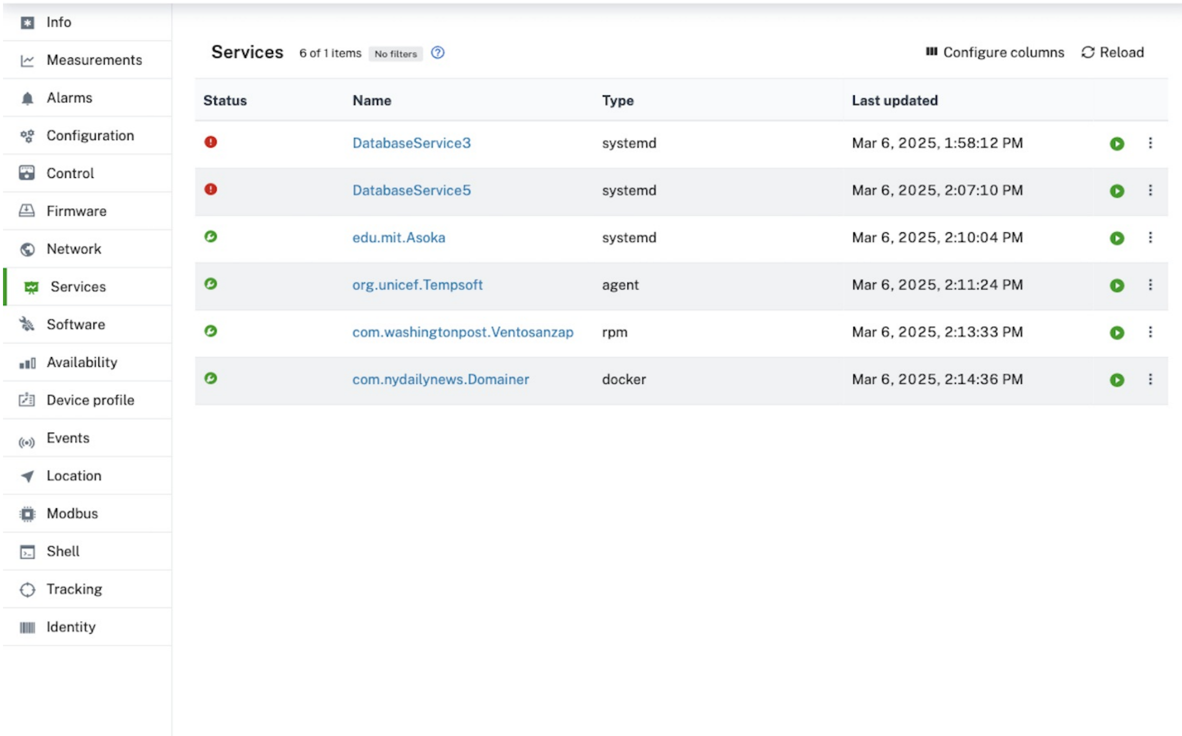
1. Enter the address range in which the connection can be established.
2. Enter the DNS.
3. Enter the DNS 2.
4. Enter the domain name.
5. Click **Save** to save your settings.

INFO

If the LAN configuration is disabled, the DHCP configuration is automatically disabled as well.

SERVICES

The **Services** tab provides a list of all services running on a device with their status, name, type and date of the last update. Every column allows services to be filtered and/or sorted by the respective value displayed.



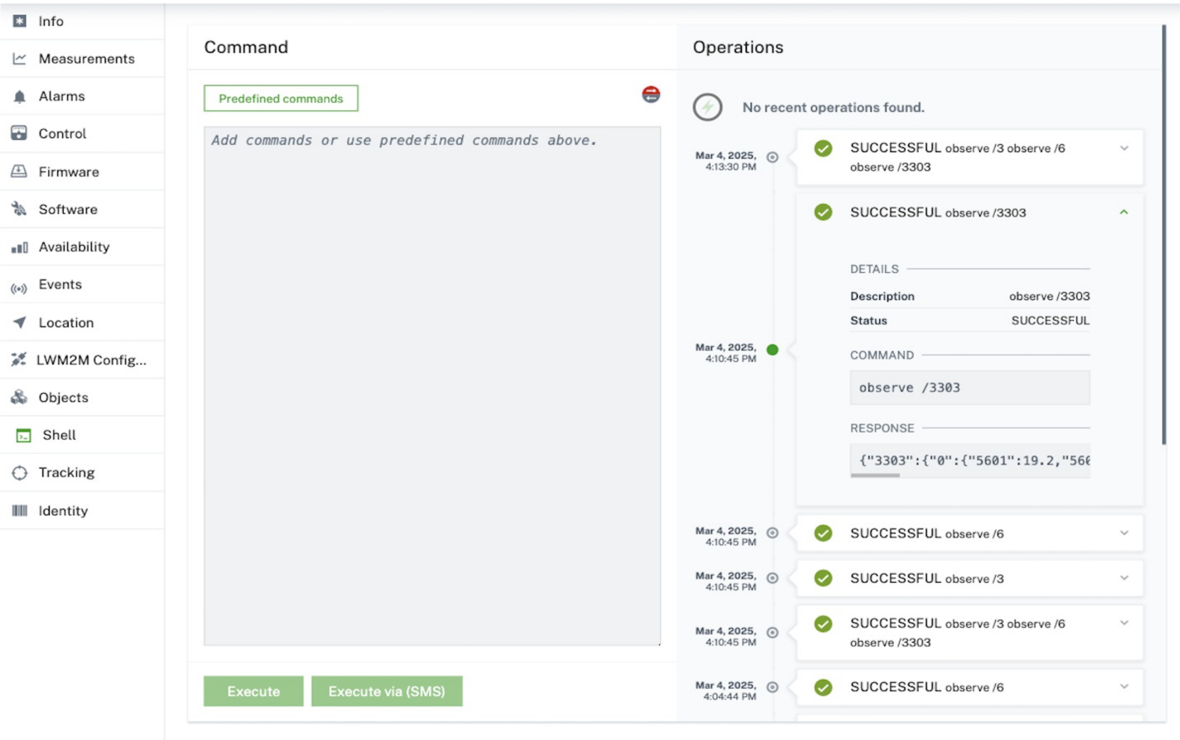
The platform defines three status options for services: **Up**, **Down** and **Unknown**. These predefined statuses have their own graphical representation and can be selected directly from the **Status** filter. Other status options are also allowed and can be freely defined, see Service module in the Web SDK documentation resources. They all share the same icon and must be entered manually in the **Status** filter. A tooltip on the status icon displays their actual value.

SHELL

The device shell enables you to interactively work with remote devices. Many industrial devices support some form of command language, like AT commands for modems, CSV-style commands for many tracking devices or elaborate scripting mechanisms provided by many gateway makers. In the shell, you can send commands in the respective language of the device and interactively view the results of the commands.

The **Shell** tab presents a command prompt to enter commands.

In the command prompt you can enter arbitrary command text. To send the command text to the device, click **Execute**. This button only is activated once the command is written in the text area.



IMPORTANT

When using Cumulocity to remotely operate machinery, make sure that all remote operations follow the safety standards and do not cause any harm.

For your convenience, Cumulocity provides several frequently used commands for some devices. Click **Predefined commands** above the command prompt area to open a window containing a list of available pre-defined commands. Select the command of your choice and click **Use**, to copy the command to the command prompt. You may also add new commands here for re-use.

SOFTWARE

See Managing software on a device for more information on how to manage and update the software installed on a device.

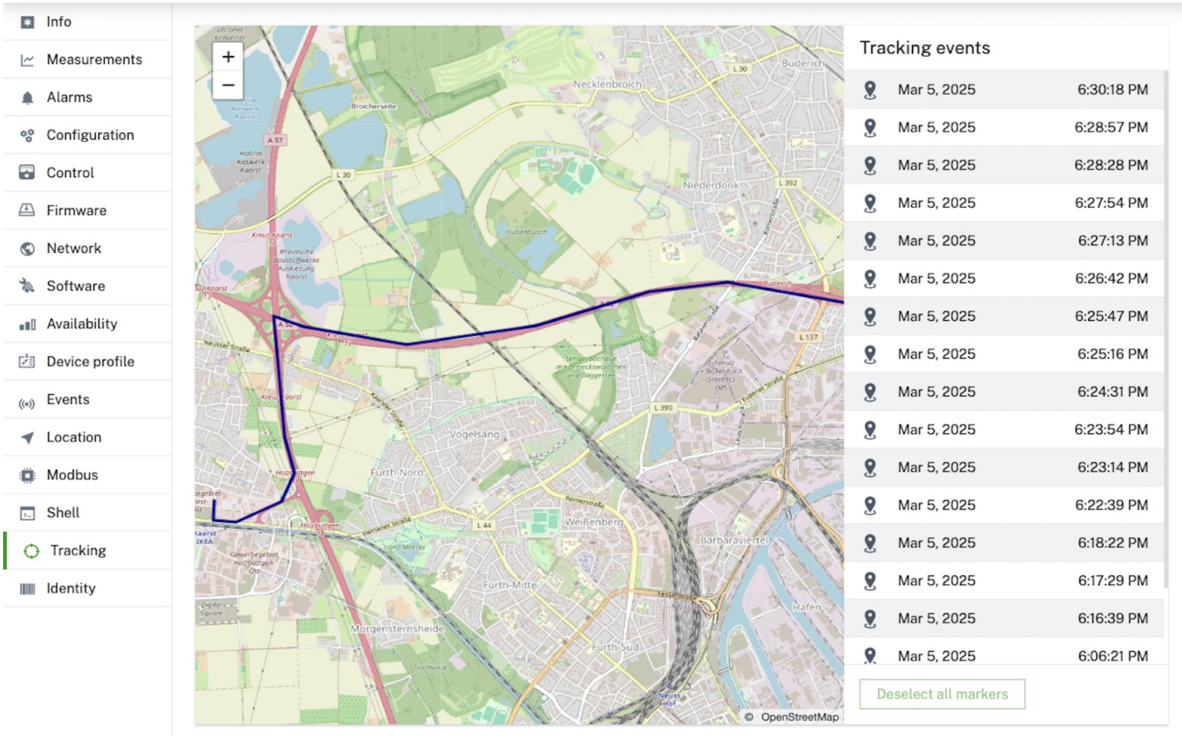
TRACKING

Devices can record the history of their movements in Cumulocity. This movements may be viewed in the **Tracking** tab.

INFO

The **Tracking** tab only shows up when a device contains `c8y_Position` property.

In the dropdown list at the top right you can select a time period (or specify one by selecting “Custom- from the list) and visualize the movements of the device during this period. Movements are shown as blue lines in the map.



Next to the map, the individual recordings with their time are listed (“location update events”). When you click a recording, a “pin” on the map will show the location at the time of the recording.

Depending on the type of device and the integration into Cumulocity, you can configure device-side geo-fencing and motion detection.

MONITORING AND CONTROLLING DEVICES

RELATED TOPICS

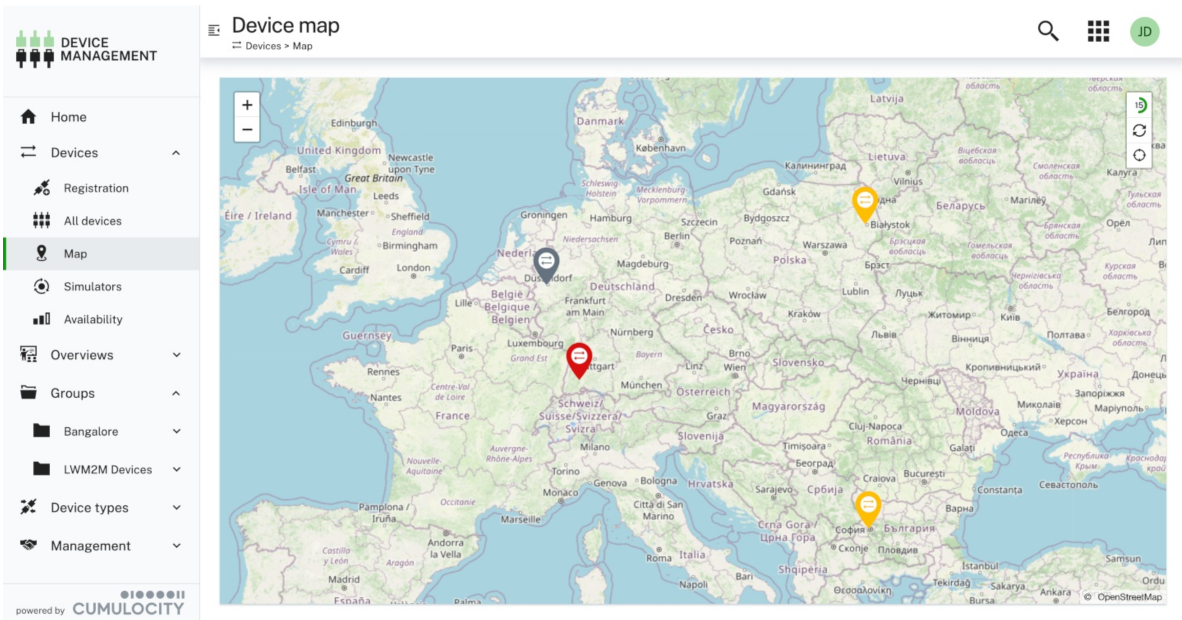
- Device management & connectivity > Device integration > Fragment library > Device availability for details on the `c8y_RequiredAvailability`, `c8y_UnavailabilityAlarm`, `c8y_Availability` and `c8y_Connection` fragments used in managed objects.
- The alarms API for REST API methods concerning alarms.
- Device management & connectivity > Device integration > Fragment library > Alarms for details on how to raise and clear alarms.
- The operations API for REST API methods concerning operations.
- The bulk operations API for REST API methods concerning bulk operations.

LOCATING DEVICES

Cumulocity provides the option to view all devices in your account on a map.

Click **Map** in the **Devices** menu in the navigator to display a map showing all devices in real time.

Devices are represented as “pins”. Click a pin to see the name of the respective device. Click the device name to switch to its device details.



CONNECTION MONITORING

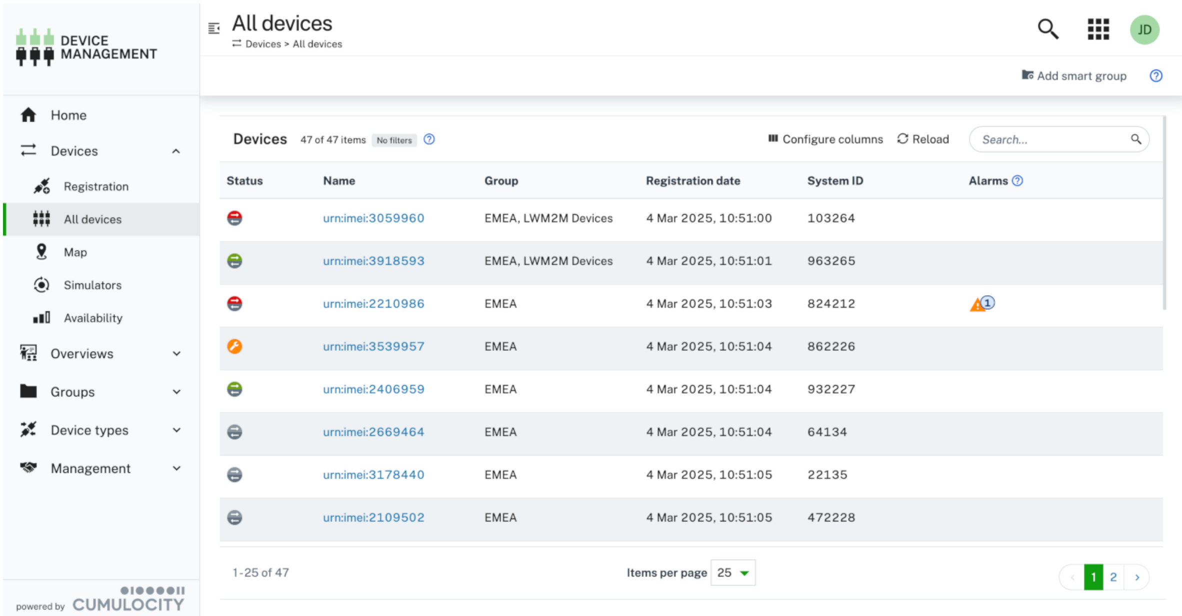
In the Device Management application you can monitor the connections to your devices.

This can be done at the level of individual devices (see below) or across multiple devices in a list.

TO MONITOR THE CONNECTION FOR MULTIPLE DEVICES

Open a device list to monitor the connections for multiple devices.

The connection status is represented by arrows in the **Status** column in the device list.



Send connections

The top arrow represents the send connection (traffic from the device to Cumulocity). The status for the send connections may be one of:

- Online (data was sent within the required interval)- indicated by a green arrow .
- Offline (data was not sent within the required interval) - indicated by a red arrow .
- Unknown or not monitored (no interval configured) - indicated by a grey arrow .

Hovering over the arrow displays the timestamp of the last request from the device to the server.

When a device is detected to be offline (stops sending data within required interval and top arrow changes to red color), an unavailability alarm is created for the device: “No data received from device within required interval”.

Send connections are updated when something is sent from the device, such as alarms, events, measurements or if a blank update is sent to the device itself. For details see Device availability.

INFO

Empty PUT requests to the managed object of the device will also update a send connection. Such requests are the recommended way of implementing a heartbeat service that monitors the server status.

Push connections

The bottom arrow represents the push connection (from Cumulocity to the device). The status for the push connections may be one of:

- Online (connection established)- indicated by a green arrow .
- Offline (connection not established) - indicated by a red arrow .
- Not monitored - indicated by a grey arrow .

A push connection is an active HTTPS long poll connection to the `/notification/operations` API endpoint or an active MQTT connection to the MQTT endpoint of Cumulocity. It is always green if the device is connected, even without data.

INFO

Connection monitoring is not real time. This means that the displayed connection status will not change immediately after switching off a device. Depending on the used protocol for push connection monitoring this can take a couple of minutes.

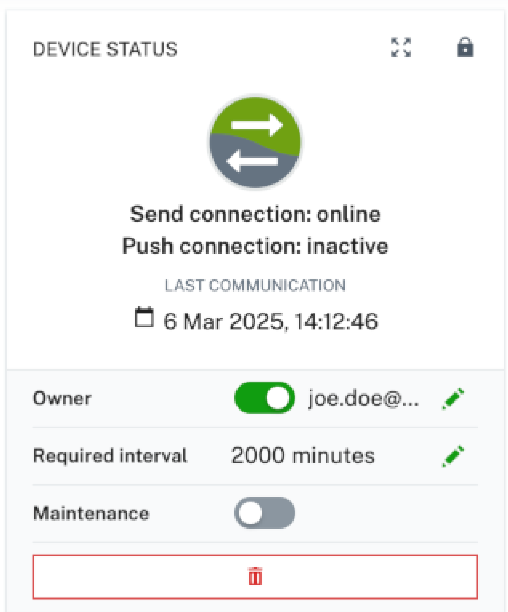
Maintenance mode

Moreover, the device may be in “Maintenance” mode, indicated by the tool icon in the **Status** column. This is a special connection status indicating that the device is currently being maintained and cannot be monitored. While a device is being maintained, no alarms for that device are raised.

You can turn the maintenance mode on or off for a device through a toggle in the **Connection monitoring** card in its **Info** tab, see below.

TO MONITOR THE CONNECTION OF A PARTICULAR DEVICE

Navigate to the **Info** tab of a particular device to monitor the connections of this device. Under **Device status**, the connection status for the device is displayed.



Below the send connection and push connection status, the time of the last communication is displayed.

INFO

“Last communication” and “Last updated” are two entirely different time stamps. “Last communication” indicates when a device has last sent data. “Last updated” indicates when the inventory entry of the device was last updated. This update may have originated from the device, from the web user interface or from another application.

In the **Required interval** field you can specify an interval. This parameter defines how often you expect to hear from the device. If, for example, you set the required interval to 60, you expect the device at least to communicate once in an hour with Cumulocity. The interval is either set by the device itself, based on the device’s knowledge how often it will try to send data, or it is set manually by you.

If an interval is set, you will find the **Maintenance** toggle below it.

With the **Maintenance** toggle you can turn the maintenance mode for the device on or off which is immediately reflected in the connection status.

AVAILABILITY

Cumulocity distinguishes between connection monitoring and availability. Connection monitoring, as described in the previous section, only indicates if the device is communicating with Cumulocity, it does not automatically indicate if it is functional or not.

Availability indicates if a device is in service. For example, a vending machine is in service if it is ready to sell goods. A vending machine can sell goods using cash money without a connection to Cumulocity. From the perspective of a merchant, it is in service. Similar, if you switch off the power on a gateway, the devices behind the gateway can still continue to work.

Cumulocity considers a device to be in service while there is no critical, unresolved alarm present for the machine. This is displayed as a share of time such an alarm was present. If a machine didn’t have any critical alarms whatsoever during a time period, it was 100% in service. If half of the time there was some critical, unresolved alarm, the machine was 50% in service.

While a machine is offline, Cumulocity assumes by default

- that the machine continues to stay in service during the connection outage, if this was the status before it lost connection.
- that the machine continues to stay out of service, if this was the status before it lost connection.

There may be exceptions from this rule. If your vending machines rely exclusively on cashless payment, losing the connection to the network means that your machine is out of service and stops selling. In this case, unavailability alarms must be set in the Administration application which have CRITICAL severity instead of MAJOR severity.

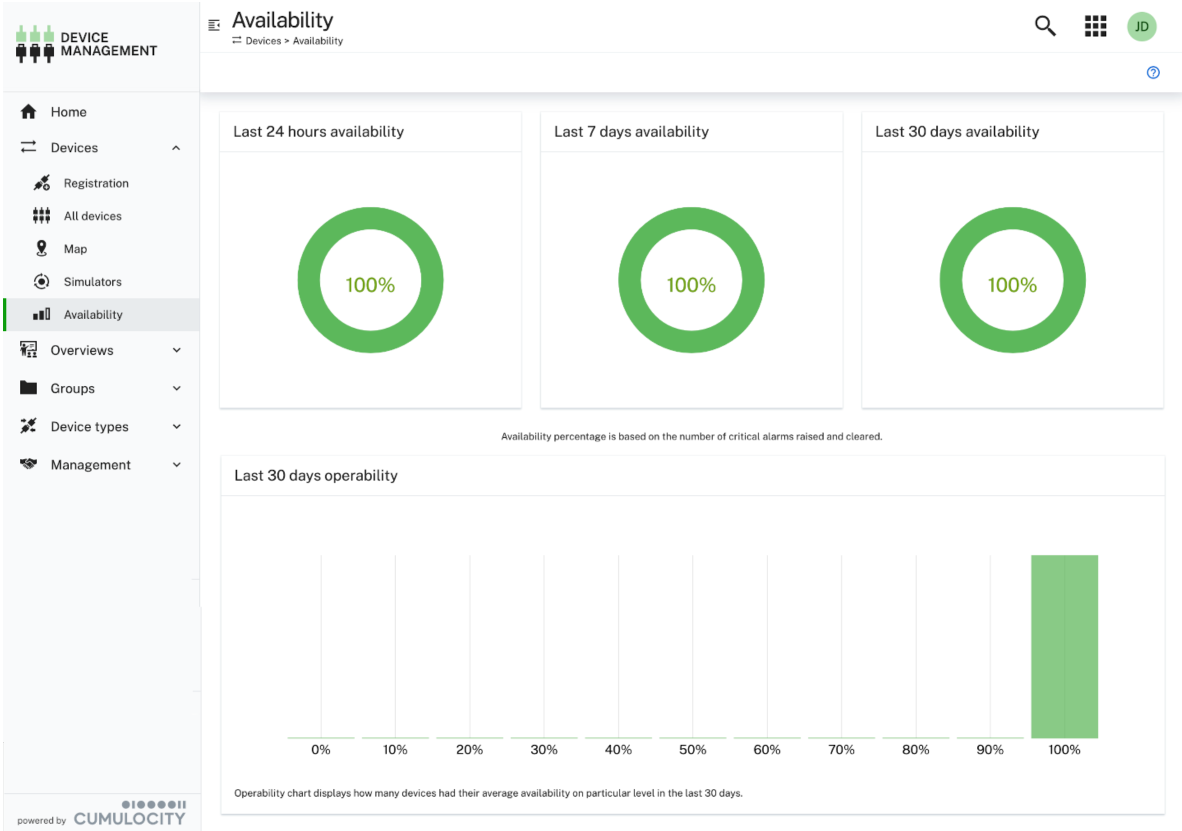
Cumulocity displays service availability at the level of individual devices and across all devices.

TO VIEW THE AVAILABILITY OF A PARTICULAR DEVICE

Click the **Availability** tab in the details of a particular device to check the availability of this device.

TO VIEW THE AVAILABILITY ACROSS ALL DEVICES

Click **Availability** in the **Device** menu in the navigator to display the overall service across all devices.



The **Availability** page shows the availability of devices for the last 24 hours, last 7 days and last 30 days in percentage.

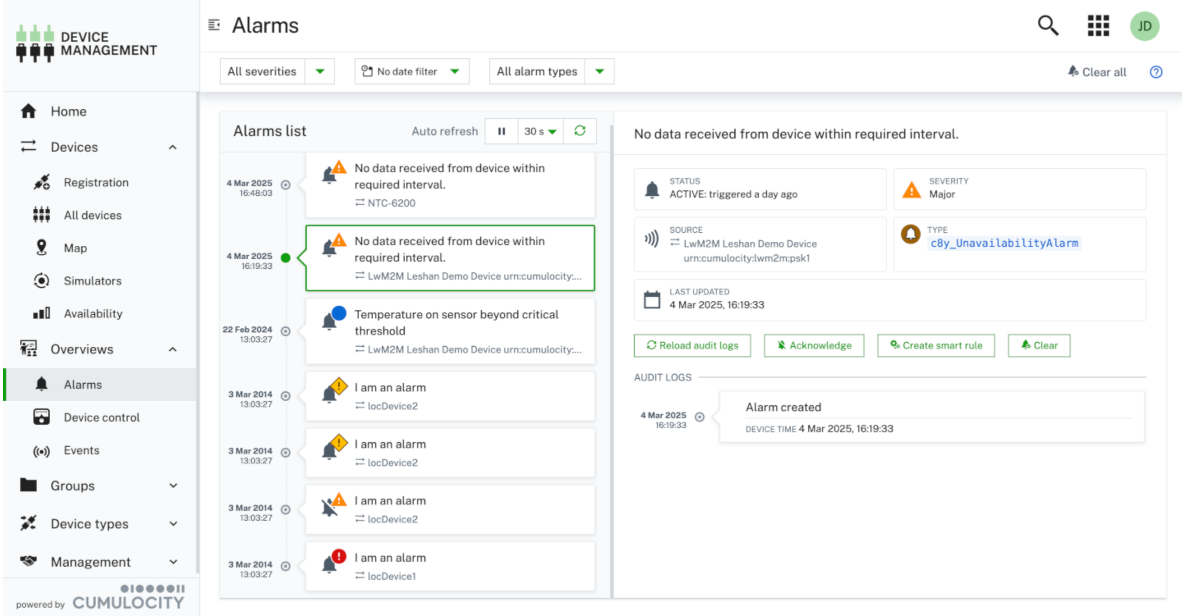
WORKING WITH ALARMS

Devices can raise alarms to indicate that there is a problem requiring an intervention.

TO VIEW ALARMS

Cumulocity displays alarms at the level of individual devices and across all devices:

- To check the alarms for all devices, click **Alarms** in the **Overview** menu in the navigator.
- To check the alarms of a particular device, switch to the **Alarms** tab in the details of this device.



By default,

- only unresolved alarms are shown. If you turn on **Show cleared alarms** in the **Filter alarms** dropdown menu at the top, you will see the entire alarm history.
- The alarms list is automatically refreshed every 30 seconds. Click **Disable auto-refresh** to stop the alarms list from refreshing automatically.

Alarms are classified according to their severity. Cumulocity includes four different alarm types:

Severity	Description
CRITICAL	The device is out of service and should be fixed immediately.
MAJOR	The device has a problem that should be fixed.
MINOR	The device has a problem that may be fixed.
WARNING	There is a warning.

In the top menu bar, a **Filter alarms** dropdown list is provided. Select the severities to be shown and then click **Apply**.

INFO

The number provided in the filter dropdown refers to the number of active alarms for the given severity, as opposed to the counter provided as blue circle next to an active alarm, which shows the number of times this same alarm has occurred (see also the table below).

Within the **Alarms list**, the alarms are sorted by their occurrence, displaying the most recent alarm first.

In the **Alarms list**, the following information for an alarm is provided:

Info	Description
Severity	One of CRITICAL, MAJOR, MINOR, WARNING (see above).

Count (provided as number in a blue circle)	The number of times this alarm was sent by the device. Only one alarm of a particular type can be active for a certain device. If another alarm of the same type is sent by the device, the number is increased by 1.
Description	An arbitrary text describing the alarm.
Status	The status of the alarm. An alarm can be: **Active**: When it was raised and nobody is so far working on the alarm. **Acknowledged**: When someone changed the status to "Acknowledged" to indicate that someone is working on the alarm. **Cleared**: When either someone manually set the status to "clear" or when the device detected by itself that the problem has gone.
Last occurrence	Timestamp of the last occurrence of the alarm (device time).
Device	The name of the device.

To display further details about an alarm, click on the respective alarm in the **Alarms list**.

- **Status**: Provides further information on the alarm status and shows the type of the alarm. The type info is used for duplicating alarms and for configuring the priority of alarms in Alarm mapping.
- **Audit Logs**: Provides the server time when the alarm was created, which may differ from the device time.
- **First Occurrence**: Provides information about when the alarm first occurred.

TO CHANGE THE STATUS OF AN ALARM

To change the status of an alarm, select the alarm and choose the desired status.

It is also possible to change the status of all alarms to “clear” at once. Click **Clear all** in the top menu bar, to clear all alarms of the selected severities.

WORKING WITH OPERATIONS

Operations are used to remotely control devices.

You can view operations at the level of individual devices and across all devices:

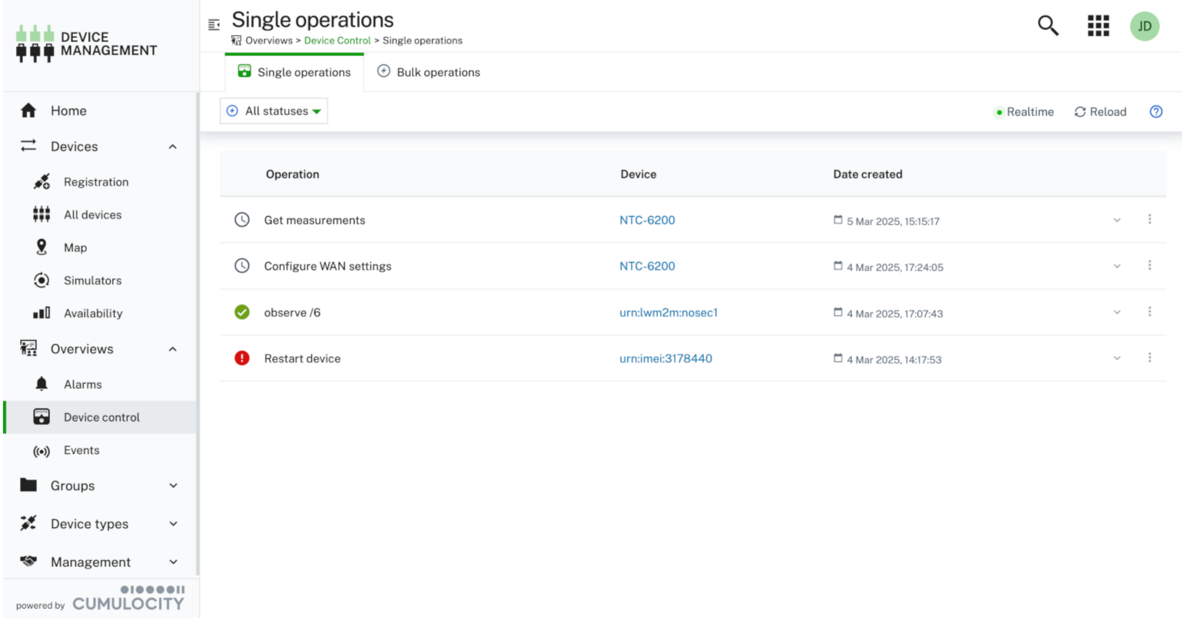
- To view the operations for all devices, click **Device control** in the **Overview** menu in the navigator.
- To view the operations of a particular device, switch to the **Control** tab in the details of this device.

There are two types of operations in **Device control**, each represented by a tab:

- **Single operations** execute on a single device, see To view single operations.
- **Bulk operations** comprise of the same single operation executed on a set of devices, see To view bulk operations.

TO VIEW SINGLE OPERATIONS

See the list of single operations in the **Single operations** tab.



Single operations can have one of the following four statuses:

Status	Description
PENDING	The operation has just been created and is waiting for the device to pick it up.
EXECUTING	The operation has been picked up by the device and is being executed.
SUCCESSFUL	The operation has been successfully executed by the device.
FAILED	The operation could not be executed by the device.

In each row, the following information for an operation is provided:

Info	Description
Status	One of PENDING, EXECUTING, SUCCESSFUL, FAILED (see above).
Name	Name of the operation.
Device	The name of the device. Clicking the name leads you to the detailed view of the device.

Clicking a row expands it and displays further details on the operation.

- **Details**: Providing information on the operation name and status. In case of status = FAILED the reason for the failure is provided. In case the single operation is part of a bulk operation, you can see the bulk operation details.
- **History of Changes**: Providing information on the past changes of the operation.

To filter the list of single operations by status, click one of the status buttons in the top menu bar. Click **All** to clear the filter.

Click **Realtime** at the right of the top menu bar to see operations coming in from the devices in realtime. Click **Reload** to update the list once manually.

INFO

Single operations are listed in descending time order. Operations are executed strictly according to this order.

TO ADD AND EXECUTE A SINGLE OPERATION

Single operations can be created either from bulk operations or via the different types of operations that the device supports: managing firmware, software, configurations and more.

When you create a bulk operation, the single operations entailed in the bulk operation are also added to the list of single operations.

Operations for a specific device can also be created and executed in the **Shell** tab of the device, see Shell.

> **IMPORTANT**
>
> When using Cumulocity to remotely operate machinery, make sure that all remote operations follow the safety standards and do not cause any harm.

TO CANCEL PENDING SINGLE OPERATIONS

To cancel a particular pending single operation, click the menu icon ⋮ at the right of the respective single operation entry and select **Cancel operation**.

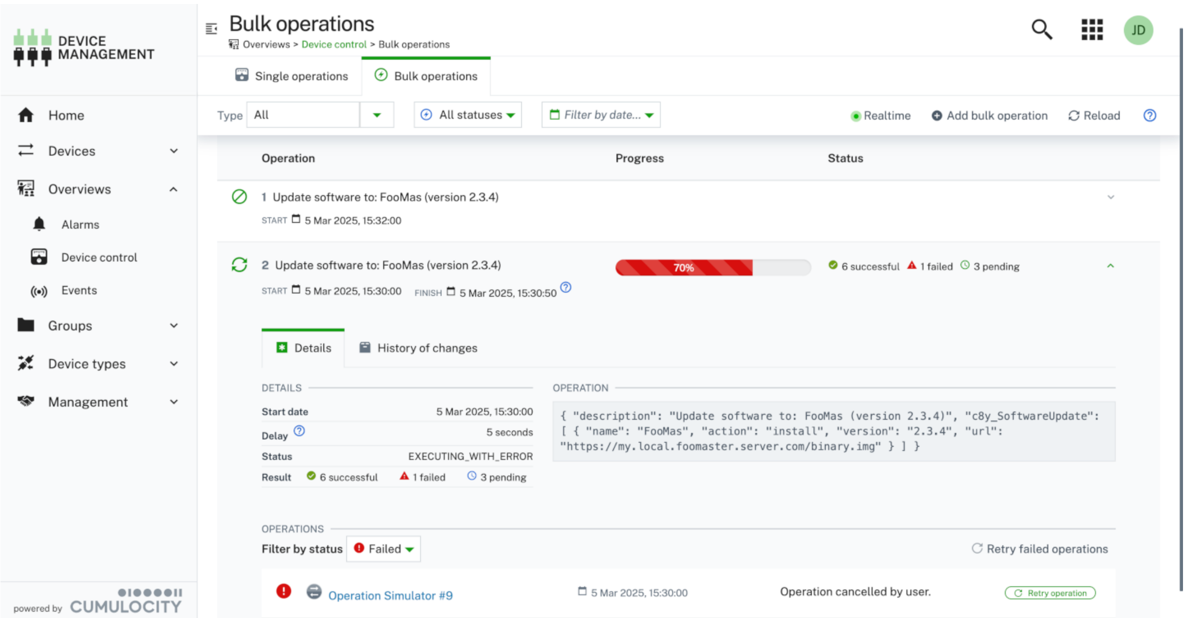
TO CREATE A SMART RULE FROM A SINGLE OPERATION

Click the menu icon ⋮ at the right of the single operation that you want to create a smart rule for, and select **Create smart rule**.

See To create a smart rule for further steps.

TO VIEW BULK OPERATIONS

See the list of bulk operations in the **Bulk operations** tab.



Bulk operations have an operation type and status.

You can add bulk operations of the following operation types with the bulk operation wizard:

Operation type	Description
Configuration update	The bulk operation updates the configuration of the selected devices.
Firmware update	The bulk operation updates the firmware on the selected devices.
Software update	The bulk operation updates the software on the selected devices.
Apply device profile	The bulk operation applies a device profile on the selected devices.

Bulk operations can have other operation types as well, for example when you schedule a single operation as bulk operation and the single operation has a different operation type.

Bulk operations can have one of the following statuses:

Status	Description
SCHEDULED	The bulk operation has been created and is on hold until the scheduled time.
EXECUTING	The bulk operation is being executed.
CANCELED	The bulk operation was created but canceled before the scheduled time.
COMPLETED WITH FAILURES	The bulk operation completed with failures for some devices.
COMPLETED SUCCESSFULLY	The bulk operation has been successfully executed on all devices.

In each row, the following information for a bulk operation is provided:

Info	Description
Status	One of SCHEDULED, EXECUTING, CANCELED, COMPLETED WITH FAILURES, COMPLETED SUCCESSFULLY (see above).
Name	Name of the operation.
Progress bar	Only for executing and completed bulk operations. Shows the operation’s progress in percent.
Start and finish dates	Only for executing and completed bulk operations. For executing bulk operations, the finish date is an approximation based on the bulk operation settings.
Refresh button	Only for executing bulk operations. Updates the progress bar.

Clicking the arrow button at the right in a row expands the row and displays further details on the bulk operation.

- **Details**: Providing information on the start date, delay, status and result of the bulk operation. The result lists the number of successful, failed and pending operations. If the bulk operation is a retry attempt for failed operations, there will be an additional row that shows the index of the bulk operation it was retried from. Click the index to scroll to that bulk operation. If a description was added when the bulk operation has been created, there will be an additional row that shows this description.
- **Operation**: Providing information on the operation in the form of a JSON object.
- **Operations**: Only present for executing or completed bulk operations. Providing information on the status and the device of single operations entailed in the bulk operation. Can be filtered by status. You may also either retry all failed operations by clicking **Retry failed operations** at the top right of the **Operations** section or retry single operations by hovering over them and then clicking the **Retry operation** button that appears right next to it. Also see To retry failed operations.
- **History of Changes**: In a second tab, providing information on the past changes of the bulk operation.

To filter the list of bulk operations by operation type, click the dropdown list in the top menu bar and select a set of operation types, then click **Apply**. To clear the filter, select **All** in the dropdown list and click **Apply** once more.

To filter the list of bulk operations by status, click one of the status buttons in the top menu bar. Click **All** to clear the filter.

To filter the list of bulk operations by date, select a date in both the **Date from** and **Date to** datepickers, then click **Apply** right next to it. To clear the filter, click **Clear** right next to it.

To clear both filters, click **Reset filters** at the bottom of the list (only visible if filters are applied).

TO ADD A BULK OPERATION

There are two ways of creating a bulk operation:

- Use the bulk operation wizard
- Schedule a single operation as bulk operation

To add a bulk operation using the wizard

Follow these steps:

1. In the **Bulk operations** tab, click **New bulk operation** at the right of the top menu bar.
2. In the resulting dialog, select an operation type.
3. The resulting wizard has four steps. The first two steps differ depending on the operation type:

- **Configuration update**
  - Select a configuration from the list. The list can be filtered by configuration type or by configuration name. Click **Next**.
  - Check the preview of the selected configuration. Click **Next**.
- **Firmware update**
  - Select a firmware from the list. The list can be filtered by firmware name. Click **Next**.
  - Expand a version and select a patch. Click **Next**.
- **Software update**
  - Expand a software from the list and select a version, then also choose to install/update or remove the software from the dropdown list. The list of available software can be filtered by device type, by software type or by software name. Click **Next**. If you selected software for multiple device types, a warning dialogue appears and informs you that some operations may fail due to unsupported software, and ask for confirmation.
  - Confirm the selection and click **Next**.
- **Apply device profile**
  - Select a device profile from the list. The list can be filtered by device type or by profile name. Click **Next**.
  - Confirm the selection and click **Next**.

4. Select target devices by applying filters to the paginated list of all devices. You can filter by status, name, type, model, group, registration date and alarms. You may apply multiple filters. To apply a filter, click the column header, make your filter option choices in the context menu and click **Apply**. The group filter also allows filtering by subgroups. To select a subgroup, if there are any, click the arrow button at the right of a group and select the desired subgroups from the dropdown. You can clear all filters by clicking **Clear filters** above the list. For the operation types “configuration update”, “software update” and “apply device profile”, the list is already filtered by the according device type. Click **Next**.
5. Enter a new title or use the preset title. Optionally enter a description. Select a start date and a delay. The delay may either be in seconds or milliseconds and is the time spent between each single operation of the bulk operation. Click **Schedule bulk operation** to create the bulk operation.

To schedule a single operation as bulk operation

You can schedule a single operation as a bulk operation either from the **Single operations** tab or from a **Control** tab of a particular device. Follow these steps:

1. Click the menu icon ⋮ at the right of the single operation that you want to schedule as a bulk operation and then click **Schedule as bulk operation**.
2. The resulting wizard is similar to the new bulk operation wizard described in To add a bulk operation using the wizard. However, there are just two steps because the operation type is inferred from the operation that is scheduled as a bulk operation. See the description of the full wizard and follow it.

TO EDIT THE SCHEDULE OF BULK OPERATIONS

You may only edit the schedule of bulk operations with status = SCHEDULED.

1. Click the menu icon ⋮ to the right of the bulk operation that you want to edit, and then click **Edit schedule**.
2. In the resulting dialog box you may change the **Start date** and **Delay** values.
3. Click **Reschedule** to apply your changes.

The changes are applied to the bulk operation accordingly.

TO CANCEL BULK OPERATIONS

You may only cancel bulk operations with status = SCHEDULED or EXECUTING. If it is executing, you may only cancel the operation until all of its single operations are created. This way, you can cancel the creation of the remaining single operations.

Click the menu icon ⋮ to the right of the bulk operation that you want to cancel, then click **Cancel bulk operation**.

TO RETRY FAILED OPERATIONS

You may retry the failed operations of a bulk operation that is either executing or completed with failures.

To do so, expand the desired bulk operation, then click **Retry failed operations** under **Operations** to create a new bulk operation with all failed operations. To retry a single operation, hover over the operation and click **Retry operation**. This will create a new single operation.

For a bulk operation that completed with failures, you may also click the menu icon ⋮ to the right of the operation, then click **Retry failed operations**.

TO MANUALLY SET FAILED BULK OPERATIONS TO SUCCESSFUL

You may manually set a failed bulk operation to the status SUCCESSFUL.

To do so, click the menu icon ⋮ to the right of the bulk operation, then click **Set operation to successful**.

This may be useful if the operation is generally a success, but contains operation failures on devices that are not too important. These failures would otherwise still leave the bulk operation in status FAILED.

> INFO
>
> **Calculation of completion percentage**
>
> The completion percentage is determined by comparing the total number of initiated operations to the number of operations that have reached a final state (success or failure). The total count is established at operation creation and remains static thereafter. This static value may lead to inaccuracies in the completion percentage if the overall number of operations subsequently changes.
>
> - **Operation removal**: Operations can be implicitly removed when their associated devices are deleted. This deletion goes unaccounted for in the completion percentage calculation.
> - **Operation addition**: Retrying failed operations can introduce new entries without modifying the original total count. This again results in a misrepresentation of the completion percentage.
>
> The completion percentage is calculated based on the information available at the time an operation is created and further changes are not applied.

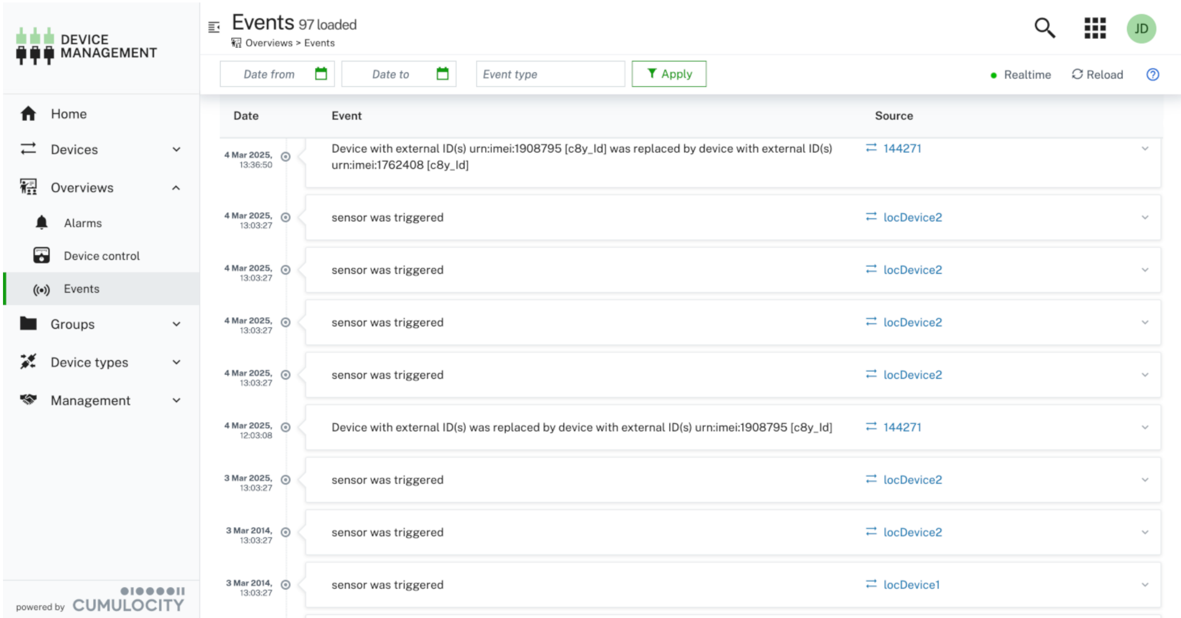
TROUBLESHOOTING DEVICES

Troubleshooting devices at a more detailed level can be done with the help of events. Events are low-level messages sent by devices that are usually used for application-specific processing. For example, a vending device sends its real-time sales in the form of events.

TO VIEW EVENTS

Cumulocity displays events at the level of individual devices and across all devices:

- To view the events for all devices, click **Events** in the **Overview** menu in the navigator.
- To view the events of a particular device, switch to the **Events** tab in the details of this device.



Per default, events are shown as coming in from the devices in real time. To disable real-time updates, click **Realtime** at the right of the top menu bar.

For each event, the following information is provided:

Info	Description
Timestamp	Timestamp when the event has been executed.
Name	Name of the event.
Device	The name of the device sending the event. Clicking the name leads you to the detailed view of the device.

In the event list, the latest entry is displayed on top.

Clicking a row expands it and displays further details on the event (as type and position of the device).

Since devices may send large amounts of event data, you can filter the data to be displayed by date.

Select a start date and an end date from the fields in the top menu bar and click **Apply** to apply the filter. Click **Clear** to clear the filter again.

MANAGING DEVICE SERVICES

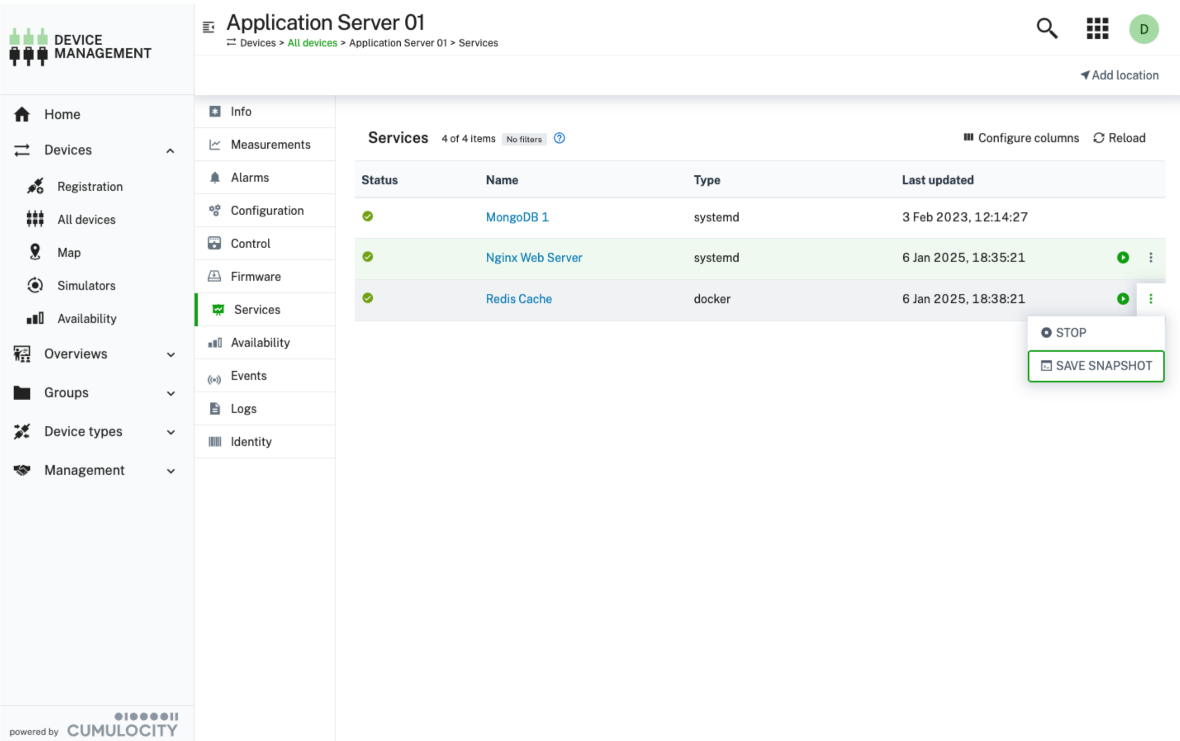
RELATED TOPICS

- The alarms API for REST API methods concerning alarms.
- Device management & connectivity > Device integration > Fragment library > Alarms for details on how to raise and clear alarms.
- The events API for REST API methods concerning events.
- The measurements API for REST API methods concerning measurements.
- Device management & connectivity > Device integration > Fragment library > Measurements for details on measurements.

The Device Management application lets you monitor the data that your devices send about the services they are running.

The Services tab on the device details view provides an overview of the services running on a given device and acts as an entry point to the service details view. There you can see detailed information about measurements, events and alarms sent for every service.

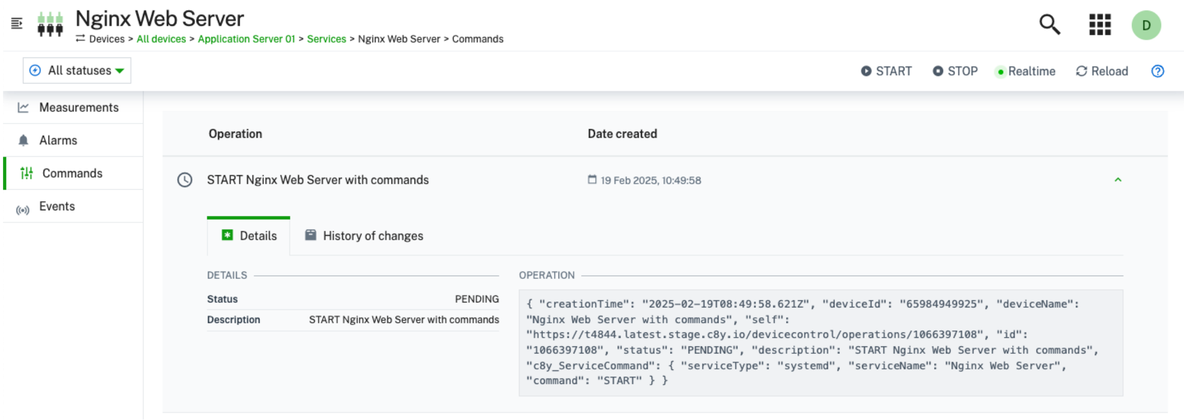
For services that support commands, actions like **Start**, **Stop**, **Restart**, or custom commands appear in the menu of each service. This allows users to quickly send commands without opening the full service details.



The following tabs make up the service details view, each described in detail in a separate sector:

Tab	Description
Alarms	Provides information on the alarms for a service. See Working with alarms. Available for each service.
Events	Displays events related to a service. Available for each service.

Measurements	Provides a default visualization of numeric data of the service in the form of charts.
Commands	Allows users to send command actions to a service and view the history of executed commands.



SERVICE COMMANDS

The **Commands** tab allows users to send available service commands and track their execution history. If a service supports commands, they will appear as action buttons in the services list and as selectable options in the service **Commands** tab.

Sending commands to services

For a service to support commands, it must include the `c8y_ServiceCommand` fragment in its supported operations.

Supported services may provide specific command actions, such as:

- Start/stop
- Restart
- Custom commands (for example, “Flush cache”, “Update”, “Reset settings”)

If a service does not specify commands, a default set (Start, Stop, Restart) is available.

Tracking service command history

The **Commands** tab displays a history of executed commands, including:

- The command type (Start, Stop, and so on).
- The execution status (Pending, Completed, Failed).
- Timestamps for sent and completed actions.

ALARMS

The **Alarms** tab provides information on the alarms of a service. See Working with alarms for detailed information on alarms.

> **INFO**
>
> The service details **Alarms** tab displays only alarms which have the particular service as a source. It does not display any alarms sourced by the device itself.

EVENTS

The **Events** tab displays events related to a service. See Troubleshooting devices for detailed information.

INFO

The service details **Events** tab displays only events which have the particular service as a source. It does not display any events sourced by the device itself.

MEASUREMENTS

The **Measurements** tab provides a default visualization of numeric data for the service in the form of charts.

INFO

The service details **Measurements** tab displays only measurements which have the particular service as a source. It does not display any measurements sourced by the device itself.

For more information about how to use the **Measurements** tab see Measurements.

MANAGING DEVICE TYPES

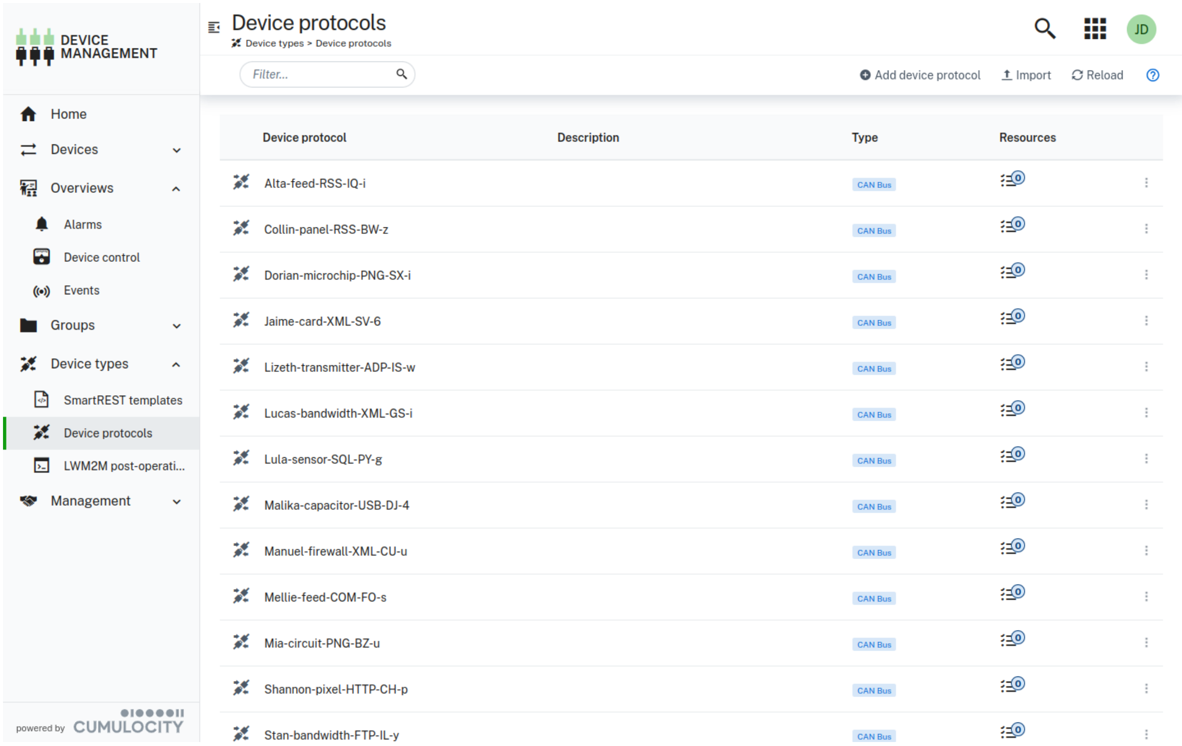
RELATED TOPICS

- Device management & connectivity > Device integration > Interfacing devices for information on the concepts relevant for interfacing IoT devices and other IoT-related data sources with Cumulocity.
- Device management & connectivity > Device integration for a list of protocols, parameters and network connectivity options for devices.

To process data from various device types, Cumulocity uses device protocols which are stored in a database.

Click **Device protocols** in the **Device types** menu in the navigator.

In the **Device protocols** page you will find a list with all device protocols available in your account.



The device protocol list shows the following information:

- the device protocol type (for example Modbus, CANOpen, LoRa)
- the device type name
- the number of resources for the device (at the right)

TO ADD A DEVICE PROTOCOL

1. Click **Add device protocol** in the top menu bar.
2. Select one of the available device protocol types from the list.
3. In the resulting dialog box, enter a name and an optional description for the device protocol and click **Create**.
4. Enter the configuration for the device protocol. The configuration of the device protocol depends on the protocol type.
   For details on configuring device protocols, follow the documentation of the particular device type you want to create, see Device integration.
5. Click **Save**.

The device protocol will be added to the device database.

TO IMPORT A DEVICE PROTOCOL

To add a device protocol from an existing protocol, perform the following steps:

1. Click **Import** in the top menu bar.
2. Either select the device protocol to be imported from a list of predefined protocols or load it from a file by browsing.
3. Provide a name for the new protocol and click **Save**.

The device protocol will be added to the device database.

TO EDIT A DEVICE PROTOCOL

To edit a device protocol, just click on the protocol or click the menu icon ⋮ at the right of the row and then click **Edit**.

Details on the fields can be found in the documentation of the particular device type, see Device integration.

TO DELETE A DEVICE PROTOCOL

To delete a device protocol, click the menu icon ⋮ at the right of the row and then click **Delete**.

The device protocol will be deleted from the device database.

TO EXPORT A DEVICE PROTOCOL

To export a device protocol, click the menu icon ⋮ at the right of the row and then click **Export**.

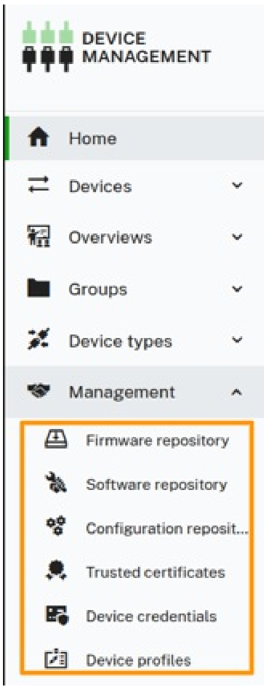
The device protocol will be exported to your file system.

MANAGING DEVICE DATA

RELATED TOPICS

- Device management & connectivity > Device integration > Fragment library > Firmware for details on the `c8y_Firmware` fragment used in managed objects.
- Device management & connectivity > Device integration > Fragment library > Software for details on legacy software management via the `c8y_SoftwareList` fragment, software updates via the `c8y_SoftwareUpdate` fragment and advanced software management via the `c8y_SupportedOperations` and `c8y_SupportedSoftwareTypes` fragments used in managed objects.
- Device management & connectivity > Device integration > Fragment library > Configuration for details on text-based configuration, file-based configuration and typed file-based configuration.
- The device credentials API for REST API methods concerning device credentials.
- Device management & connectivity > Device integration > Fragment library > Device profile for details on the `c8y_DeviceProfile` fragment used in managed objects.
- Device management & connectivity > Device integration > Device certificates for information on device certificates in the context of MQTT.

The Device Management application provides features that support you in efficiently managing your devices. They are accessible through the **Management** menu in the navigator:



MANAGING FIRMWARE

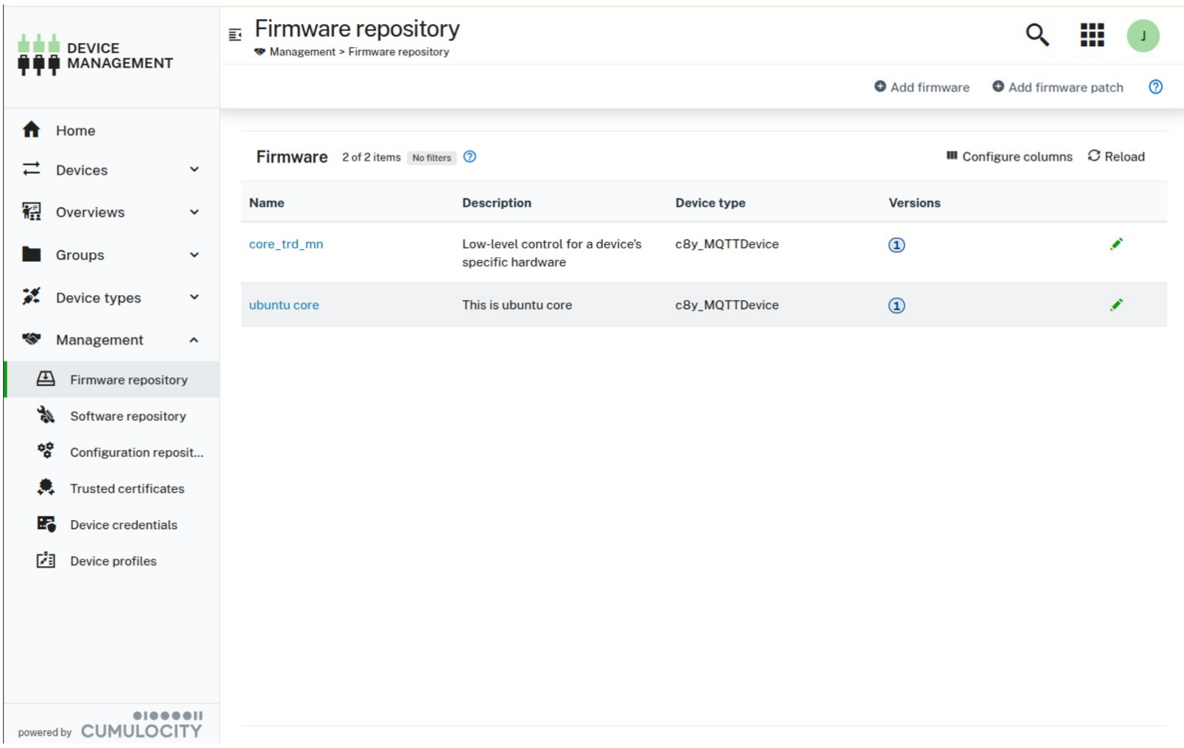
In the firmware repository, Cumulocity offers to collect reference firmware for devices.

Only one firmware package version can be applied per device.

TO VIEW FIRMWARE

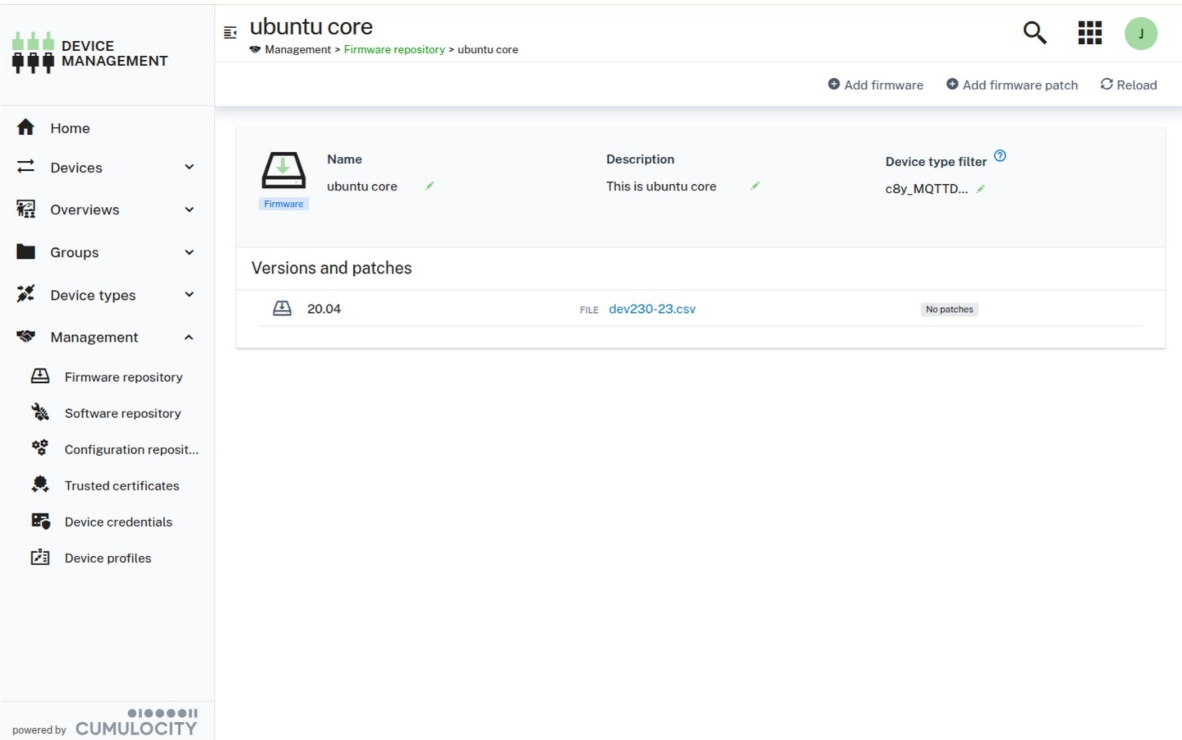
Click **Firmware repository** in the **Management** menu in the navigator.

The available firmware objects will be displayed as a list.



Each entry shows the firmware name, the device type it is applicable for (if set), and a label indicating if and how many versions are available for a particular firmware. At the left in the top menu bar, you can filter the repository entries by name, description or device type. For details on the filtering functionality, see Search and filter functionality.

When clicking on an entry, the details for this firmware are displayed along with all available versions and patches.

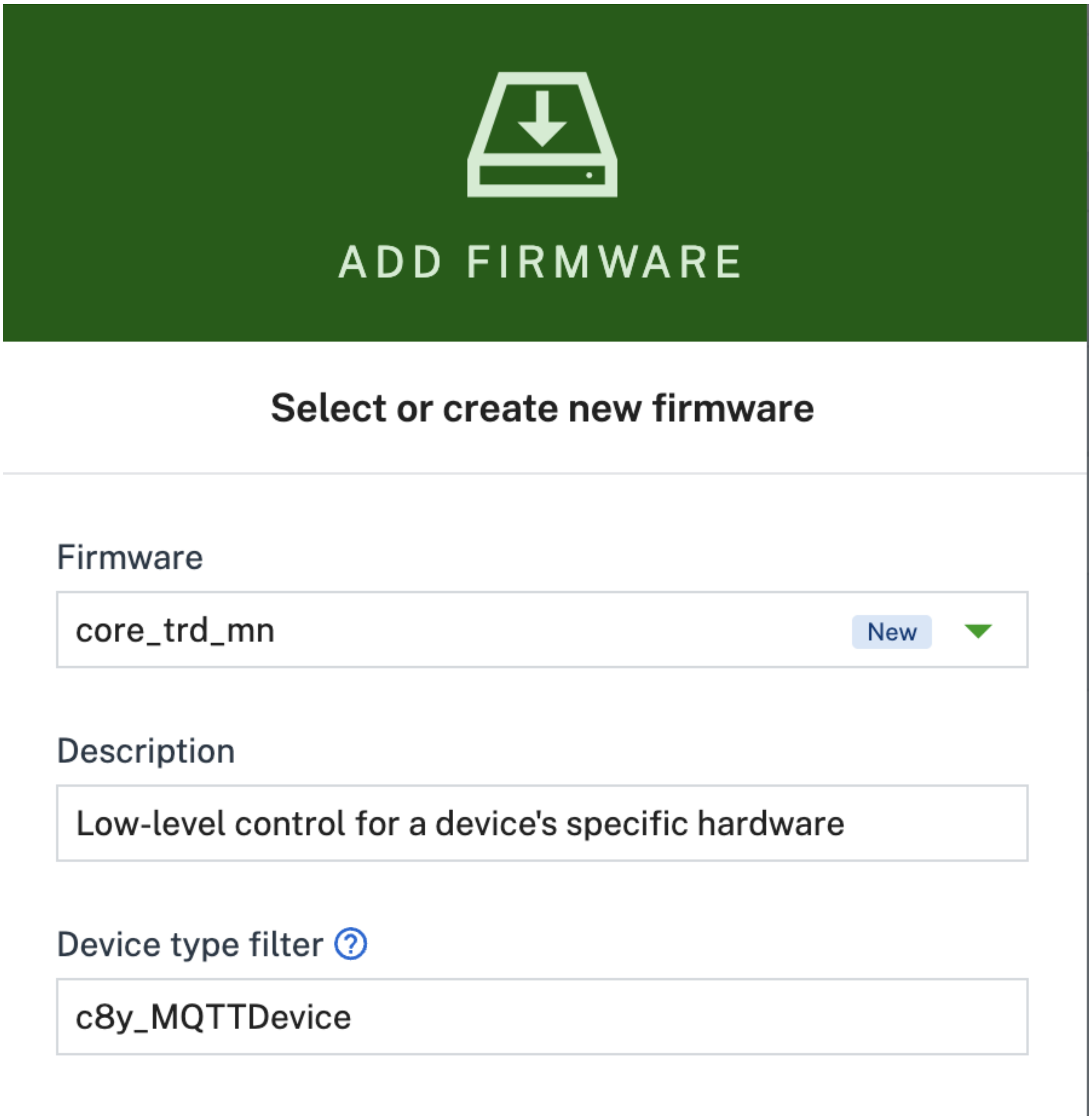


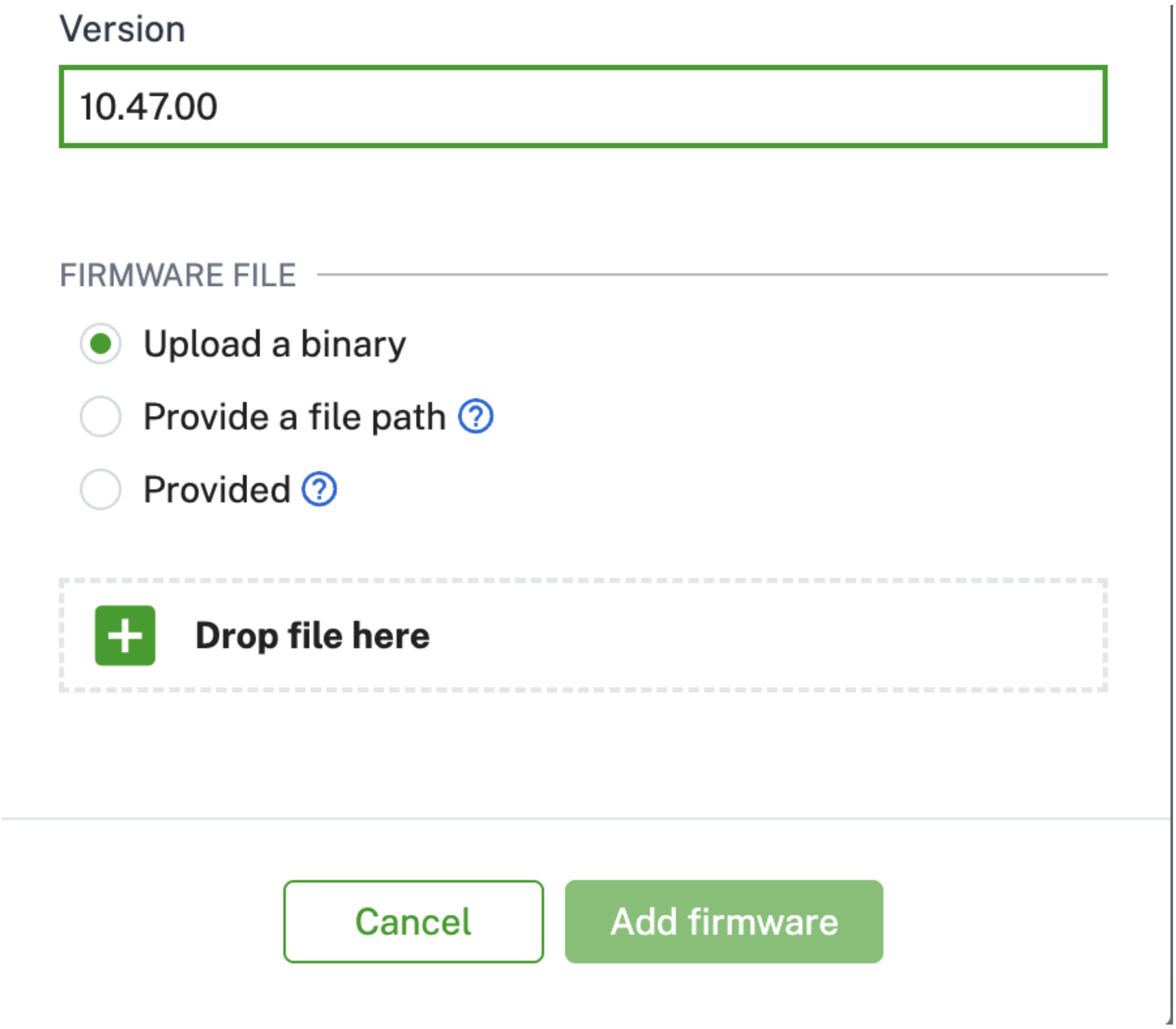
At the top, the firmware name, a description, and optional device type filter(s) are shown. If a filter is set, the firmware will show up for installation only for devices of that type. If no filter is set, it will be available for all devices.

The list of versions and patches shows the version name and the name of the firmware binary. Moreover, the list indicates if a firmware version has patches, which can be viewed by expanding the version entry. The versions and patches are ordered by their creation time (descending).

TO ADD A NEW FIRMWARE OR FIRMWARE VERSION

1. In the **Firmware repository** page, click **Add firmware** at the right of the top menu bar.
2. In the resulting dialog box,
   - to add a new firmware, enter a name for the firmware (and confirm it by clicking **Create new** in the resulting window), add a description and its version (all required).
   - to add a new version, select the firmware for which you want to add a new version from the dropdown list in the **Firmware** field and enter a version.
3. Optionally, you can define the device type filter when adding a new firmware.
4. Select one of the following options to define the patch binary:
   - **Upload a binary** from your file system.
   - **Provide a file path (URL)** to download the binary from.
   - **Provided** – Select this option if the device is expected to resolve the binary itself. No file or URL will be defined, and a `$PROVIDED` placeholder is stored.
5. Click **Save**.





The firmware object will be added to the firmware list or the firmware version will be added to the firmware details and the version label will be updated accordingly.

If you click **Add firmware** from within the details of a specific firmware, the dialog box looks slightly different as the firmware is already selected. Enter the new version number and upload a binary or provide a file path.

TO ADD A NEW FIRMWARE PATCH

1. In the **Firmware repository** page, click **Add firmware patch** at the right of the top menu bar.
2. In the resulting dialog box, select the firmware, for which you want to add a patch, from the dropdown list in the **Firmware** field.
3. in the **Version** field, select the version, for which you want to add a patch.
4. In the **Patch** field, enter a name for the patch.
5. Select one of the following options to define the patch binary:
   - **Upload a binary** from your file system.
   - **Provide a file path (URL)** to download the binary from.
   - **Provided** – Select this option if the device is expected to resolve the binary itself. No file or URL will be defined, and a `$PROVIDED` placeholder is stored.
6. Click **Save**.

As with adding versions, if you click **Add firmware patch** from within the details of a specific firmware, the dialog box looks slightly different as the firmware is already selected.

The firmware patch will be added to the version details within the firmware details.

TO EDIT A FIRMWARE

1. Click the menu icon ⋮ at the right of a specific firmware entry and in the context menu click **Edit**.
2. Update the name, description or device type filter by clicking the edit icon next to it. Make the desired changes and click **Save**.

The firmware will be updated accordingly.

TO DELETE A FIRMWARE

Click the menu icon ⋮ at the right of a specific firmware entry and in the context menu click **Delete**.

The object will be deleted from the firmware repository.

TO DELETE A FIRMWARE VERSION OR PATCH

In the details of a specific firmware, hover over the version or patch entry you want to delete and click the delete icon. The firmware version or patch will be deleted from the firmware details.

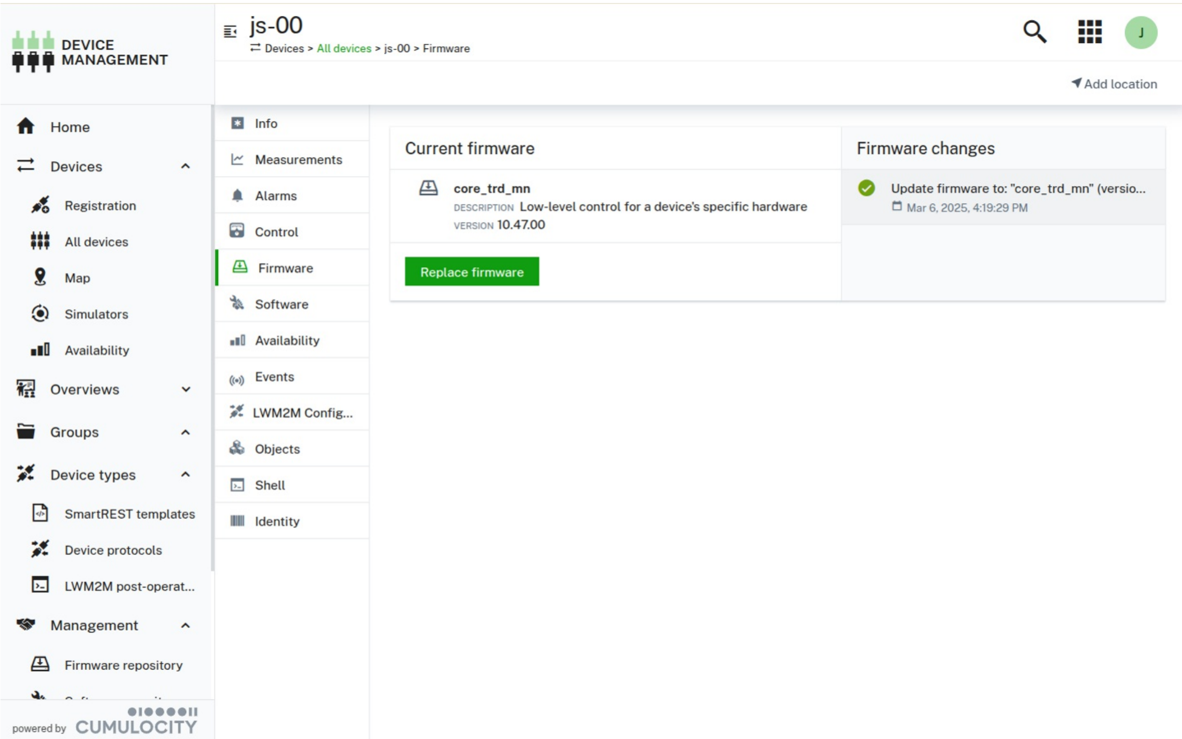
TO MANAGE FIRMWARE ON A DEVICE

In the **Firmware** tab of a device you can manage the installed firmware for the device.

> **INFO**
>
> The **Firmware** tab shows up for a device if the device supports `c8y_Firmware` operations.

Click **All devices** in the **Devices** menu in the navigator, select the desired device from the device list and open its **Firmware** tab.

The **Firmware** tab shows the current firmware installed on the device.



Additionally, it shows the operation status for the last operation (one of SUCCESSFUL, PENDING, EXECUTING, FAILED). Clicking on the operation will show you the operation details.

To install/replace firmware on a device

1. In the **Firmware** tab, click **Install firmware** (or **Replace firmware** if there is already firmware installed on the device).
2. Select a firmware and the desired version from the list, which contains all firmware available for the particular device type (or the ones that have no device type) in the firmware repository.
3. Click **Install**.

The install operation to be executed by the device will be created. The firmware installation is completed as soon as the device has executed the operation.

Click on the operation to view its details. The status of the last operation is also shown on the **Firmware** tab.

To install/update firmware on multiple devices

Cumulocity offers the option to execute firmware updates for multiple devices at once.

1. Execute the firmware operation (install or replace) on a single device to test that the new version works.
2. Navigate to the operation in the **Control** tab and in the context menu select **Schedule as bulk operation**.
3. Fill in the fields to schedule the bulk operation and click **Create**. For details on bulk operations, see Monitoring and controlling devices.

The status of the bulk operation is shown in the **Bulk operations** tab under **Device control**.

Moreover, the operation details are shown in the **Control** tab of the selected devices.

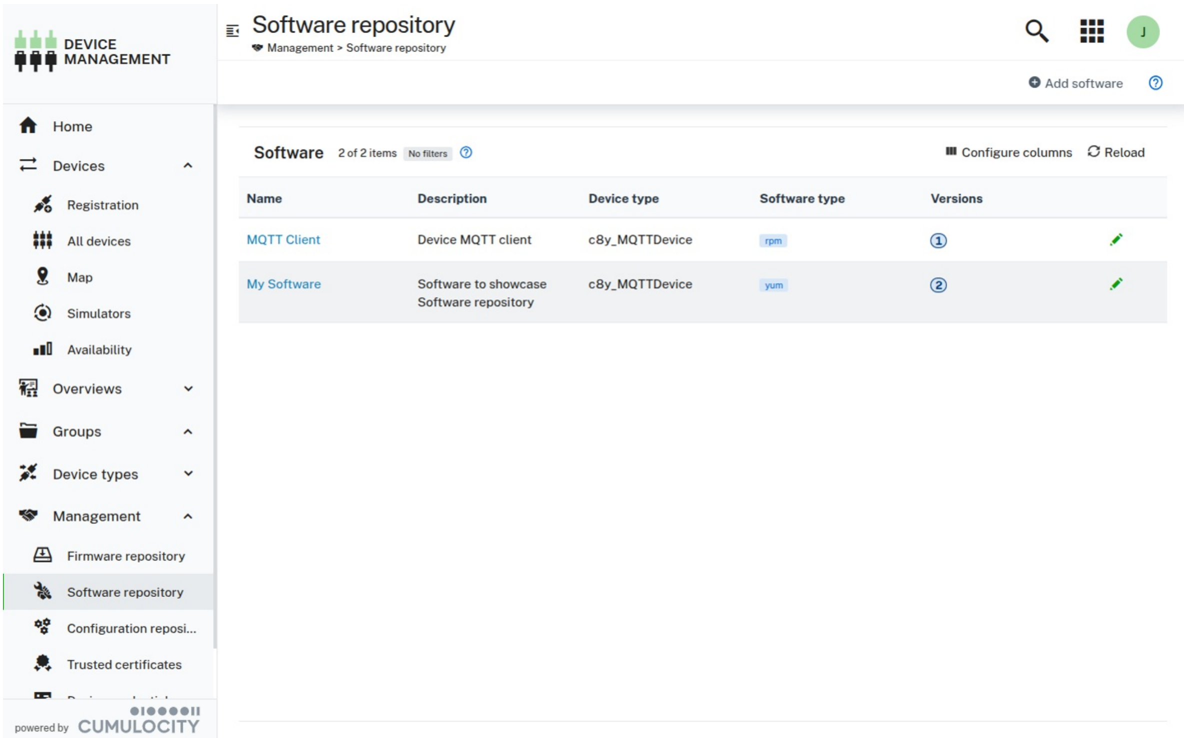
MANAGING SOFTWARE

In the software repository, Cumulocity offers to collect reference software for devices. Multiple software packages can be installed on a device.

TO VIEW SOFTWARE

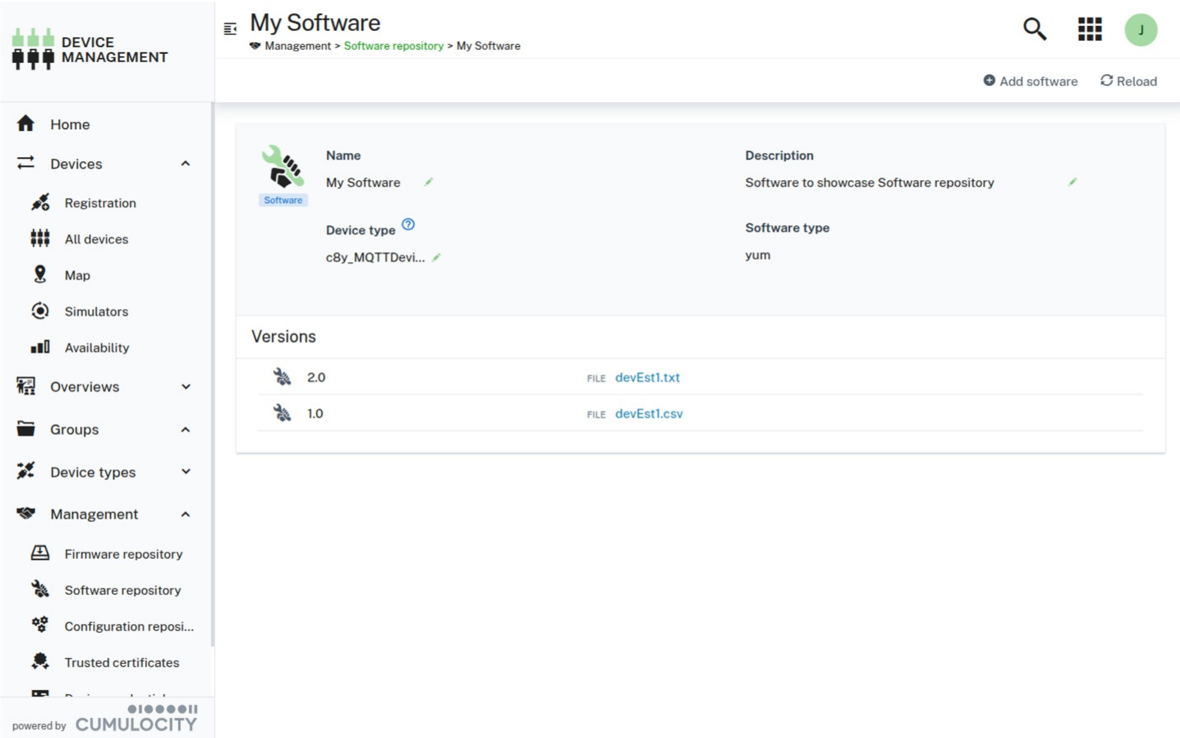
Click **Software repository** in the **Management** menu in the navigator.

The available software objects will be displayed as a list.



Each entry shows the software name, the device type it is applicable for (if set), the software type (if set), and a badge indicating if and how many versions are available for a particular software. The values in every column except for the **Versions** column can be filtered and sorted by clicking the filter and sort icons in the column header.

When clicking on an entry, the details for this software are displayed along with all available versions.



At the top, the software name, a description, an optional device type filter(s), and a software type are shown. If a device type filter is set, the software will show up for installation only for devices of that type. If no filter is set, it will be available for all devices. The software type will make the software installable only on devices that specifically support the particular software type.

INFO

The **Software type** field suggests you a list of types already used in your software repository. Before you consider defining a new software type (the field accepts new values directly) check if the type you need has already been defined before for another software by looking at the suggestions in the dropdown. This will help you keep software types consistent within your organization. If you use, for example, container images you may look for `container` or `image` , or try to search for more specific types like `docker` , `lxc` , and so on. This may prevent you from scattering your software types and using different names for effectively the same software type.

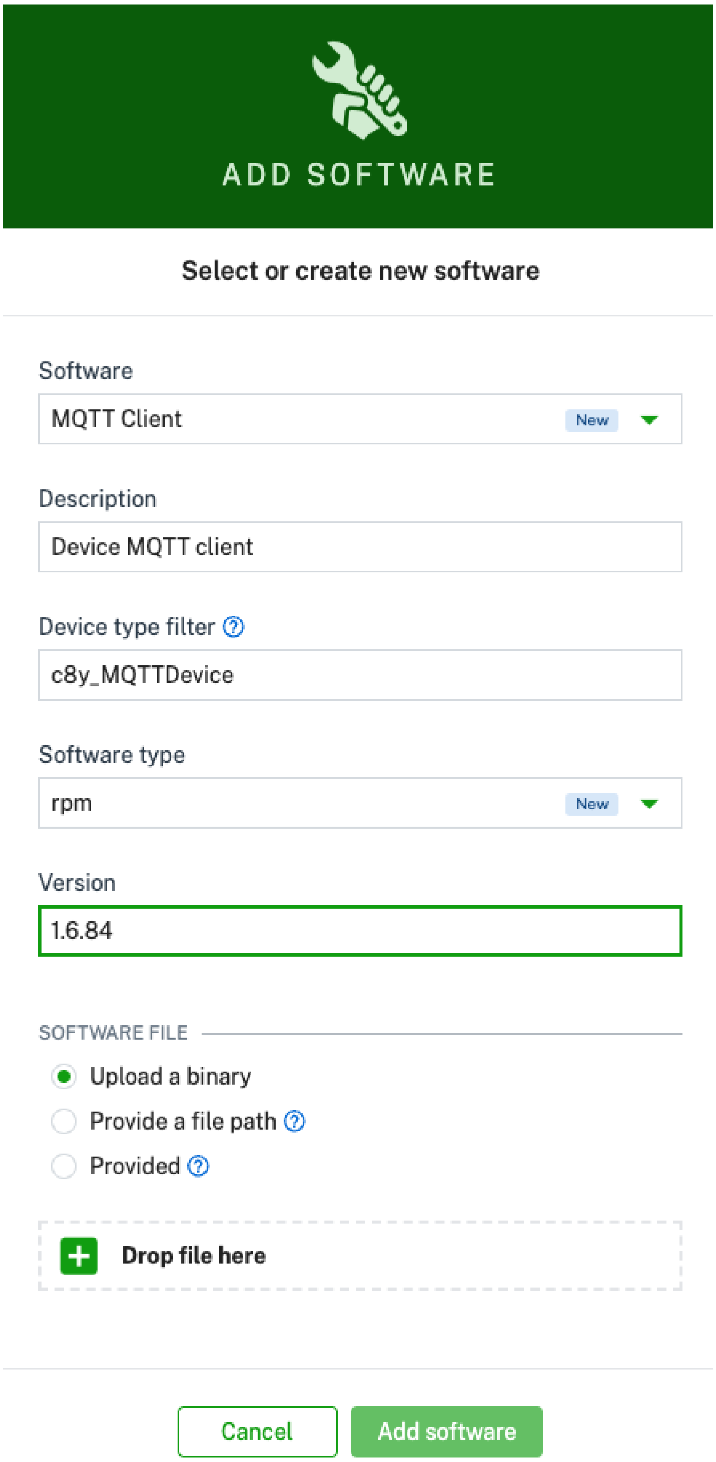
The list of versions shows the version name and the name of the software binary. The versions are ordered by their creation time (descending).

TO ADD A NEW SOFTWARE OR SOFTWARE VERSION

1. In the **Software repository** page, click **Add software** at the right of the top menu bar.
2. In the resulting dialog box,
    - to add a new software, enter a name for the software (and confirm it by clicking **Create new** in the resulting window), a description, and its version (all required).
    - to add a new version, select the software for which you want to add a new version from the dropdown list in the **Software**

field and enter a version.

3. Optionally, you can define the device type filter when adding a new software.
4. Define the software type. It will make the software installable only on devices that have declared to support the particular software type.
5. Select one of the following options to define the binary:
   - **Upload a binary** from your file system.
   - **Provide a file path (URL)** to download the binary from.
   - **Provided** – Select this option if the device is expected to resolve the binary itself. No file or URL will be defined, and a `$PROVIDED` placeholder is stored.
6. Click **Add software**.



The software object will be added to the software list or the software version will be added to the software details and the version count label will be updated accordingly.

If you click **Add software** from within the details of a specific software, the dialog box looks slightly different as the software is already selected. Enter the new version number and upload a binary or provide a file path.

TO EDIT A SOFTWARE

1. Click the edit icon at the right of a specific software item.
2. Update the name, description, device type filter or software type by hovering over the edit icon next to it. Make the desired changes and click **Save**.

The software will be updated accordingly.

TO DELETE A SOFTWARE

Click the delete icon at the right of a specific software item and in the context menu click **Delete**.

The software and all its versions will be deleted from the software repository.

TO DELETE A SOFTWARE VERSION

In the details of a specific software, hover over the version entry you want to delete and click the delete icon . The software version will be deleted from the software details.

If the last version of a software is deleted, the software is entirely removed from the repository.

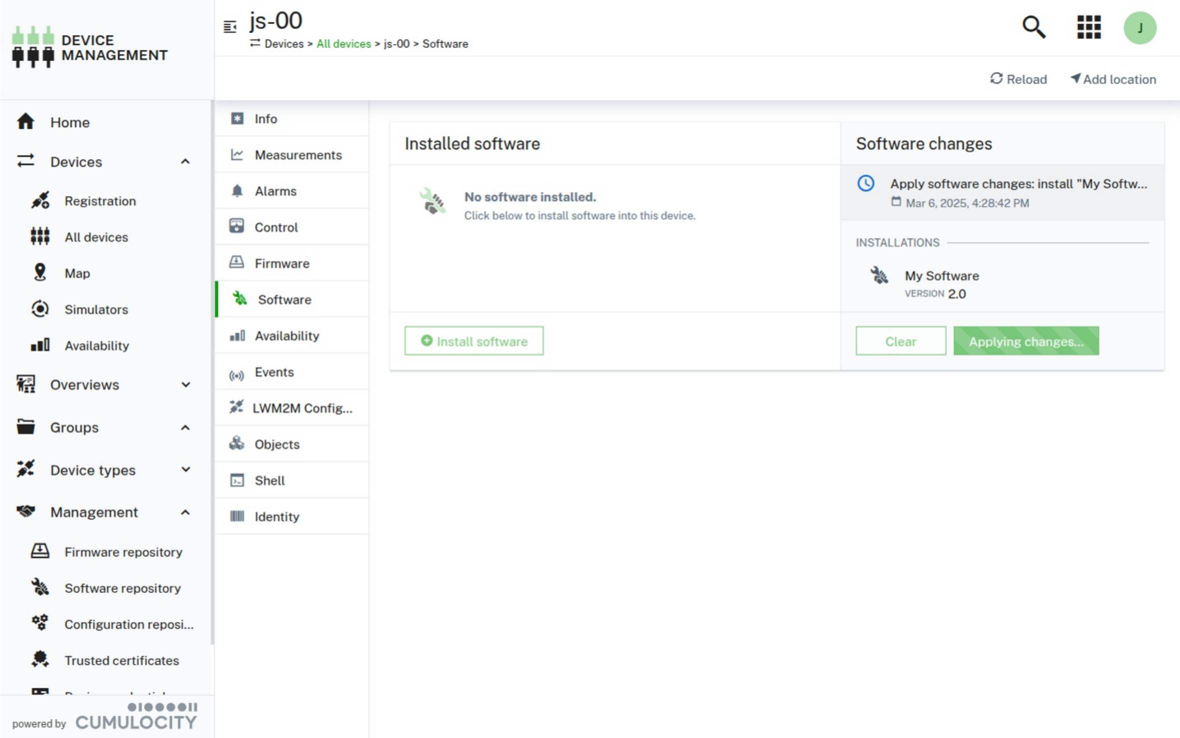
TO MANAGE SOFTWARE ON A DEVICE

In the **Software** tab of a device you can manage the software for the particular device.

> INFO
>
> The **Software** tab shows up for a device if the device supports one of the following operations: c8y_SoftwareUpdate, c8y_SoftwareList, c8y_Software.

Click **All devices** in the **Devices** menu in the navigator, select the desired device from the device list and open its **Software** tab.

The **Software** tab shows a list of all available software installed on the device. If a given software has a type, it will be displayed next to its name. You can search for a particular software by its name or filter the list by software type.



Additionally, it shows the operation status for the last operation (one of SUCCESSFUL, PENDING, EXECUTING, FAILED). Clicking on the operation will show you the operation details.

To install software on a device

1. In the **Software** tab, click **Install software**.

> **INFO**
>
> The **Install software** dialog will only display software items which match the device type or have no device type specified. Additionally, if the device has any `c8y_SupportedSoftwareTypes` declared the dialog will only display the software items matching the supported software types.

2. Select one or multiple software items by selecting the respective version from the list which contains all software items for the particular device type available in the software repository.
   For devices supporting advanced software management features, already installed software items cannot be pre-filtered from the list of available software items. Thus, after a particular software version has been selected, a check is done if the selected software is already installed on the device. If this is the case, a warning next to the selected version indicates that this software version is already present on the device.
   You can remove the already installed software item under **Software changes** or leave it and apply it as part of the changes. It is up to the device agent to decide how to handle such an update.
3. Click **Install**.
4. Under **Software changes** at the right, review your planned changes and confirm the software update operation by clicking **Apply changes**.

The install operation to be executed by the device will be created. The software installation is completed as soon as the device has executed the operation.

Click on the operation to view its details. The status of the last operation is also shown on the **Software** tab.

To update software on a device

Hover over the software item which you want to update and click **Update**. Select a version from the list and click **Update** again.

The software will be updated with the selected version.

To delete software from a device

Hover over the software item which you want to delete and click the delete icon.

To install software on multiple devices

Cumulocity offers the option to execute software updates for multiple devices at once.

1. Execute the software operation (install or update) on a single device to test that the new version works.
2. Navigate to the operation in the **Control** tab and in the context menu select **Schedule as bulk operation**.
3. Fill in the fields to schedule the bulk operation and click **Create**. For details on bulk operations, see Monitoring and controlling devices.

The status and details of the bulk operation are shown in the **Bulk operations** tab under **Device control**.

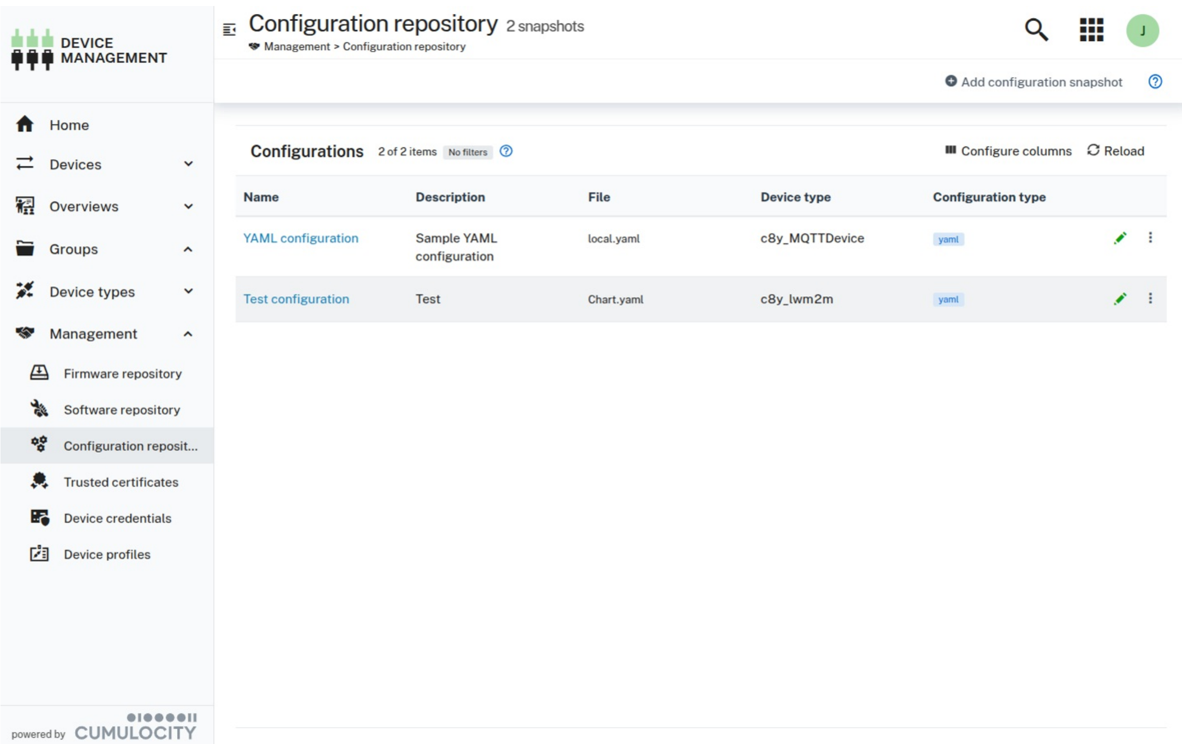
Moreover, the operation details are shown in the **Control** tab of the selected devices.

MANAGING CONFIGURATIONS

Cumulocity allows to retrieve configuration data and store and manage it in a configuration repository. The configuration data contains the parameters and the initial settings of your device.

Configuration snapshots help you, for example, to apply the same configuration to multiple devices as described below.

Click **Configuration repository** in the the **Management** menu in the navigator. In the **Configuration repository** page, all available configuration snapshots are listed. Each entry shows the configuration name, the description of the configuration, the device type, and the configuration type.



TO ADD A CONFIGURATION SNAPSHOT

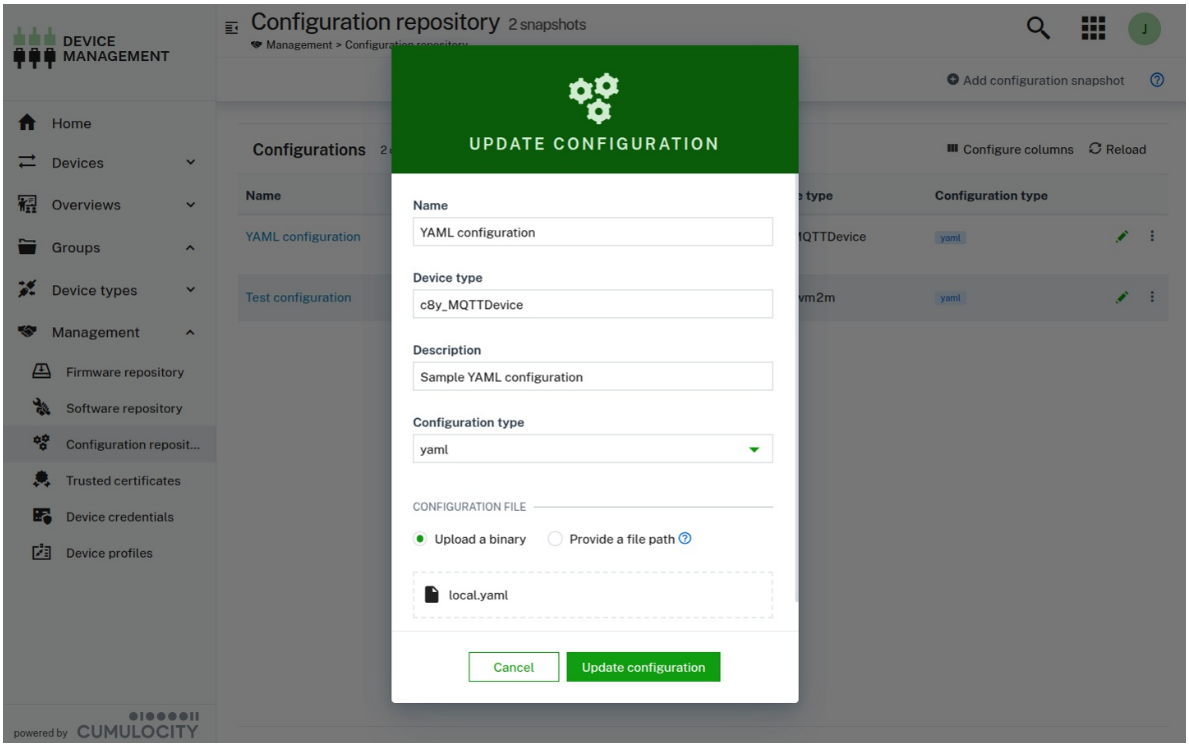
1. Click **Add configuration snapshot** at the right of the top menu bar.
2. In the resulting dialog box, enter a unique name.
3. In the **Device type** field, enter a device type. The device type can be found in the **Info** tab of the target device.
4. Optionally enter a description for the configuration.
5. Enter the configuration type, for example “ssh”.
6. Specify the configuration snapshot file by either uploading it from the file system, specifying a URL from where the configuration snapshot can be obtained or choosing a file.
7. Click **Add configuration**.

The configuration snapshot will be added to the configuration repository.

TO EDIT A CONFIGURATION SNAPSHOT

To edit a configuration snapshot, click on the menu icon ⋮ at the right of the row and then click **Edit**.

For details on the fields, see To add a configuration snapshot.



Click **Update configuration** to save your changes.

TO DELETE A CONFIGURATION SNAPSHOT

To delete a configuration snapshot, click on the menu icon ⋮ at the right of the row and then click **Delete**.

The configuration snapshot will be deleted from the configuration snapshot repository.

TO RETRIEVE AND APPLY A CONFIGURATION SNAPSHOT

Managing configurations, that is requesting a configuration from a device and sending a configuration to a device, can be done in multiple ways. Depending on user permissions and device settings, you can work with text based, typed file-based or legacy file-based configuration. Refer to Configuration for more detailed and technical information.

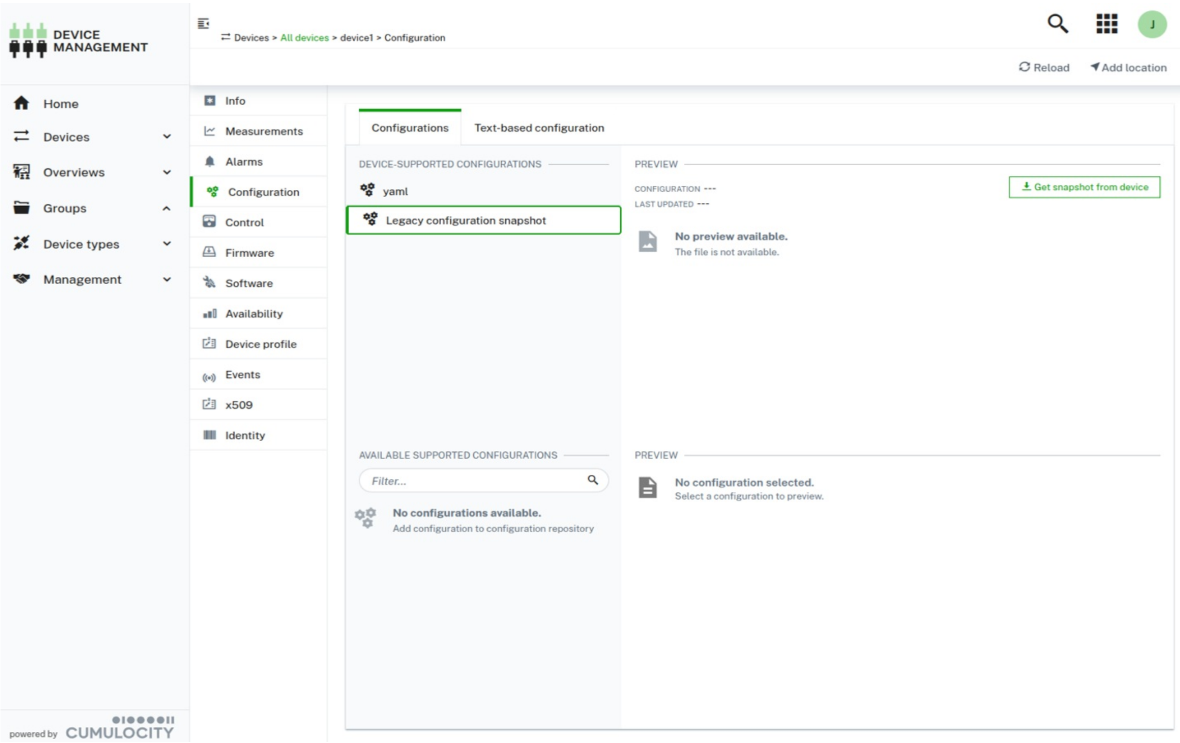
TO RETRIEVE AND APPLY A CONFIGURATION SNAPSHOT TO A DEVICE WHICH SUPPORTS TYPED FILE-BASED

CONFIGURATION

We recommend you to use typed file-based configuration. With typed file-based configuration, devices can manage multiple configurations at the same time. You can upload or retrieve different configurations for different types. Using this approach is more versatile because the configurations are handled as events rather than as files, which is more efficient.

1. Navigate to the desired device in **Devices** > **All devices** and open its **Configuration** tab.
2. Under **Device-supported configurations**, select the desired configuration type and click **Get snapshot from device** at the right.

Once retrieved, you can save or download the snapshot in the **Preview** section. The snapshot will be added to the **Configuration repository**, accessible from the **Management** menu in the navigator.

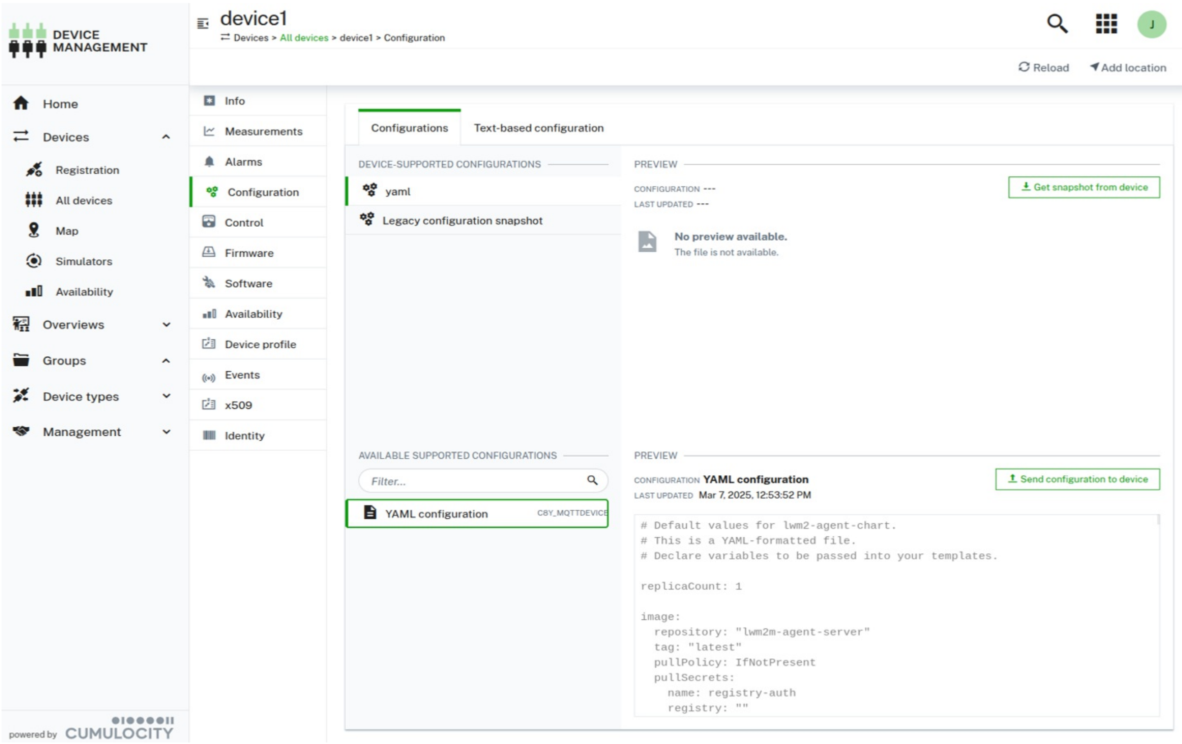


INFO

Clicking **Get snapshot from device** creates a new operation. If the operation is in status PENDING or EXECUTING, it is not possible to trigger another configuration request for the configuration type. Navigate to the **Control** tab of a device to cancel the operation or view the history of operation changes.

To apply a configuration snapshot to a device which supports multiple configuration types:

1. Navigate to the desired device and open its **Configuration** tab.
2. Under **Device-supported configurations**, select the desired configuration type.
3. Under **Available supported configurations**, select a configuration file.
4. Click **Send configuration to device** at the right to apply the selected snapshot to the device.

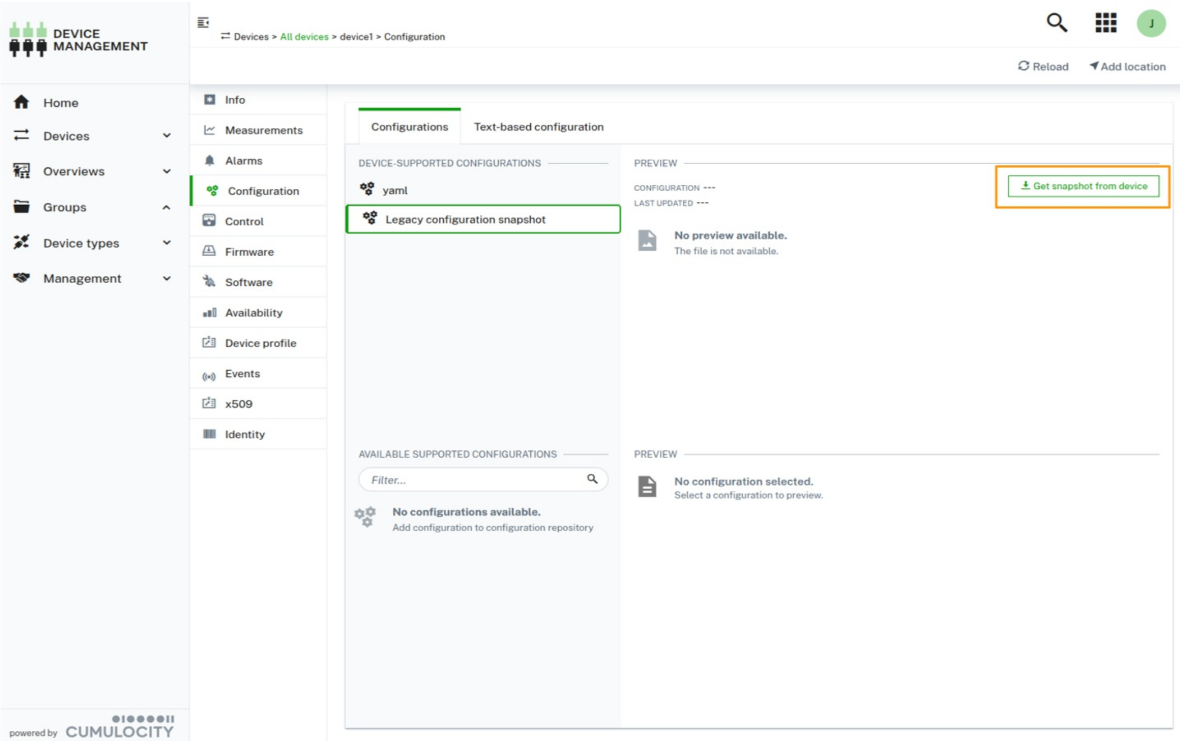


INFO

Under **Available supported configurations**, only configuration files with a matching configuration type property or without a configuration type defined are displayed. Also, configuration files are filtered based on the device type (ones that match the device type or have no device type specified).

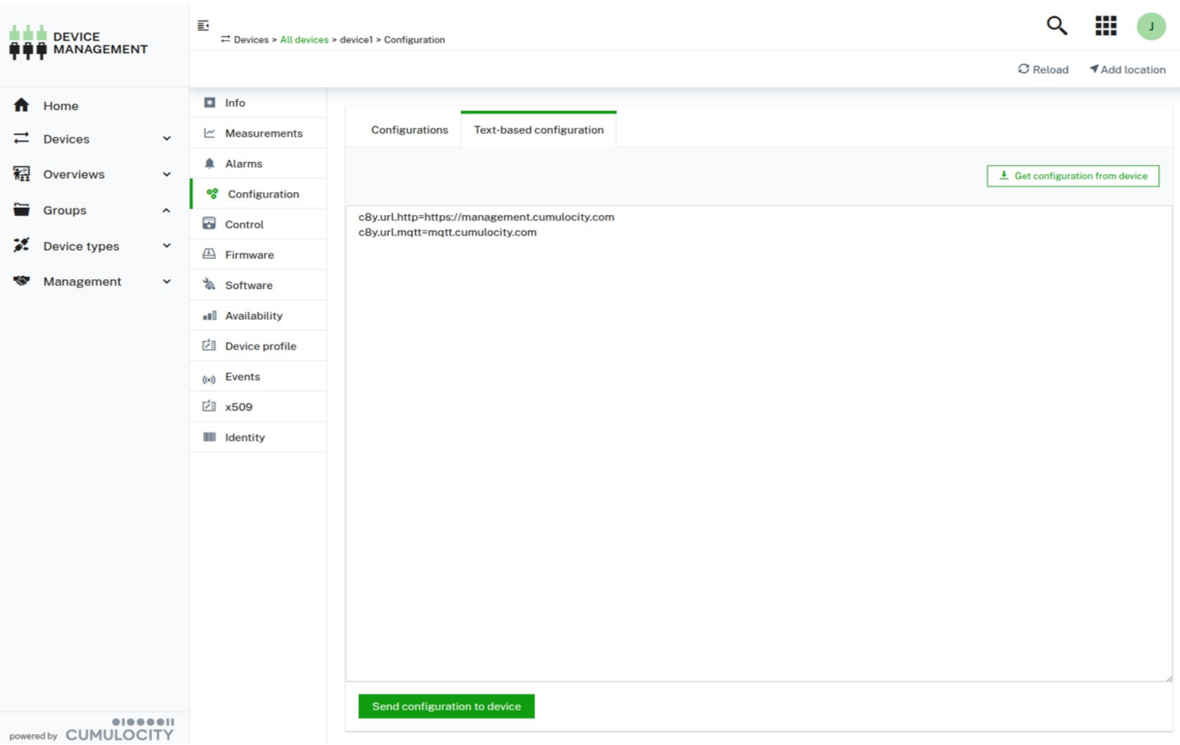
TO RETRIEVE AND APPLY A CONFIGURATION SNAPSHOT TO A DEVICE WHICH SUPPORTS LEGACY FILE-BASED CONFIGURATION

Devices managing configuration as files can do so in a basic form using legacy file-based configuration. Legacy file-based configuration only allows a single configuration to be set per a device.



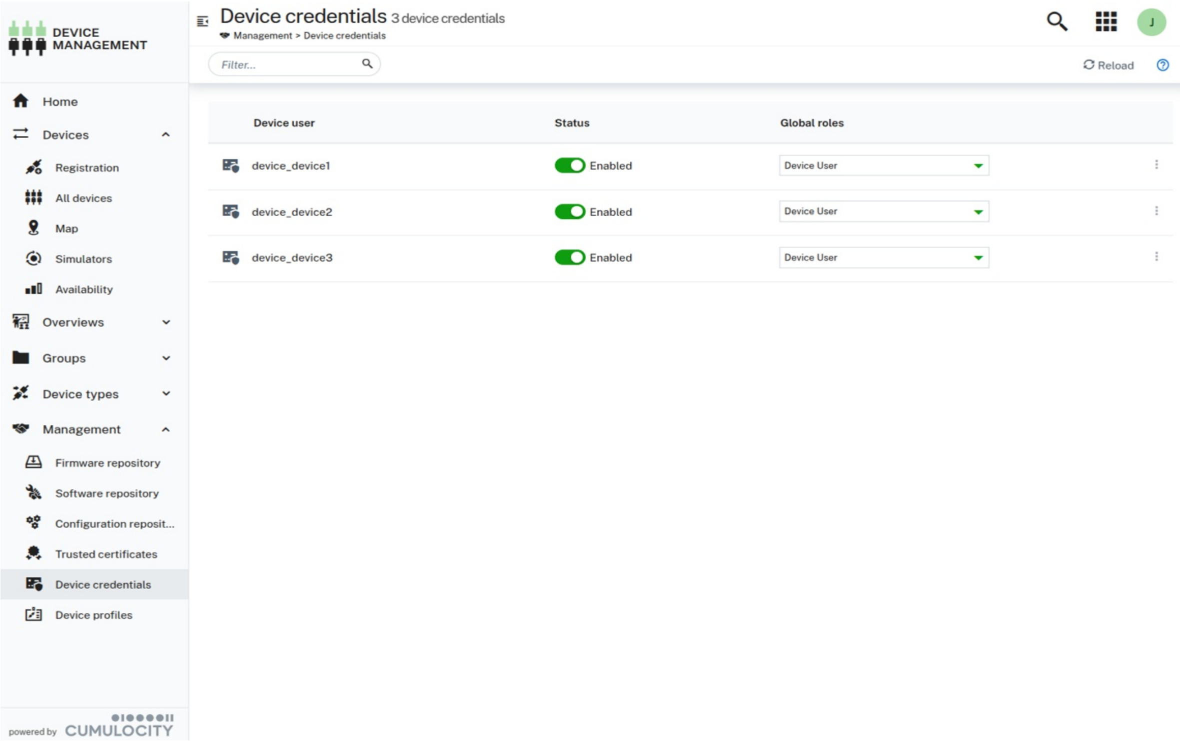
TO RETRIEVE AND APPLY A CONFIGURATION SNAPSHOT TO A DEVICE WHICH SUPPORTS TEXT-BASED CONFIGURATION

The most basic form of configuration is text-based configuration. A text command can be sent or received from a device. We recommend you to use text-based configuration for short human readable configuration files only.



MANAGING DEVICE CREDENTIALS

The **Device credentials** page lists all credentials that have been generated for your connected devices. Each device that has been registered shows up here with the naming convention “device_<id>”.



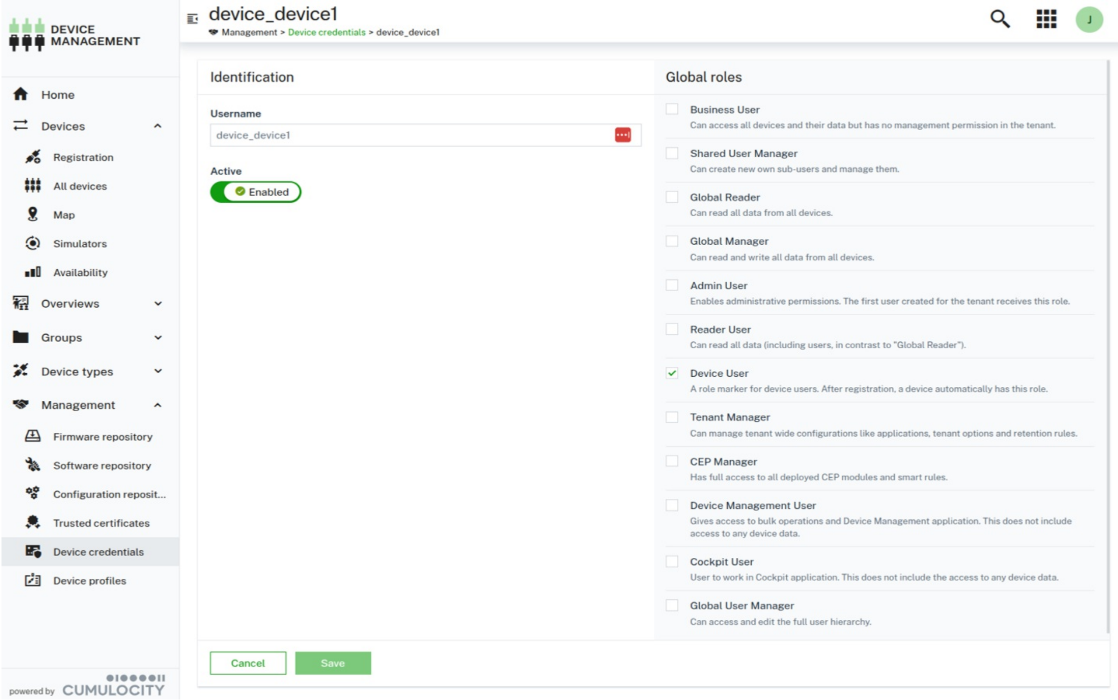
TO MANAGE PERMISSIONS FOR A DEVICE

1. Click the arrow in the **Global roles** column of a device to open a list with available global roles.
2. Assign or remove permissions for an individual device by selecting/deselecting roles.
3. Click **Apply**.

The roles for the device will be updated accordingly.

TO EDIT DEVICE DETAILS

1. Click the menu icon ⋮ at the right of a device credentials entry and then click **Edit** to open the device details.
2. In the details page, you may disable/enable a device by clicking the **Active** toggle, or assign or remove permissions by selecting/deselecting roles in the **Global roles** list.



3. Click **Save**.

The device details will be updated accordingly.

TO DISABLE DEVICE CREDENTIALS

Click the menu icon ⋮ at the right of a device credentials entry and then click **Disable**.

The device credentials will be temporarily disabled.

TO DELETE DEVICE CREDENTIALS

Click the menu icon ⋮ at the right of a device credentials entry and then click **Delete**.

The device credentials will be permanently deleted.

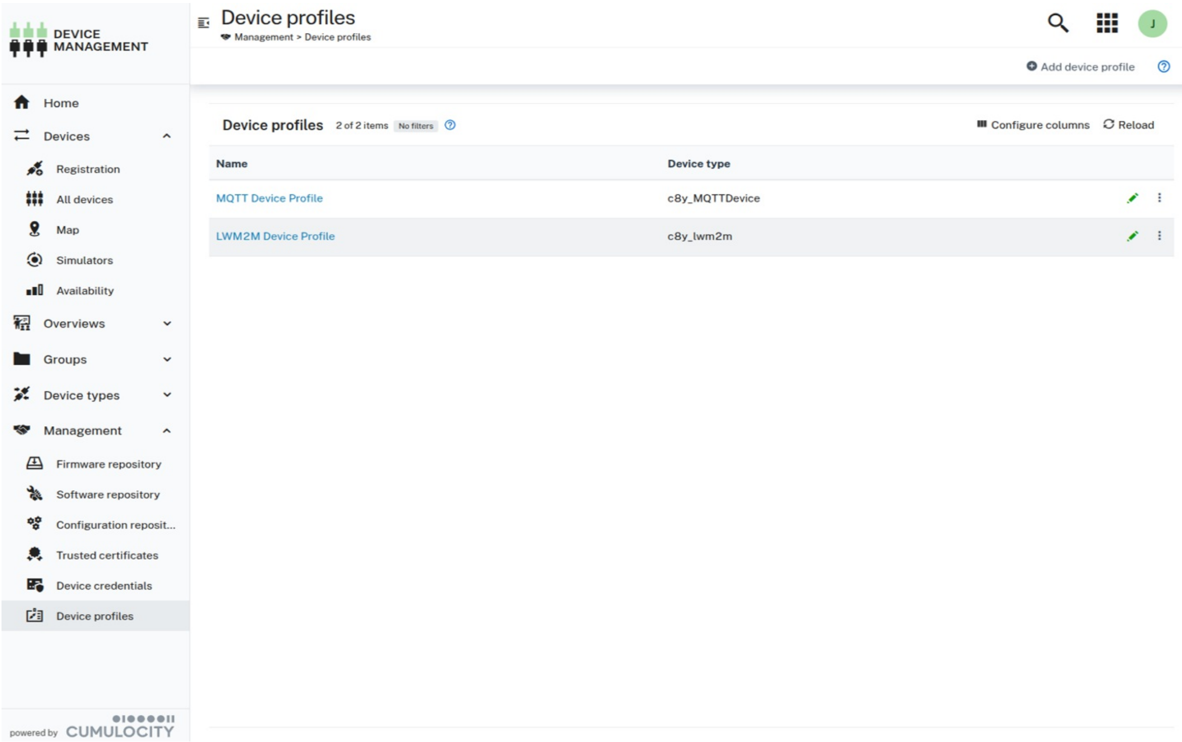
Deleting device credentials might be required if you have carried out a factory reset on a device. In this case, the device will often loose its assigned credentials. Delete the device credentials and continue with the normal registration process to re-register the device.

MANAGING DEVICE PROFILES

Device profiles represent a combination of a firmware version, one or multiple software packages, and one or multiple configuration files which can be deployed on a device. Based on device profiles, users can deploy a specific target configuration on devices by using bulk operations.

TO VIEW DEVICE PROFILES

Click **Device profiles** in the **Management** menu in the navigator to access the **Device profiles** page, which lists all available device profiles.

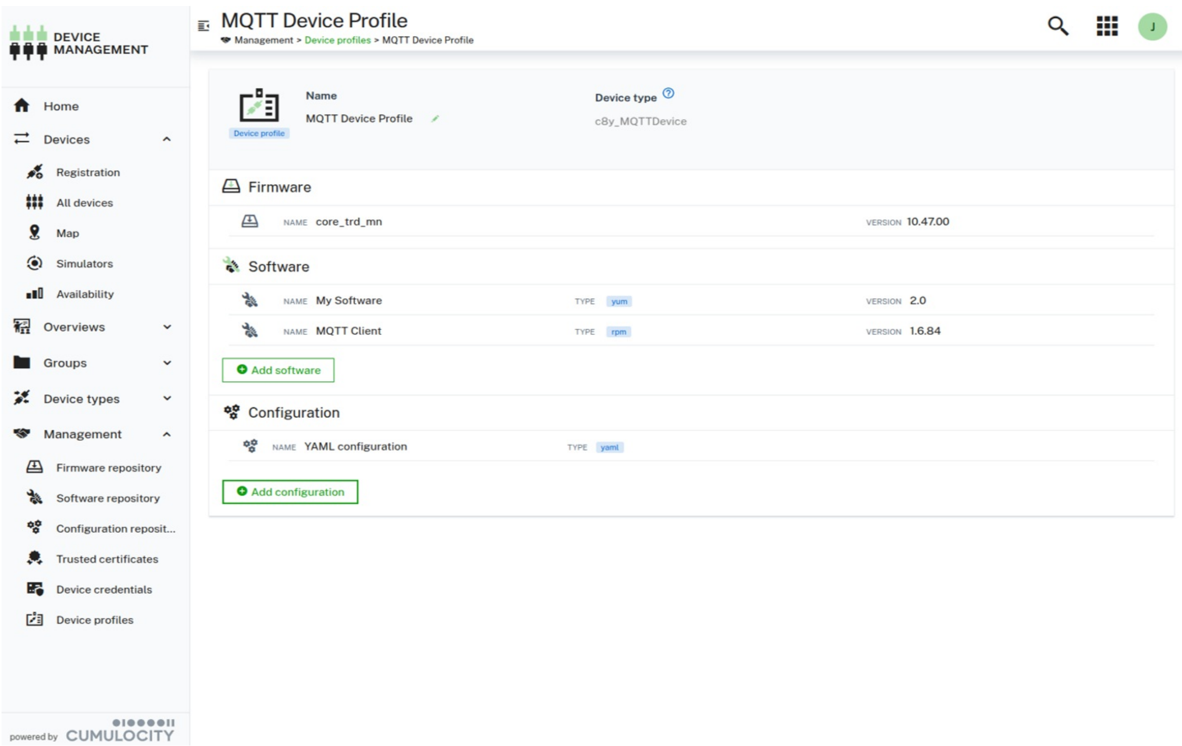


Each device profile entry shows the profile name and the selected device type(s), if any.

Click a device profile name to view its details.

The **Name and device type** section shows the name of the profile and optionally selected device types.

The sections below list the firmware version, software packages, and configuration files for this particular device profile.



TO ADD A DEVICE PROFILE

Click **Add device profile** at the right of the top menu bar, to add a new device profile.

In the **Add device profile** window, provide a name for the profile and optionally enter one or more device types. If a device type is provided, the device profile can only be assigned to devices of the specified type. If left empty, it will be available for all device types.

TO ADD ITEMS TO A DEVICE PROFILE

In the device profile details, you can add firmware versions, software packages, and configuration files.

Click **Add firmware** to add a firmware version to the profile. Select a firmware and a version from the list and click **Save** to add the selection to the profile. If a device type has been defined for the profile, only those firmware versions can be selected that match the device type. Only one firmware version can be added to a profile.

For details on firmware, see Managing firmware.

Click **Add software** to add a software package to the profile. Select a software and a software version from the list and click **Save** to add the selection to the profile. If a device type has been defined for the profile, only those software versions can be selected that match the device type. You can add multiple software packages to a profile.

For details on software, see Managing software.

Click **Add configuration** to add a configuration file to the profile. Select a configuration file from the list and click **Save** to add the selection to the profile. You can add multiple configuration files to a profile.

For details on configuration snapshots, see Managing configurations.

TO UPDATE DEVICE PROFILES

To update a device profile click the menu icon ⋮ at the right of the respective device profile entry and then click **Edit**.

You may edit the name and the device types by clicking the edit icon next to the respective fields. Make the desired changes and click **Save** to save your edits.

Moreover, you can delete firmware, software or configuration items or add new ones.

To delete an item, hover over it and click the delete icon.

See To add items to a device profile for details on how to add firmware, software or configuration items.

Note that in case of firmware, only one item is allowed in a profile at a time.

TO DUPLICATE DEVICE PROFILES

To duplicate a device profile, click the menu icon ⋮ at the right of the respective device profile entry and then click **Duplicate**.

Duplicating a profile creates another instance of the profile with the same content. Per default, the original profile name is extended with “Copy of”. You may give the profile a more appropriate name by clicking the edit icon next to the name field and editing it.

TO DELETE DEVICE PROFILES

To delete a device profile, click the menu icon ⋮ at the right of the respective device profile entry and then click **Delete**.

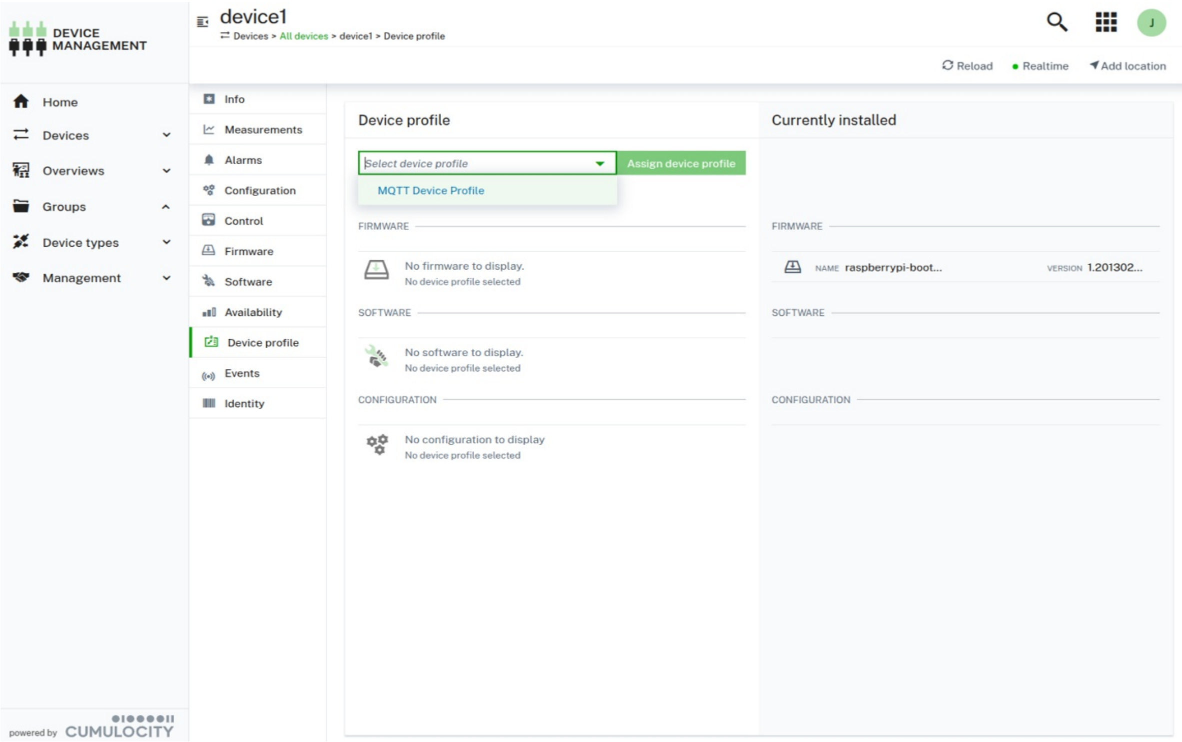
> **INFO**
>
> Deleting a profile deletes the entry from the device profile repository. It has no affect towards the devices that currently use the profile.

TO APPLY DEVICE PROFILES TO DEVICES

Device profiles can be assigned to:

- Individual devices
- Multiple devices through bulk operations

The **Device profile** tab of a particular device shows the details of the currently installed profile on a device.



INFO

The **Device profile** tab shows up for a device if the device supports `c8y_DeviceProfile` operations.

To apply device profiles to a single device

Device profiles can be applied to individual devices in the **Device Profile** tab of the particular device.

1. In the **Device profile** tab, select a device profile from the dropdown list. Only profiles that match the device type (if specified) or have no device type specified are displayed.
2. Click **Assign device profile** to start the update operation.

To apply device profiles to multiple devices

Device profiles can be applied to multiple devices by using bulk operations.

1. Click **Device control** in the **Overview** menu to navigate to the **Device control** page. In the **Device control** page, a new bulk operation can be created to apply a device profile.
2. In the **Bulk operations** tab, click **New bulk operation** at the right of the top menu bar and in the resulting dialog select **Apply device profile**.
3. Follow the steps described in To add a bulk operation to schedule a bulk operation which applies a device profile.

The devices will install the firmware, software, and configurations items of the profile and report back the status of the operation. After applying the profile, the device objects in the platform are updated accordingly with the new profile information.

INFO

When creating bulk operations, it is possible to use filters, and by this create bulk operations only for those devices where a profile has not been applied yet.

SMARTREST TEMPLATES

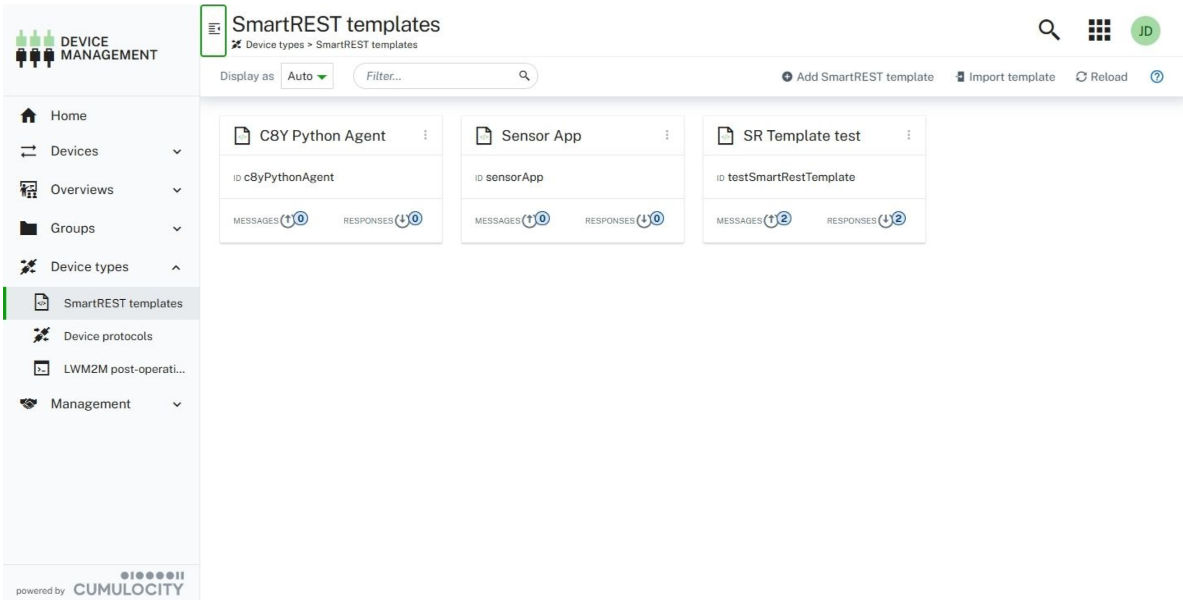
SmartREST templates are a collection of request and response templates used to convert CSV data and Cumulocity REST API calls. For example, you can use SmartREST templates to easily add devices to the platform instead of manually writing the requests each time.

To ease the device integration, Cumulocity supports static templates that can be used without the need for creating your own templates. These templates focus only on the most commonly used messages for device management. For further information on static templates, refer to the MQTT static templates.

RELATED TOPICS

- Device management & connectivity > SmartREST > SmartREST 2.0 for details on the SmartREST protocol, the data format used, as well as the anatomy and registration of SmartREST templates.
- Device management & connectivity > Device integration > MQTT for information on integrating devices via MQTT.

Open the **SmartREST template** page from the **Device Types** menu in the navigator.



For each template, the following information is provided:

- Template name
- Template ID
- Number of send messages
- Number of responses

There are two ways to add a SmartREST template:

- Import an already existing template
- Create a new template

TO IMPORT AN EXISTING SMARTREST TEMPLATE

1. Click **Import template** at the right of the top menu bar.
2. In the resulting dialog box, select a file to upload by browsing for it.
3. Enter a template name and a unique template ID (both mandatory fields).

4. Click **Import** to import the template.

TO CREATE A NEW SMARTREST TEMPLATE

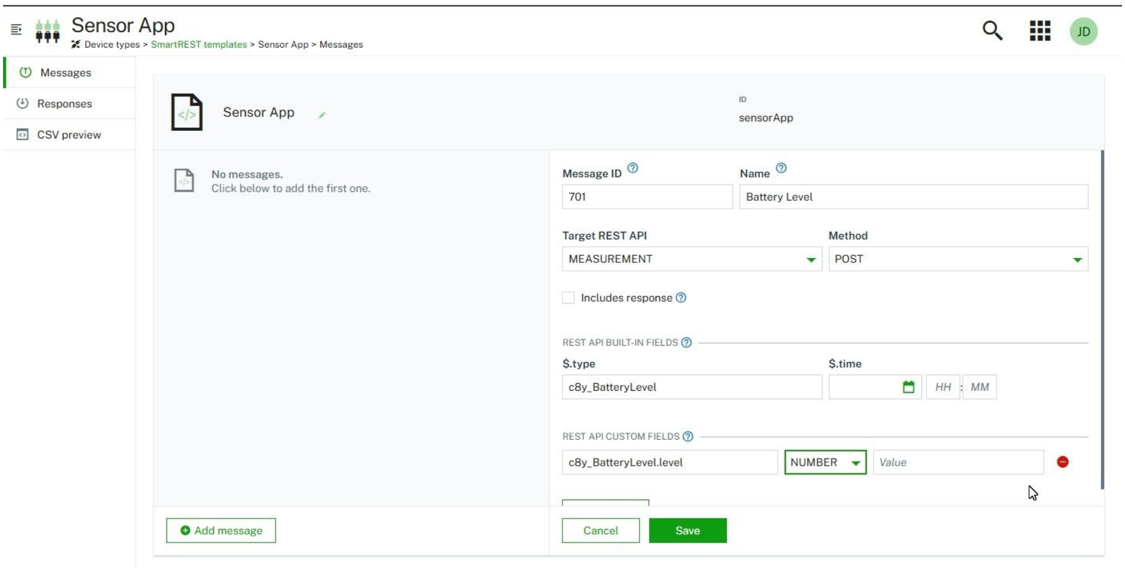
1. Click **Create template** at the right of the top menu bar.
2. In the resulting dialog box, enter a template name and a unique template ID (both mandatory fields).
3. Click **Continue** to proceed adding messages or responses.

To add a message

The message template contains all necessary information to convert the SmartREST request into a corresponding REST API call which is then sent to the platform.

1. To add a new message, navigate to the **Messages** tab in your desired SmartREST template and click **Add message**.
2. Complete the following fields:

Field	Description
Message ID	Unique integer that will be used as a message identifier. It must be unique among all message and response templates.
Name	Name for the message. Mandatory.
Target REST API	REST API for the target. Dropdown list. May be one of Measurement, Inventory, Alarm, Event, Operation.
Method	Request method. May be one of POST, PUT, GET, depending on the selected Target REST API.
Include Responses	Select this checkbox if you want to process the results of the request with response templates.
REST API built-in fields	These fields are optional and vary depending on the target REST API selected. In case no value is provided, a device will be able to set it when sending an actual message.
REST API custom fields	Additional fields can be added by clicking **Add field**. Enter the API key and select the desired data type.



Under **Preview** you can see a preview of your request message.

3. Click **Save**.

The message will be added to the SmartREST template.

To remove a message

To remove a message, open it and click **Remove** at the bottom.

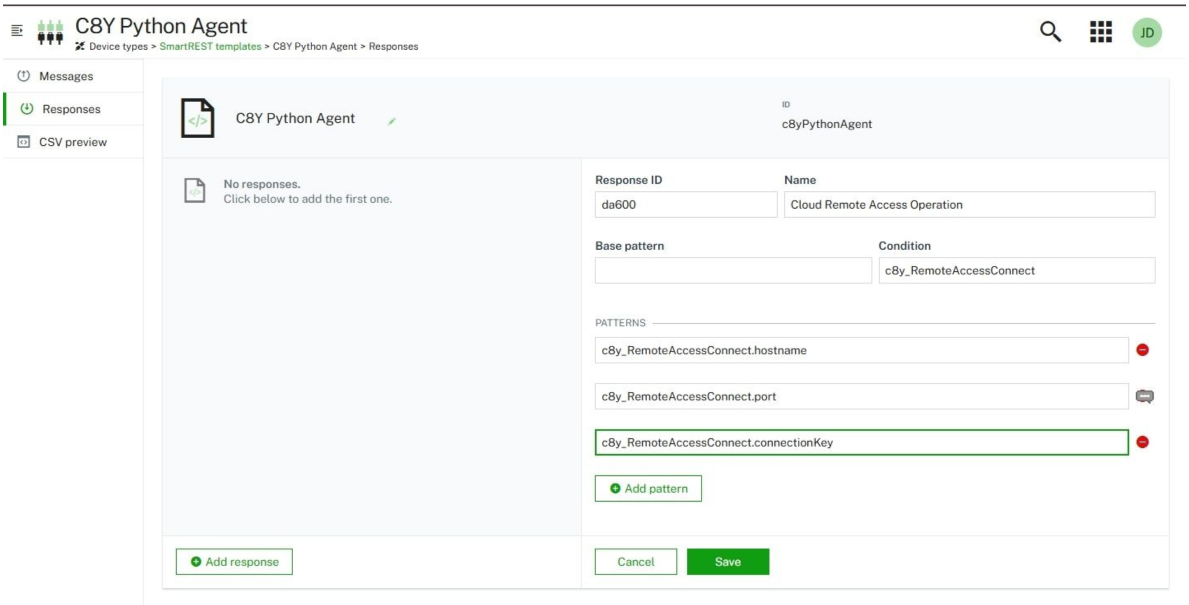
The message will be removed from the SmartREST template.

To add a response

A response template contains the necessary information to extract data values from a platform REST API call response, which is then sent back to the client in a CSV data format.

1. To add a new response, navigate to the **Response** tab in your desired SmartREST template and click **Add response**.
2. Complete the following fields:

Field	Description
Response ID	A unique string that will be used as a response identifier.
Base Pattern	Path in a JSON document. The base pattern acts as a prefix to all patterns. You can enter either a base pattern here and add patterns with only the subpath below the base pattern, or leave this field empty and provide patterns with the full path.
Condition	Condition value of the response.
Pattern	At least one pattern is required. Click **Add pattern** and enter a pattern value.



3. Click **Save**.

The response will be added to the SmartREST template.

For further information see Response templates.

To remove a response

To remove a response, open it and click **Remove** at the bottom.

TO EDIT A SMARTREST TEMPLATE

Either click the desired template or click the menu icon ⋮ at the top right of the respective template card and then click **Edit**.

After editing the template, click **Save** to save your settings.

TO DELETE A SMARTREST TEMPLATE

Click the menu icon ⋮ at the top right of the respective template card and then click **Remove**.

TO EXPORT A SMARTREST TEMPLATE

Click the menu icon ⋮ at the top right of the respective template card and then click **Export**.

The template will automatically be downloaded to your file system.

To export a SmartREST template as CSV file follow these steps:

1. Open the template you want to export and select the **CSV preview** tab.
2. In the resulting dialog box, specify the preferred options for the field separator, decimal separator and character set.
3. In the **CSV preview** tab, which provides additional information on messages and responses, click **Copy to clipboard**.

The SmartREST template will be exported as CSV file.

WORKING WITH SIMULATORS

With the Cumulocity simulator all aspects of IoT devices can be simulated, such as:

- Setting up a simulated device or a network of simulated devices.
- Specifying the operations which a device can process.
- Creating work instructions based on predefined message templates or user-defined templates and scheduling work steps.
- Creating up to ten devices of a defined type.
- Generating messages for measurements, alarms, events and inventory.
- Viewing simulation problems as alarms.

ABOUT SIMULATORS

With the simulator you can create devices that simulate the same level of functionality as connected hardware devices.

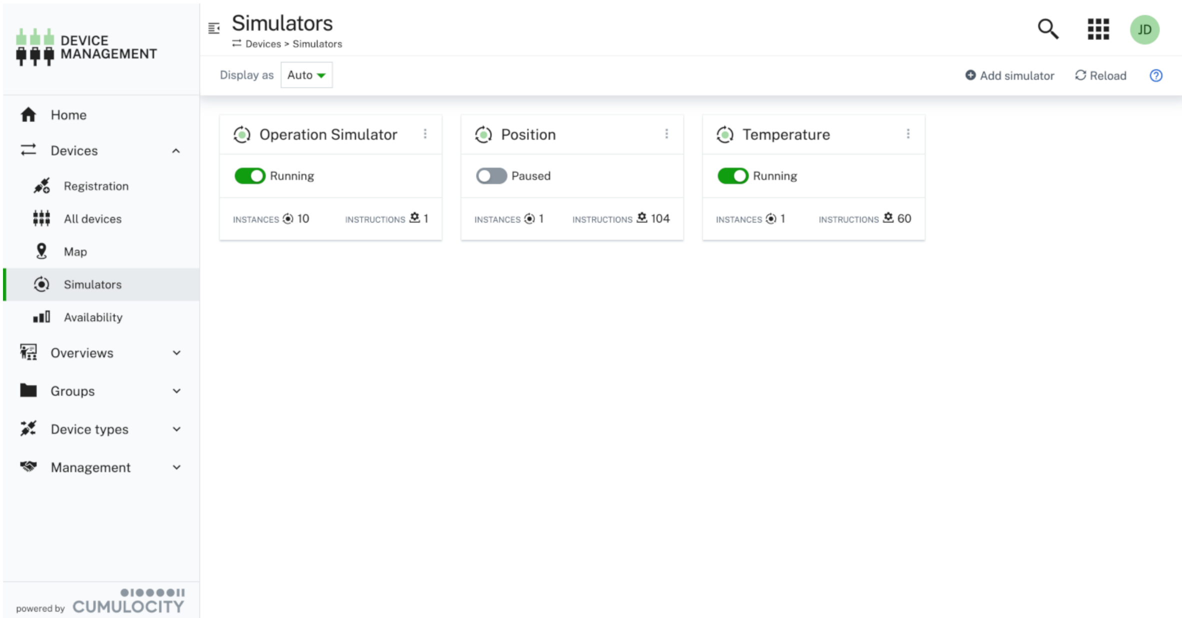
A simulator uses a playlist to simulate messages that the device sends to the Cumulocity platform. A playlist is a series of instructions that the simulator executes one after the other. When the last instruction is reached, the simulator starts again with the first one.

An instruction can either send a message (measurements, alarms, events and inventory) or wait for a specified time (sleep).

A message is defined by choosing a message template (like sending a temperature) and providing the values for this template (for example 23.0 degrees). Many predefined message templates are provided, for example, for creating a measurement, sending an event, creating and cancelling an alarm. These templates are based on MQTT static templates. Additionally, custom message templates can be defined using the SmartREST template editor.

TO VIEW SIMULATORS

Click **Simulators** in the **Devices** menu in the navigator to open the **Simulators** page.



All simulators which you can access will be listed here.

TO CREATE A SIMULATOR

1. Click **Add simulator** at the right of the top menu bar.
2. In the resulting dialog box, select a simulator type from the dropdown list in the **Presets** field. Select “Empty simulator” to create a simulator from scratch or select one of the sample simulators.
3. Enter a name for the simulator.
4. Select the number of instances for this simulator (up to ten).
5. Click **Create**.

The simulator will be created and added to the list.

TO EDIT A SIMULATOR

1. Click the menu icon ⋮ at the top right of a simulator card and then click **Edit**, or simply click the simulator card.
2. In the resulting dialog box, make the relevant changes.
3. Click **Save** to apply your changes.

TO DUPLICATE A SIMULATOR

1. Click the menu icon ⋮ at the top right of a simulator card and then click **Duplicate**.
2. In the resulting dialog box, provide a name for the new simulator.
3. Click **Duplicate**.

The new simulator will be added to the list.

TO REMOVE A SIMULATOR

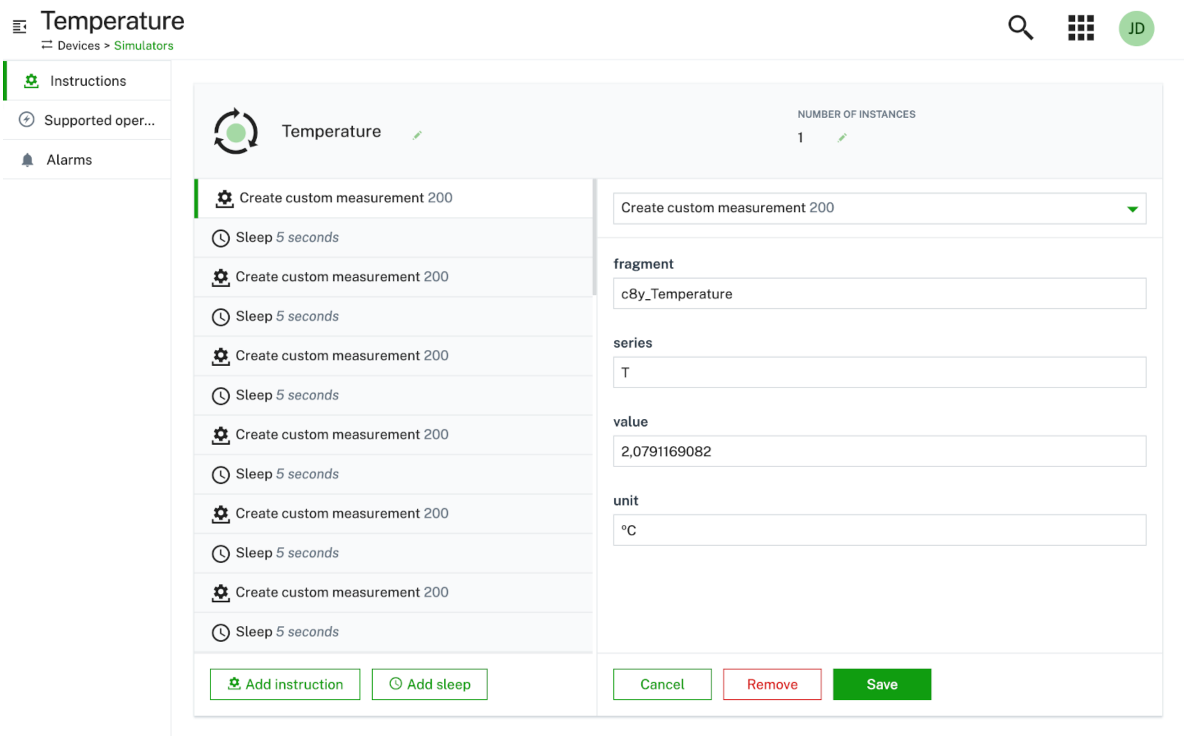
1. Click the menu icon ⋮ at the top right of a simulator card and then click **Remove**.
2. In the resulting dialog box, confirm to remove the simulator.
3. Click **Save**.

The simulator will be removed from the list.

INSTRUCTIONS

For each simulator, you can create instructions specifying what the simulator is supposed to do. Instructions are single tasks added to a playlist through which the simulator will work.

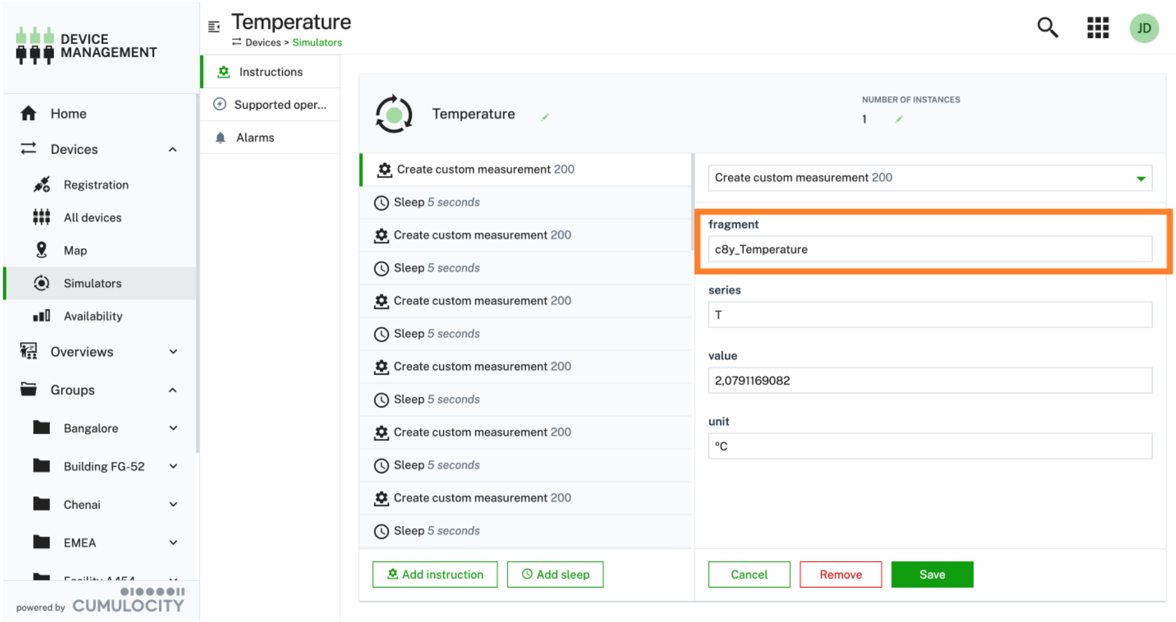
Instructions can be viewed and edited on the **Instructions** tab of the simulator.



Examples

The simulator presets already contain sample instructions. For example, the “Temperature measurement” preset contains instructions for the steps “Create measurement” and “Sleep”, see image above.

The panel at the right changes according to the type of instruction selected at the left.



The measurement instruction refers to a fragment. Fragments are used to identify capabilities of a managed object. Find more details about fragments in the fragment library.

The “Sleep” instruction requires one value for its duration in seconds.

To add an instruction

1. Click **Add instruction** to add a new instruction to the simulator.
2. In the resulting dialog box, select a message from the dropdown list.
3. Specify the required parameters, depending on the message type.
4. Click **Save**.

The new instruction will be added to the simulator.

To add a sleep

1. Click **Add sleep** to add a new instruction to the simulator.
2. In the resulting dialog box, specify the duration.
3. Click **Save**.

The new sleep instruction will be added to the simulator.

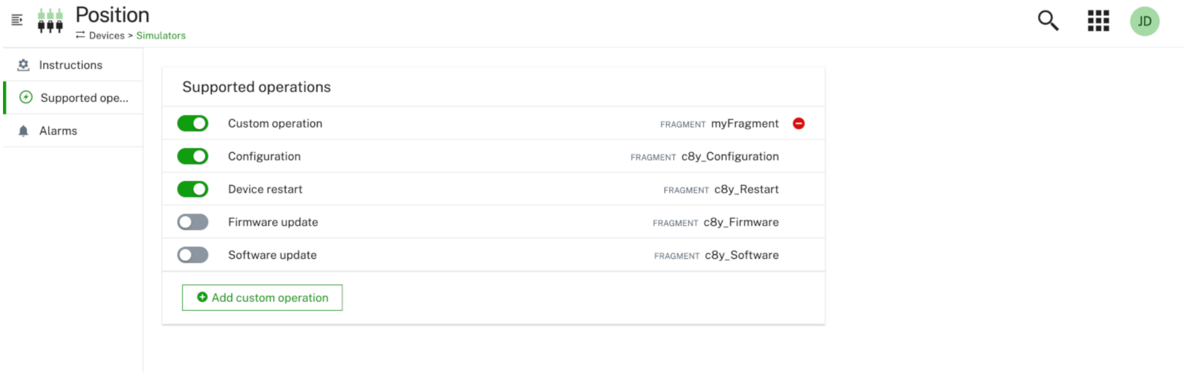
To remove an instruction

Hover over the instruction or the sleep you want to remove and click the remove icon .

The instruction will be removed from the simulator.

SUPPORTED OPERATIONS

In the **Supported operations** tab of a simulator you can find specific operations like configurations or software/firmware updates.



Click the toggle to turn the respective operation on or off.

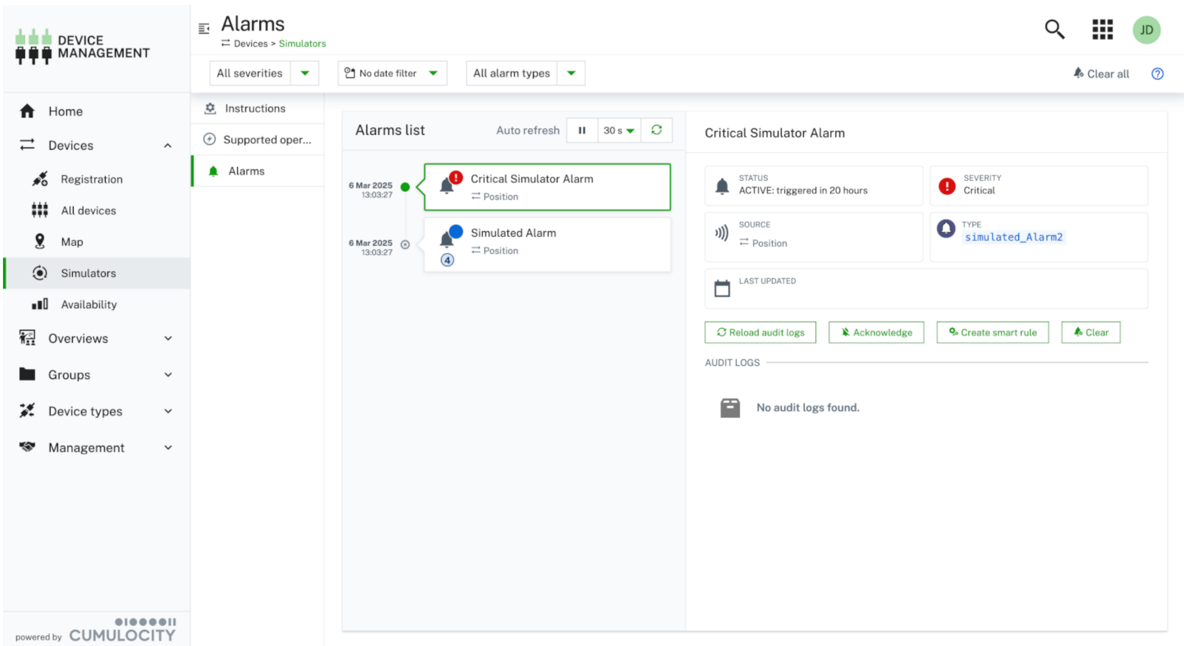
To add a custom operation

1. Click **Add custom operation** to specify a customized operation.
2. In the resulting dialog box, Enter the custom operation type to be supported by the simulator.
3. Click **Add**.

The custom operation will be added to the operation list.

ALARMS (SIMULATOR)

The **Alarm** tab of a simulator displays alarms related to the simulator itself, not related to the simulated device, that is, if the simulator itself does not work correctly. See Working with alarms for information on alarms.



CONNECTIVITY

RELATED TOPICS

- Device management & connectivity > Device integration > Fragment library > Connectivity for details on the `c8y_Mobile` fragment used in managed objects.
- Platform administration > Standard tenant administration > Changing settings > Connectivity for information on how to change the connectivity settings via the UI.

The Connectivity agent, which works from within the Cumulocity Device Management application, provides basic information on mobile devices and additional connectivity details.

The Cumulocity platform integrates with the SIM connectivity platforms Comarch, Ericsson, Jasper and Kite.

The following features are supported by these providers:

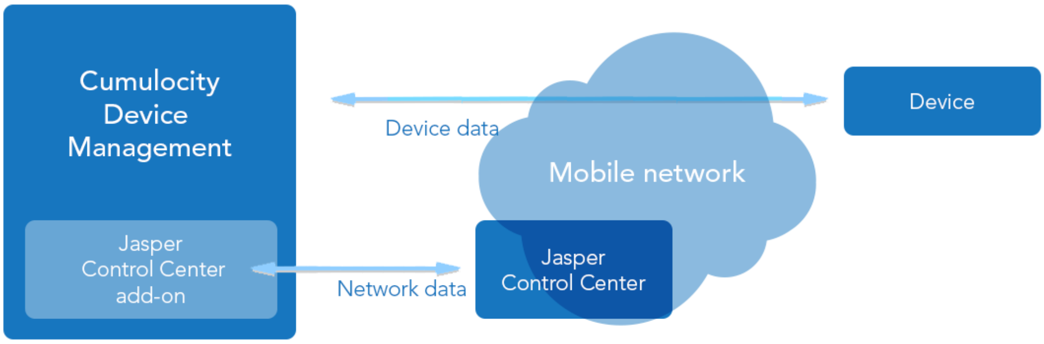
Feature	Comarch	Ericsson	Jasper	Kite
Check the status of the SIM card in the device	x	x	x	x
Check the online status of the device as reported by the network	x	x	x	x
Change SIM card status, for example activate or deactivate it	x	x	x	x
Disconnect SIM card from current session	x			
Communicate with the device through text messages, for example, to set APN parameters	x		x	x
View summary usage of data traffic, text messages and voice calls	x	x	x	x
View usage details of data traffic, text messages and voice calls		x	x	
View the history of data sessions and any changes to the SIM card or traffic			x	

As you can see, Jasper currently is the most feature-rich provider.

Each provider requires either ICCID or MSISDN to be set in the c8y_Mobile fragment of the managed object. This is used to properly map the managed object in Cumulocity to the associated SIM on the respective provider's platform.

Requires	Comarch	Ericsson	Jasper	Kite
ICCID	x		x	x
MSISDN		x		

The following description is primarily based on Jasper, but the same configuration and usage also applies to the other providers. If there are any differences, they will be stated explicitly.



The following sections describe:

- How to set up your Jasper Control Center account (examplarily).
- How to configure the connectivity for the SIM provider in your Cumulocity tenant.
- How to link SIMs and mobile devices.
- Which information is shown in the Connectivity tab.
- How to manage connectivity from the Device Management application.

SETTING UP YOUR JASPER CONTROL CENTER ACCOUNT

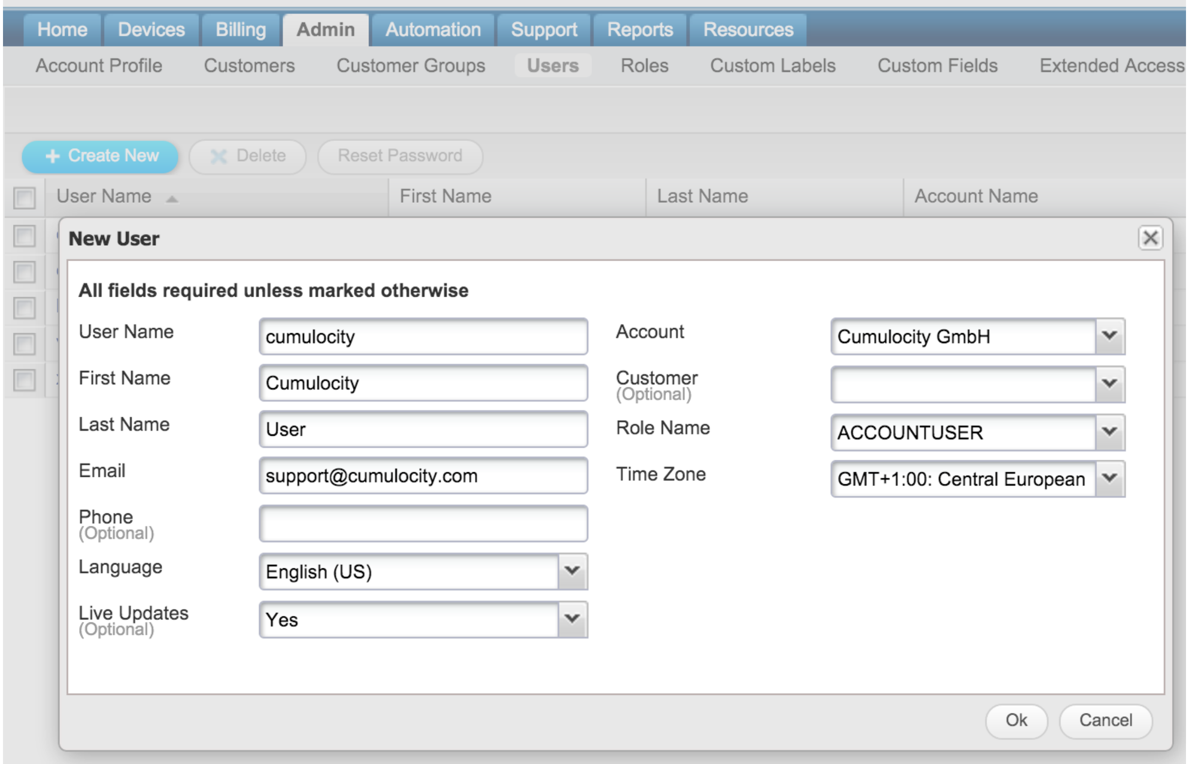
The following steps describe how to create a dedicated user in the Jasper Control Center. This user is used for all access from Cumulocity to Jasper Control Center, so the permissions of the user have influence on functionalities available in Cumulocity.

> **INFO**
>
> In a similar way, we recommend you to set up a dedicated user for Comarch, Ericsson or Kite to get the credentials required to connect to Cumulocity. Ask your administrator or our product support for further information.

Besides the user, you also need a so-called API license key (only required for Jasper) and API server URL. To determine your API license key and API server URL, use a Control Center administrator user to log in to your Control Center account and click **API integration** on the Control Center home page. Your API license key and the API server URL are displayed on the top left.

To create a user in Jasper Control Center perform the following steps:

1. As an admin user, navigate to **Admin** and **Users**.
2. Click **Create New**.
3. Enter the username and further details of the user.
4. If you want to be able to activate and deactivate SIM cards from Cumulocity, or to send SMS from Cumulocity, use the role ACCOUNTUSER. Otherwise, use the role ACCOUNTREADONLY.
5. Click **OK** to create the user, then enter your admin password and click **OK** again.



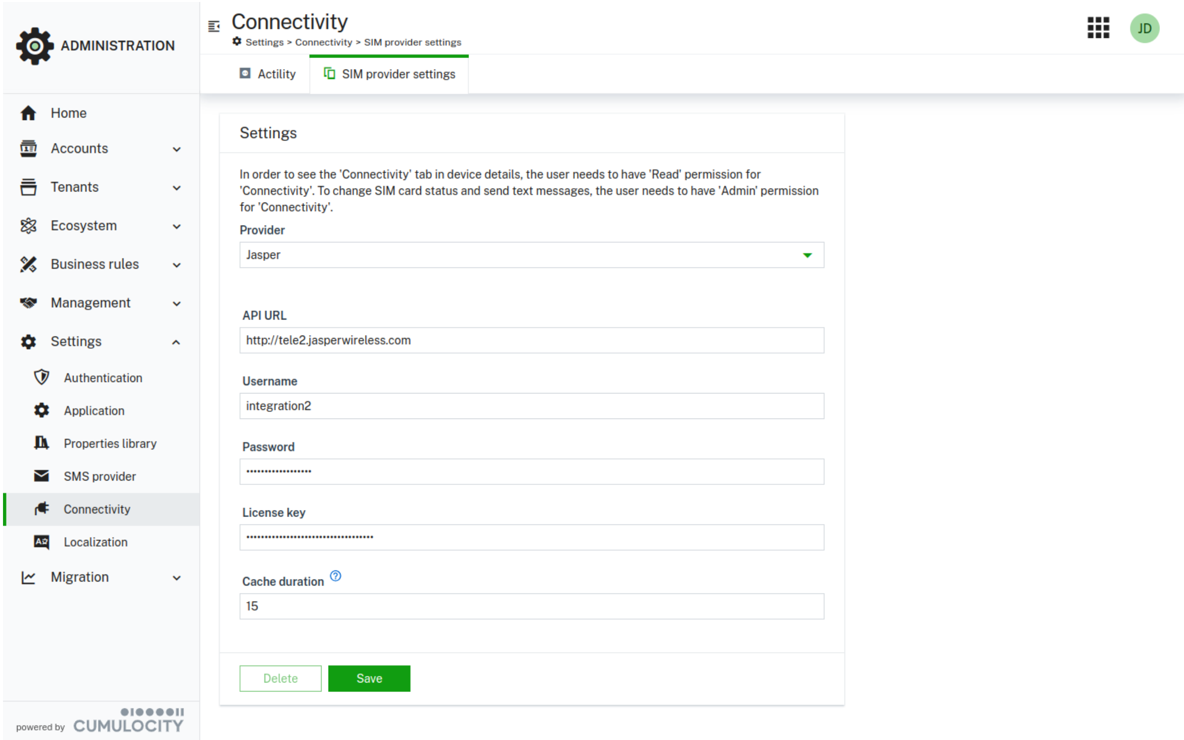
The user is now created but does not have a password yet. Follow the instructions emailed to you by Control Center to set a password.

CONFIGURING THE CONNECTIVITY FOR THE SIM PROVIDER

Process the following step to configure the connectivity in Cumulocity:

1. Use a Cumulocity administrator user to log into the Cumulocity platform.
2. Switch to the Administration application.
3. Click **Connectivity** in the **Settings** menu of the navigator. If the menu item is not displayed, make sure that your user has READ and ADMIN permissions for Connectivity. If the menu item is still not available, please contact product support to make the Connectivity agent available in your tenant.
4. Switch to the **SIM provider settings** tab.
5. Select a provider from the drop-down list.
6. Enter the credentials for the respective SIM provider account. If you do not have any credentials, ask your administrator.
7. Set a **Cache duration** in seconds to determine how long information from the provider is cached. This will prevent timeout issues.
8. Click **Save** to save your settings.

Depending on the connectivity provider you choose you must provide parameters specific to the provider. The parameter **Cache duration** defines how long SIM data returned by the provider is cached before fresh data is requested again, in seconds. This reduces traffic and the number of requests to SIM providers for billing and reliability purposes. A longer cache duration means less traffic to your SIM provider while a shorter duration means that more recent data is displayed.

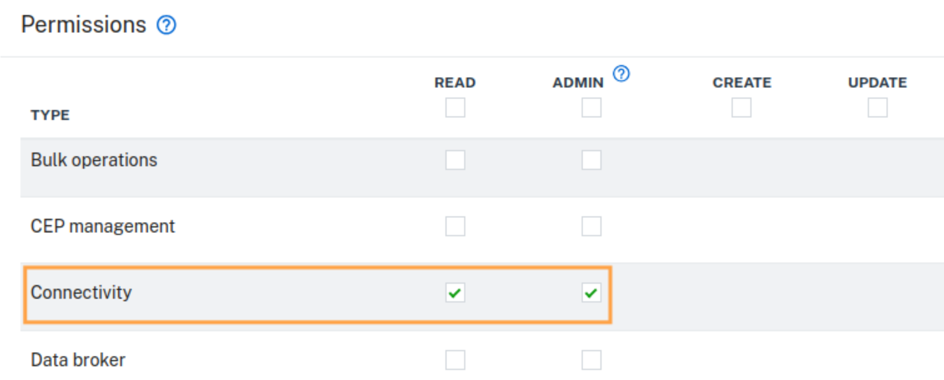


LINKING SIMS AND MOBILE DEVICES

Switch to the Device Management application and navigate to a device that is connected through a SIM card managed by the SIM provider of your choice. The device should have a **Connectivity** tab. If this tab is not shown, one of the following applies:

- Your user does not have permissions for Connectivity.
- The device is not linked to a SIM card.
- The device is linked to a SIM card, but the card is not managed by the respective SIM provider account.

To assign permissions, navigate to the Administration application and make sure that your user has a role assigned with READ or ADMIN permission for Connectivity.



Jasper and Comarch identify SIM cards through their ICCID (Integrated Circuit Card Identifier). Ericsson is using MSISDN (Mobile Station International Subscriber Directory Number) instead. In most cases, devices will report the ICCID and MSISDN of their SIM card automatically to Cumulocity.

If the ICCID is not shown automatically check the following:

- Determine the ICCID of the SIM card. It is printed on the SIM card and is visible in Control Center.
- Enter the ICCID in the **Info** tab, then click **Save**.
- Click **Reload** in the top menu bar to make the **Connectivity** tab appear.

INFO

Note that it may take a few seconds until the tab appears for the first time on a device, as Cumulocity checks if the particular SIM card is managed by the SIM provider.

The Kite provider requires the following device configuration: ICCID (Integrated Circuit Card Identifier) and MSISDN (Mobile Station International Subscriber Directory Number).

CONNECTIVITY TAB

In the **Connectivity** tab you will find the following sections:

- Status
- SMS
- Sessions
- Audit logs

INFO

Some sections may not appear or may be empty. For example, if there have been no SMS sent and you do not have permission to send SMS, you will not see the SMS section.

The **Status** section lists summary information for the SIM card.

The first row shows if the device is currently running a data session. If it is, the start of the session and the current WAN IP address of the device is displayed.

The second row shows further status information: The ICCID of the SIM card, the activation state of the SIM card and, if set, the fixed IP address assigned to the SIM card. Provided you have ADMIN permission for Connectivity, you can change the activation state by using the drop-down menu.

At the bottom you will find usage information for the current month, that is, from the first of the month till today. Hovering over the tooltip shows the covered time period, including the usage during the past month.

The **SMS** section shows the text messages sent to the device and received from the device, including the following information:

- When the message was sent or received.
- Where it was sent from and where it was sent to.
- The delivery status of the message:
  - For messages to the device: “Pending”, if it was not yet received by the device, or “Delivered”, if it was received by the device.
  - For messages from the device: “Received”, if it was received by the SIM provider, or “Canceled”, if it was not yet received by the SIM provider.
- What the direction of the message is: MT (“Mobile terminated”), if it went to the device, or MO (“Mobile originated”) if it came from the device.

Provided you have ADMIN permission for Connectivity, you can also send text messages to the device by entering the text and clicking **Send SMS**.

The **Sessions** section shows the log of data sessions carried out by the device. It lists when the session started, how long it took and how much data traffic was consumed.

The **Audit logs** section lists all changes to the SIM card and its tariff. It shows the type of change, old and new values when the change was carried out by whom, and if it was successful.

The **Connectivity** tab does not update in real-time. To show current data, click the **Reload** in the top menu bar.

CHECKING CONNECTIVITY

If you suspect that a device is not correctly reporting to Cumulocity, or it is not receiving commands, you can verify the connectivity status of the device.

In the **Connectivity** tab, check the following conditions:

- The SIM is activated. If the SIM card is not activated, you can activate it selecting “ACTIVE” from the “SIM status” drop-down menu. It may take a while until the SIM card is activated in the network. There may be a reset of the device needed to make it dial up to the network again.
- The device is connected to the network. If the device is not connected to the network, this may have several reasons:
    - The device is in a location without mobile network coverage. If the device reports network quality parameters, you can navigate to the **Measurements** tab of the device and verify the last reported signal strength and error rate parameters.
    - There is a network or hardware problem (antenna, modem). For the Jasper Control Center, for example, click the settings icon on the top right and select **SIM details**, then open the Jasper Control Center diagnostics tool. If the device is not attempting to connect to the network, it may be broken.
    - The device is in a data session. If the device is not in a data session, this may, again, have several reasons:
    - The APN settings are incorrectly configured in the device.
    - The SIM card is over traffic limit.
    - Data roaming is disabled on the device and the device is not in the SIM card’s home network.
    - Data roaming for the particular network is not included in the SIM card’s plan.
    - The SIM configuration was changed.

Data connectivity can be analyzed in various places:

- If the device reports its network configuration, navigate to the **Network** tab and verify, potentially edit, APN settings.
- If the device supports shell, navigate to the **Shell** tab and verify, potentially edit, APN settings and roaming configuration.
- Check the **Sessions** section on the **Connectivity** tab to see if the device has been communicating earlier and how much traffic it used.
- Check the **Audit logs** section on the **Connectivity** tab to see if there were any recent changes to the SIM card.
- Finally, click the cogwheel on the top right and select **SIM details** to navigate to the SIM configuration in Jasper Control Center.

INFO

The **SIM details** menu item requires you to have a login for Jasper Control Center. This login is independently provided by your administrator.

If the device is still not reporting to Cumulocity, there may be a configuration or software problem on the device.

- The device may have lost its credentials, for example, due to a factory reset or full loss of power. In this case, you can re-register the device.
- There may be a configuration or software problem with the device, which must be analyzed in a device-specific way.